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FED-NIDS: A Bio-Inspired Federated Deep Learning-Based Network Intrusion Detection System for IoT Ecosystem Security

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To all those who have supported me throughout my journey.” **- SUNDAY**

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-MALLEM BOURHANE EDDINE ISLAM

Abstract

The exponential growth of the Internet of Things (IoT) has transformed various industries, enabling unprecedented connectivity and innovation. However, the rapid proliferation of IoT devices, coupled with their heterogeneous and resource-constrained nature, has introduced significant security vulnerabilities. These challenges necessitate robust and scalable solutions to safeguard IoT networks from an evolving cyber threat landscape. This thesis addresses this critical need by introducing FED-NIDS, a bio-inspired federated learning-based deep learning network intrusion detection system designed to enhance IoT ecosystem security.

FED-NIDS leverages Federated Learning (FL) to decentralize model training, ensuring data privacy and regulatory compliance by retaining sensitive information on local devices. This approach overcomes the limitations of centralized Network Intrusion Detection Systems (NIDS), including privacy risks, scalability bottlenecks, and single points of failure. By integrating bio-inspired algorithms, FED-NIDS optimizes deep learning models for efficiency and adaptability, catering to the dynamic and resource-constrained environments of IoT networks.

This thesis begins with a comprehensive analysis of IoT security challenges and the limitations of traditional NIDS solutions. It explores the potential of federated learning and bio-inspired optimization techniques to address these issues. The design and implementation of FED-NIDS are detailed, highlighting the integration of federated learning protocols with bio-inspired algorithms to enhance detection accuracy, scalability, and resource efficiency. The system's performance is rigorously evaluated using the real-world Aposemat IoT-23 datasets, demonstrating its superiority over conventional NIDS in detection accuracy, scalability, and privacy preservation.

Key findings of this research include the development of efficient federated aggregation protocols tailored for heterogeneous IoT devices and the application of bio-inspired optimization methods to enhance model performance in dynamic environments. Additionally, FED-NIDS successfully balances the trade-offs between computational efficiency, privacy preservation, and detection effectiveness, making it a viable solution for real-world IoT security applications.

This thesis contributes to the field of IoT cybersecurity by presenting a scalable, adaptive, and privacy-preserving intrusion detection system that addresses the unique challenges of IoT ecosystems. The findings underscore the potential of federated learning and bio-inspired algorithms in revolutionizing IoT security. Future work will explore the deployment of FED-NIDS in large-scale IoT infrastructures and its integration with advanced threat intelligence mechanisms to further enhance its capabilities.

Résumé

La croissance exponentielle de l'Internet des objets (IoT) a transformé diverses industries, permettant une connectivité et une innovation sans précédent. Cependant, la prolifération rapide des appareils IoT, associée à leur nature hétérogène et aux ressources limitées, a introduit d'importantes vulnérabilités en matière de sécurité. Ces défis nécessitent des solutions robustes et évolutives pour protéger les réseaux IoT d'un paysage de cybermenaces en constante évolution. Cette thèse répond à ce besoin critique en introduisant FED-NIDS, un système de détection d'intrusion réseau basé sur l'apprentissage fédéré et inspiré de la biologie, conçu pour améliorer la sécurité de l'écosystème IoT.

FED-NIDS s'appuie sur l'apprentissage fédéré (FL) pour décentraliser la formation des modèles, garantissant la confidentialité des données et la conformité réglementaire en conservant les informations sensibles sur les appareils locaux. Cette approche surmonte les limites des systèmes centralisés de détection d'intrusion réseau (NIDS), notamment les risques liés à la confidentialité, les goulots d'étranglement de l'évolutivité et les points de défaillance uniques. En intégrant des algorithmes bio-inspirés, FED-NIDS optimise les modèles d'apprentissage profond pour l'efficacité et l'adaptabilité, répondant aux environnements dynamiques et aux ressources limitées des réseaux IoT.

Cette thèse commence par une analyse complète des défis de sécurité de l'IoT et des limites des solutions NIDS traditionnelles. Elle explore le potentiel de l'apprentissage fédéré et des techniques d'optimisation bio-inspirées pour résoudre ces problèmes. La conception et la mise en œuvre de FED-NIDS sont détaillées, mettant en évidence l'intégration de protocoles d'apprentissage fédéré avec des algorithmes bio-inspirés pour améliorer la précision de détection, l'évolutivité et l'efficacité des ressources. Les performances du système sont rigoureusement évaluées à l'aide des ensembles de données Aposemat IoT-23 du monde réel, démontrant sa supériorité par rapport aux NIDS conventionnels en termes de précision de détection, d'évolutivité et de préservation de la confidentialité.

Les principales conclusions de cette recherche comprennent le développement de protocoles d'agrégation fédérée efficaces adaptés aux appareils IoT hétérogènes et l'application de méthodes d'optimisation bio-inspirées pour améliorer les performances des modèles dans des environnements dynamiques. De plus, FED-NIDS parvient à équilibrer les compromis entre

l'efficacité informatique, la préservation de la confidentialité et l'efficacité de la détection, ce qui en fait une solution viable pour les applications de sécurité IoT du monde réel.

Cette thèse contribue au domaine de la cybersécurité IoT en présentant un système de détection d'intrusion évolutif, adaptatif et préservant la confidentialité qui répond aux défis uniques des écosystèmes IoT. Les résultats soulignent le potentiel de l'apprentissage fédéré et des algorithmes bio-inspirés pour révolutionner la sécurité de l'IoT. Les travaux futurs exploreront le déploiement de FED-NIDS dans des infrastructures IoT à grande échelle et son intégration avec des mécanismes avancés de renseignement sur les menaces pour améliorer encore ses capacités.

ملخص

لقد أدى النمو الهائل لإنترنت الأشياء (IoT) إلى إحداث تحول في العديد من الصناعات، مما أتاح اتصالاً غير مسبوق وابتكاراً واسع النطاق. ومع ذلك، فإن الانتشار السريع لأجهزة إنترنت الأشياء، إلى جانب طبيعتها غير المتجانسة ومحدودية مواردها، قد أدى إلى ظهور ثغرات أمنية كبيرة. تستلزم هذه التحديات حلولاً قوية وقابلة للتوسع لحماية شبكات إنترنت الأشياء من مشهد التهديدات السيبرانية المتطور. تتناول هذه الأطروحة هذه الحاجة الملحة من خلال تقديم نظام الكشف عن التسلل في الشبكات (FED-NIDS) القائم على التعلم المتدرج والملهم بيولوجياً لتحسين أمن بيئة إنترنت الأشياء.

يستفيد نظام FED-NIDS من التعلم الفيدرالي (FL) لتوزيع تدريب النماذج مع الحفاظ على خصوصية البيانات والامتثال التنظيمي عن طريق إبقاء المعلومات الحساسة على الأجهزة المحلية. يتغلب هذا النهج على قيود أنظمة الكشف عن التسلل الشبكي المركزية(NIDS) ، بما في ذلك مخاطر الخصوصية واختناقات القابلية للتوسع ونقاط الفشل الفردية. من خلال دمج الخوارزميات المستوحاة من الطبيعة، يقوم FED-NIDS بتحسين نماذج التعلم العميق من حيث الكفاءة والقدرة على التكيف، بما يتماشى مع بيئات إنترنت الأشياء الديناميكية والمحدودة الموارد.

تبدأ هذه الأطروحة بتحليل شامل لتحديات أمن إنترنت الأشياء وقيود الحلول التقليدية لأنظمة الكشف عن التسلل. كما تستكشف إمكانات التعلم الفيدرالي وتقنيات التحسين المستوحاة من الطبيعة لمواجهة هذه المشكلات. يتم تفصيل تصميم وتنفيذ نظامFED-NIDS ، مع تسليط الضوء على تكامل بروتوكولات التعلم الفيدرالي مع الخوارزميات المستوحاة من الطبيعة لتحسين دقة الكشف، والقابلية للتوسع، وكفاءة الموارد. يتم تقييم أداء النظام بشكل دقيق باستخدام مجموعات بيانات Aposemat IoT-23 الواقعية، مما يُظهر تفوقه على أنظمة NIDS التقليدية في دقة الكشف، والقابلية للتوسع، والحفاظ على الخصوصية.

تشمل النتائج الرئيسية لهذا البحث تطوير بروتوكولات تجميع فيدرالية فعالة مصممة خصيصاً لأجهزة إنترنت الأشياء غير المتجانسة، وتطبيق أساليب التحسين المستوحاة من الطبيعة لتحسين أداء النماذج في البيئات الديناميكية. بالإضافة إلى ذلك، ينجح نظام FED-NIDS في تحقيق توازن بين الكفاءة الحسابية، والحفاظ على الخصوصية، وفعالية الكشف، مما يجعله حلاً قابلاً للتطبيق لتطبيقات أمن إنترنت الأشياء الواقعية.

تساهم هذه الأطروحة في مجال أمن إنترنت الأشياء من خلال تقديم نظام كشف عن التسلل قابل للتوسع، متكيف، ويحافظ على الخصوصية، يعالج التحديات الفريدة لبيئات إنترنت الأشياء. وتؤكد النتائج على إمكانات التعلم الفيدرالي والخوارزميات المستوحاة من الطبيعة في إحداث ثورة في أمن إنترنت الأشياء. ستستكشف الأعمال المستقبلية نشر نظام FED-NIDSفي بنى تحتية واسعة النطاق لإنترنت الأشياء ودمجه مع آليات استخبارات التهديدات المتقدمة لتعزيز قدراته بشكل أكبر.

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List of acronyms

Acronym	Meaning
6LoWPAN	IP6 Low power Wireless Personal Area Networks
AIDS	Anomaly-based Intrusion Detection System
AE	Auto-encoder
ANN	Artificial Neural Network
BLE	Bluetooth Low Energy
CIA	Confidentiality, Integrity, Availability
CNN	Convolutional Neural Network
DL	Deep Learning
DoS	Denial of Service
DT	Decision Tree
DTG	Discrete-Time Dynamic Graph
FNN	Feedforward Neural Network
FL	Federated Learning
GNN	Graph Neural Network
GRU	Gated Recurrent Unit
HIDS	Host Intrusion Detection System
IDS	Intrusion Detection System
IoE	Internet of Everything
loT	Internet of Things
IOTWF	loT World Forum
LoRaWAN	Long-Range Wide-Area Network
LSTM	Long Short-Term Memory
ML	Machine Learning
MLP	Multilayer Perceptron
NIDS	Network Intrusion Detection System
PCA	Principal Component Analysis
RFID	Radio Frequency Identification
RNN	Recurrent Neural Network
SIDS	Signature-based Intrusion Detection System
SVD	Singular Value Decomposition
SVM	Support Vector Machine
TCP/IP	Transmission Control Protocol/Internet Protocol
WSN	Wireless Sensor Networks

General Introduction

The Internet of Things (IoT) has revolutionized our interconnected world, permeating various aspects of modern life, from smart homes and cities to industrial applications and healthcare systems. This proliferation of IoT devices has led to an unprecedented surge in data generation and network traffic, opening new avenues for innovation while simultaneously presenting significant security challenges. As IoT ecosystems become increasingly complex and ubiquitous, ensuring their security has evolved into a critical concern for individuals, businesses, and governments alike.

Projections estimate that by 2025, there will be over 75 billion connected IoT devices worldwide demonstrate IoT's rapid growth, and the number of IoT devices worldwide is forecast to rise to almost close to 30 billion in 2030 [1]. This exponential growth, coupled with the heterogeneous nature of IoT devices and their often resource-constrained characteristics, creates a unique and challenging landscape for cybersecurity professionals [2] .

Problem Statement

The rapid expansion and inherent complexity of IoT networks have outpaced the development of adequate security measures, leaving these systems vulnerable to a wide array of cyber threats. Traditional Network Intrusion Detection Systems (NIDS) face significant limitations when applied to IoT environments, struggling to cope with the massive scale of networks that can comprise thousands or even millions of devices. The heterogeneous nature of IoT ecosystems, consisting of diverse devices with varying capabilities, protocols, and security features, further complicates the implementation of a unified security solution.

Many IoT devices operate under severe resource constraints, with limited computational power, memory, and energy resources. This restricts the direct deployment of complex security measures on the devices, necessitating more efficient and lightweight approaches. Additionally, the dynamic nature of IoT networks, where devices frequently join and leave, demands adaptive security solutions capable of responding to changing network topologies and emerging threats in real-time.

The sensitive nature of data collected by IoT devices raises serious privacy concerns, making centralized security solutions problematic and potentially non-compliant with data protection regulations. The need for real-time threat detection and response in many critical IoT applications, which is challenging to meet given the volume and velocity of data generated in IoT networks, exacerbates this privacy challenge. Furthermore, the IoT threat landscape is constantly evolving, with new attack vectors and techniques emerging regularly, necessitating security solutions that can adapt to and learn from new threat patterns.

While the application of machine learning and deep learning techniques to IoT security shows promise, it introduces additional challenges. The effectiveness of these models often depends on large datasets, but privacy concerns and regulations can prevent the centralization of IoT data, resulting in data silos that hinder model training. This creates a tension between model accuracy and privacy preservation, as improving model performance typically requires more data, which can conflict with privacy goals. Moreover, the computational overhead of sophisticated deep learning models can potentially overwhelm resource-constrained IoT devices, making their direct implementation impractical in many scenarios.

In this complex landscape, there is a pressing need for innovative approaches to IoT security that can address these multifaceted challenges while respecting privacy constraints and resource limitations. Such solutions must be scalable to accommodate the vast number of IoT devices, adaptable to diverse and dynamic network environments, and capable of providing robust protection against both known and emerging threats. They must also strike a delicate balance between leveraging the power of advanced machine learning techniques and operating within the constraints of IoT ecosystems.

Federated Learning (FL) has emerged as a promising paradigm that allows for collaborative model training without centralizing data, potentially addressing many of the privacy and scalability concerns inherent in IoT security. When combined with deep learning techniques and bio-inspired algorithms, FL offers the potential to create robust, adaptive, and privacy-preserving NIDS for IoT ecosystems [3]. To reach this potential, however, we need to solve some big technical problems. These include creating federated learning protocols that work well,

developing ways to protect privacy, and developing algorithms that can work well in the resource-limited and diverse environment of IoT networks.

Objectives and Organization of the Thesis

The main goal of this thesis is to create FED-NIDS, a bio-inspired federated deep learning-based network intrusion detection system that aims to keep the IoT ecosystem safe. The research has the following primary objectives:

1. Conduct a comprehensive analysis of the unique security challenges posed by IoT ecosystems, and critically evaluate the limitations of current NIDS solutions in addressing these challenges.
2. We aim to explore the potential of federated learning, deep learning techniques, and bio-inspired algorithms in augmenting IoT security, emphasizing their capacity to offer scalable, privacy-preserving, and adaptive intrusion detection.
3. We aim to create and execute a cutting-edge NIDS architecture that seamlessly combines bio-inspired algorithms, federated learning, and deep learning techniques to surmount the recognized constraints of current solutions.
4. The goal is to develop efficient federated learning protocols and aggregation algorithms tailored for the heterogeneous and resource-constrained nature of IoT devices.
5. We aim to develop bio-inspired optimization methods that can improve the efficiency and flexibility of federated deep learning models in constantly changing IoT settings.
6. We're implementing privacy-preserving mechanisms within the FED-NIDS framework to ensure compliance with data protection regulations and address privacy concerns in IoT data processing.
7. We aim to carry out comprehensive tests and assessments to gauge the efficacy, efficiency, scalability, and privacy-protection capabilities of the suggested FED-NIDS system in diverse IoT scenarios and datasets.
8. We aim to evaluate the effectiveness of FED-NIDS in comparison to traditional NIDS and other cutting-edge methods, offering a thorough examination of its advantages and possible drawbacks.

9. We aim to investigate the practicality of FED-NIDS in the real world by taking into account the challenges of deployment and suggesting methods for incorporating it into current IoT infrastructures.

Structure of the Thesis

The state-of-the-art review and the contribution are the two main sections of this thesis.

Part 1: The State-of-the-Art Review

Chapter 1 focuses on the Internet of Things (IoT) and the challenges associated with ensuring security. This chapter provides an overview of IoT technology, its applications, and the unique security challenges it presents. It explores the layers and components of the IoT communication architecture and discusses common attacks, threats, and vulnerabilities in IoT systems.

Chapter 2 explores the use of network intrusion detection systems (NIDS) on the Internet of Things. This chapter delves into the fundamentals of NIDS, including different types and detection techniques. It examines the challenges of implementing traditional NIDS in IoT environments and explores recent advancements in IoT-specific NIDS solutions.

Chapter 3 delves into the application of machine learning and deep learning techniques for NIDS. This chapter investigates the application of machine learning and deep learning techniques in network intrusion detection. It discusses various algorithms, their advantages, and their limitations in the context of IoT security.

Chapter 4 presents Federated Learning for Secure and Privacy-Preserving IoT. This chapter introduces the concept of federated learning and its potential applications in IoT security. It explores how federated learning can address privacy concerns while enabling collaborative model training across distributed IoT devices.

Part 2: Contribution

Chapter 5 focuses on FED-NIDS: System Architecture and Design. This chapter presents the proposed FED-NIDS architecture, detailing its components, data flow, and the integration of bio-inspired algorithms with federated deep learning techniques.

Chapter 6. Implementation and Experimental Setup: This chapter describes FED-NIDS's implementation details, including the federated learning framework, deep learning models, and bio-inspired optimization algorithms used.

Chapter 7. Performance Evaluation and Results: This chapter presents a comprehensive evaluation of FED-NIDS, comparing its performance against traditional NIDS and other state-of-the-art approaches. It analyzes the system's detection accuracy, efficiency, scalability, and privacy preservation capabilities.

Chapter 8. Conclusion and Future Work: The final chapter summarizes the thesis's key contributions, discusses the implications of the findings, and proposes directions for future research in federated learning-based IoT security.

By addressing the critical need for advanced, privacy-preserving security solutions in IoT ecosystems, this thesis aims to contribute significantly to the development of more resilient and trustworthy IoT technologies. The FED-NIDS system is a new idea that combines the strengths of bio-inspired algorithms, federated learning, and deep learning to solve the complicated and always-changing security problems that come up in modern IoT network settings. This research not only advances the field of IoT security but also paves the way for future innovations in privacy-preserving, distributed intelligence for cyber-physical systems.

PART 1

STATE-OF-THE-ART

Chapter 1: The Internet of Things (IoT) and the Challenges of Ensuring Security

The Internet of Things (IoT) is a notable technological advancement in which physical things can connect and communicate with each other through the internet. This network of networked devices allows for the interchange of data, the execution of computations, and the coordination of operations without the need for human interaction [4]. Many industries, including housing, healthcare, transportation, agriculture, and smart cities, utilize Internet of Things (IoT) applications. These applications improve efficiency and enable more intelligent operations [4] [5].

1.1 Internet of Things (IoT) devices

The Internet of Things (IoT) refers to a large and interconnected system of objects and networks that communicate with each other through the Internet [6]. These devices, commonly known as "things" or "entities," have certain essential characteristics that allow them to engage in the Internet of Things (IoT):

- **Identification**: IoT devices have unique identifying codes, like electronic product codes (EPC) or universal codes (uCode), that allow for individual address and monitoring [6].
- **Sensing**: Numerous IoT devices incorporate sensors to gather data about their surroundings or conditions [6].
- **Communication**: IoT devices employ a range of protocols and technologies, such as Wi-Fi, Bluetooth, ZigBee, and cellular, to communicate and share data with other devices and systems [6].
- **Computation**: IoT devices commonly have computation capabilities, which can vary from basic microcontrollers to more advanced systems-on-chip (SoC) [6].
- **Actuation**: Certain IoT devices have the capability to influence their surroundings using actuators or other mechanisms [6].
- **Energy efficiency**: Many IoT devices specifically use batteries or harvested energy to reduce power consumption [6].

- **Security**: IoT devices necessitate the inclusion of inherent security measures to safeguard against unauthorized entry and data breaches [6].

IoT devices encompass a diverse array of examples, including consumer items such as intelligent thermostats and fitness trackers, as well as industrial sensors and self-driving vehicles. Small-scale Internet of Things (IoT) projects frequently use development boards like Arduino, Raspberry Pi, BeagleBone, CubieBoard, and radio frequency identification (RFID) tags for prototyping and implementation [6]. These boards often consist of a microcontroller (MCU) or microprocessor, a read-only memory (ROM), a random-access memory (RAM), and a range of digital and analog general-purpose input/output interfaces and connectivity choices.

Sensors are essential components in numerous IoT applications, enabling devices to collect data regarding their surroundings. Typical sensor types present in IoT devices include temperature, accelerometers, light, moisture, vibration, air quality, and proximity sensors [6]. For example, a smart home system could include temperature sensors to regulate the climate, motion sensors for security purposes, and air quality sensors to monitor indoor contaminants.

In essence, RTOS is important for tasks like information processing, memory management, and offering utility services that facilitate messaging and other types of communication [6]. The selection of a real-time operating system (RTOS) is based on the performance, security, and functional requirements of the product. Some of the noteworthy operating systems for the Internet of Things (IoT) are TinyOS, Contiki, Mantis, FreeRTOS, BrilloOS, and ARM's mbedOS [7] [6].

The wide range of Internet of Things (IoT) devices and their various functionalities offer advantages as well as difficulties for manufacturers, developers, and users alike. As the Internet of Things (IoT) progresses, researchers and industry professionals are placing significant emphasis on guaranteeing interoperability, security, and efficient resource use within this diverse collection of devices [6].

1.2 The Layers and Components of IoT Communication Architecture

Multiple layers, each with unique functions crucial for the overall operation of the system, typically organize the architecture of the Internet of Things (IoT) [7] [8] [9]. Academics and

industry widely accept a four-layer model, despite the lack of a widely recognized architecture. This model offers a complete framework for comprehending the movement of data and processes in Internet of Things (IoT) systems.

- **Perception Layer**: The IoT architecture is based on the perception layer, also known as the physical or sensory layer [7] [9]. It consists of a variety of sensors, actuators, and gadgets that directly interact with the physical environment. These devices gather unprocessed data from their environment, like temperature, humidity, motion, or light intensity. Recent technological breakthroughs have resulted in the creation of highly advanced and energy-efficient sensors, such as MEMS (Micro-Electro-Mechanical Systems) sensors, which provide enhanced accuracy and lower power usage [8]. Furthermore, the perceptual layer, RFID (Radio-Frequency Identification) tags, and readers are all essential components of IoT systems for object identification and tracking [7].
- **Network Layer**: The network layer, also known as the transmission layer, is responsible for the reliable transmission of data acquired by the perception layer to higher layers. References [7] [8] have discussed the architecture. This layer utilizes a diverse range of communication technologies. Both wired and wireless connections are used to provide uninterrupted connectivity. Notable advancements in this field involve the use of low-power wide-area networks (LPWAN), such as LoRaWAN, SigFox, and NB-IoT. These technologies enable long-distance communication while consuming little power [5] [9]. 5G networks are gaining significance due to their ability to offer high-speed, low-latency connections that are essential for real-time Internet of Things (IoT) applications [5] [8]. The network layer is responsible for data routing, utilizing protocols such as 6LoWPAN and RPL (Routing Protocol for Low-Power and Lossy Networks). We tailor the networks for IoT contexts, focusing on power efficiency and minimizing data loss [5] [7].
- **Middleware Layer**: The middleware layer, sometimes referred to as the processing or service management layer, is a critical component in data processing and management [7] [8]. This layer serves as a connection between the unprocessed data obtained by the perception layer and the applications that make use of it. The system uses data filtering and aggregation to reduce network traffic and improve system efficiency [8]. The middleware

layer also incorporates edge computing principles, which involve pushing processing closer to data sources and reducing the term "latency" refers to the time lag or delay in a system or process [9]. The primary roles of this layer are semantic analysis and data abstraction, which convert unprocessed data into practical information [1]. Recent developments in this field involve the utilization of artificial intelligence and machine learning algorithms to enhance data analysis and decision-making [9].

- **Application Layer**: The application layer serves as the intermediary between the IoT system and its end users. Individuals who use data to generate valuable services and perform actions are called users [7] [8] [9]. This layer contains a diverse range of uses in various fields, such as smart homes and buildings, healthcare monitoring, industrial IoT (IIoT) applications, and smart cities. Current developments in this field include the advancement of more user-friendly and easy-to-understand user interfaces, the incorporation of voice assistants for seamless integration, and the use of augmented reality to improve user interaction with Internet of Things (IoT) devices [9].

Each layer in IoT systems is essential for collecting, transmitting, processing, and utilizing data from the physical environment. As Internet of Things (IoT) technologies persist, we can anticipate additional advancements at each of these levels, resulting in enhanced efficiency. We design IoT ecosystems to be both secure and simple to use. Figure 1 illustrates the several layers included in an Internet of Things (IoT) system.

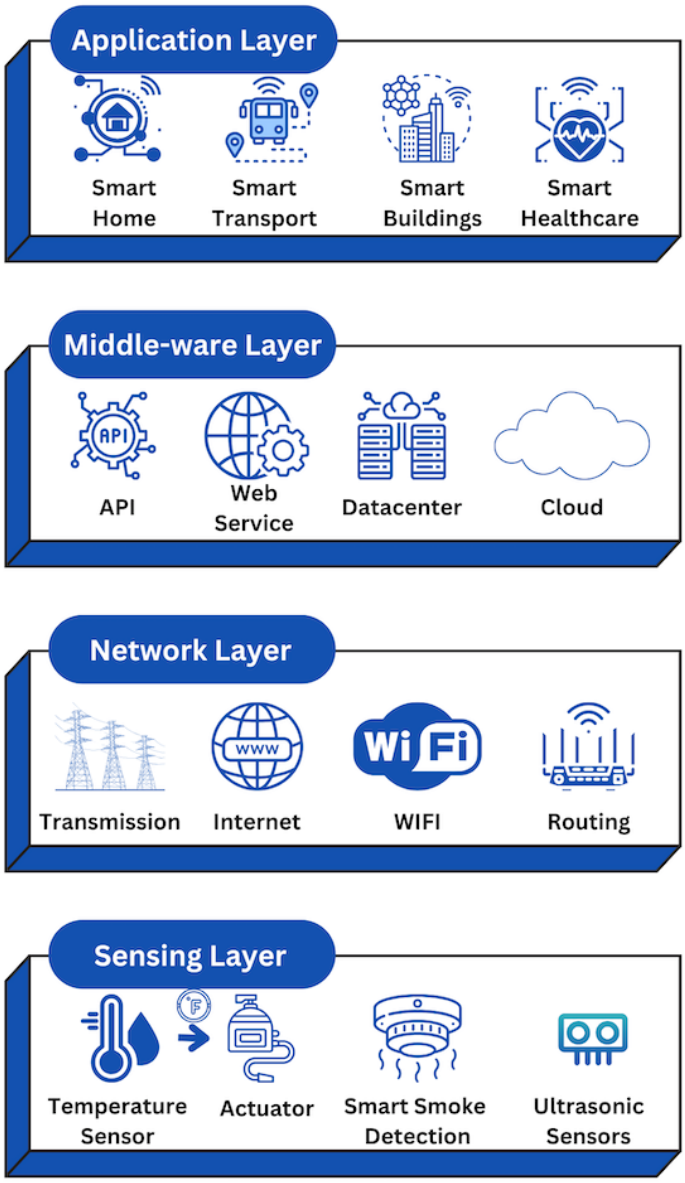


Figure 1 : Layers in IoT System

1.3 Applications and Utilization of the Internet of Things (IoT)

IoT applications are extensive and diverse, catering to various requirements in different sectors. Every application utilizes the distinct capabilities of the Internet of Things (IoT) to address issues and enhance effectiveness. We can classify the applications of IoT into several categories, such as healthcare, medical, military, industrial, automotive, environmental, agricultural, retail, and consumer. Furthermore, there exist specific areas of focus within the field of technology, such

as transportation, logistics, and smart environments, encompassing smart homes and offices [10]. In the subsequent sections, we will examine a few of these uses.

- **Smart Homes**: IoT facilitates the remote control and monitoring of household appliances, security systems, and energy usage, thereby improving convenience and effectiveness. Smartphones or voice assistants can manage smart home gadgets such as thermostats, lighting systems, security cameras, and smart locks [8]. These systems can acquire user preferences and automate procedures to conserve energy and enhance convenience.
- **Healthcare**: Wearable gadgets and remote monitoring systems enhance healthcare by providing immediate health data, resulting in improved patient care and reduced hospital visits. IoT applications in healthcare encompass the use of continuous glucose monitors, smart inhalers, and connected scales to aid in the management of chronic diseases [8] [9]. In addition, remote patient monitoring systems have the capability to notify healthcare providers of impending problems, allowing for prompt action and lowering healthcare expenses.
- **Smart Cities**: In smart cities, the IoT enables effective urban administration by supporting tasks such as traffic regulation, garbage handling, and energy allocation. Smart city applications utilize sensors to monitor the flow of traffic, air quality, and energy usage. This allows city planners to maximize resources and enhance the quality of life for people [8] [9]. Smart traffic signals can adjust their behavior based on real-time traffic circumstances to minimize traffic congestion and pollution.
- **Agriculture**: We use Internet of Things (IoT) sensors to monitor soil moisture, weather conditions, and crop health. This allows for the optimization of farming techniques, resulting in increased crop output. Smart agricultural technologies encompass precision irrigation systems, crop monitoring using drones, and automated tractors. These technologies assist farmers in making data-based decisions, reducing resource usage, and improving production, thus promoting sustainable agricultural practices [8] [10].
- **The Industrial Internet of Things (IIoT) and Industrial Automation**: IoT improves production processes by utilizing predictive maintenance, real-time monitoring, and automation. By utilizing connected sensors and machines, it is possible to identify equipment faults in advance, thereby minimizing downtime and lowering maintenance

expenses [10]. The Internet of Things (IoT) enables the seamless integration of production lines, enhancing efficiency and flexibility in industrial Internet of Things (IIoT) operations.

- **Retail, Consumer, and Logistics Supply Chain**: The Internet of Things (IoT) improves merchandise monitoring and control across retail, consumer, and logistics supply chains. Connected sensors and GPS devices offer immediate insight into the whereabouts and state of shipments, aiding firms in optimizing inventory levels, minimizing losses, and guaranteeing prompt delivery of goods [10].

Whether an IoT application is already in use or in the process of implementation, security is crucial. The use of IoT is rapidly increasing and spreading across a variety of industries. Operators enable these IoT applications over existing networking. To meet the demands of various applications, it is necessary to include stronger security measures in the utilized technology.

1.4 Security Challenges on the Internet of Things (IoT).

While the Internet of Things (IoT) provides several benefits, it also poses significant security challenges. Regrettably, like the modern Internet and its precursor, the ARPANET, the majority of commercial IoT devices do not give importance to security in their design. This is particularly accurate for inexpensive, consumer-focused, widely manufactured IoT devices. To achieve both competitive pricing and significant profits, manufacturers often implement cost reduction strategies, which may result in the delivery of devices with default passwords and old or publicly known vulnerabilities. Attackers can readily exploit these vulnerabilities [9]. Internet of Things (IoT) devices, especially those connected to the internet, are inherently more vulnerable to security breaches due to their overall lack of strong security mechanisms and infrequent software updates. Manufacturers usually design these devices to be cost-effective, which leads to restricted processing capabilities that make it difficult to include robust security features. Furthermore, manufacturers sometimes prioritize cost reduction over ensuring security. Factors influencing this include a lack of knowledge, a reluctance to allocate resources for security measures, limited funding, pressure to meet production targets, concerns about the time and complexity of implementing advanced security features, and a greater willingness to take risks to save money or allocate funds elsewhere. These challenges can also arise because of the proliferation of devices, their varied and heterogeneous characteristics, their extensive range of capabilities, and the

sensitivity of the data they handle [7] [9] [11]. We've discussed several of these security concerns here:

- **Device Heterogeneity**: Device heterogeneity refers to the wide variety of capabilities and security features included in IoT devices, making it difficult to establish uniform security solutions. Systems differ in their capacities, ranging from simple sensors with limited processing power to complex systems with substantial computing capability [10]. The existence of heterogeneity poses a challenge when attempting to enforce consistent security standards across all devices [7] [10].
- **Scalability**: Ensuring secure and efficient communication becomes more complex as the number of IoT devices grows. Scalability difficulties stem from the need to efficiently handle numerous devices, each of which generates data and requires secure connections [7] [10]. Expanding the functionality of existing systems and ensuring that new devices can work together is a difficult task, especially when faced with a diverse array of hardware platforms and communication protocols [345]. Traditional security measures may not be sufficiently adaptable to the scale of the Internet of Things (IoT) network, thus requiring the adoption of novel approaches.
- **Ensuring Data Privacy**: Internet of Things (IoT) devices often collect sensitive personal information, necessitating robust privacy protection methods to thwart illegal access. The collected data may include health information, geographic data, and personal preferences. The unauthorized acquisition of this data might result in substantial violations of privacy [7] [9] [10]. To ensure the confidentiality of data, it is imperative to include strong encryption, access controls, and anonymization technologies.
- **Lack of Interoperability**: The absence of interoperability is a significant challenge in the IoT business. The existence of a wide range of items on multiple platforms adds complexity to the task of ensuring secure communication and collaboration among different devices and systems. Devices from various manufacturers sometimes utilize multiple protocols and standards, which might present difficulties for integration [4] [7] [9] [10]. Interoperability issues can lead to security risks when devices are unable to securely exchange information.
- **Physical Security and Natural Disasters**: IoT devices are frequently located in unsafe or remote regions, making them susceptible to physical tampering or theft, especially in the

event of natural catastrophes. Natural disasters may expose mobile or sensing devices to theft, while stationary equipment may undergo relocation or damage [7]. Implementing tamper-evident designs, secure enclosures, and location-based access controls are essential physical security methods for protecting equipment from unauthorized physical access.

- **Resource Constraints**: Resource constraints pose a challenge for the application of comprehensive security mechanisms to IoT devices. These constraints include limited processor capabilities, low memory capacity, and finite battery life [7] [9] [10]. Devices with limited resources require the adoption of lightweight cryptographic algorithms and efficient security methods to guarantee protection without compromising speed.
- **Software Vulnerabilities**: Software vulnerabilities arise when IoT devices run outdated or unpatched software, making them susceptible to known security weaknesses [9]. Moreover, multiple investigations indicate that unencrypted channels on the Internet transmit a staggering 98% of all user data, commands, and sensor readouts. The absence of encryption makes them extremely vulnerable to even the most basic types of man-in-the-middle attacks, allowing attackers to intercept, read, or modify important plaintext data without the knowledge of the sender or receiver [349, 350, 351]. Fixing security vulnerabilities and preventing device exploits requires regular software updates and patches.

1.5 Common Attacks, Threats, and Vulnerabilities on IoT Systems and Structure

As outlined in subsection 1.2, an IoT application can be divided into four distinct layers: (1) the perception layer; (2) the network layer; (3) the middleware layer; and (4) the application layer. Each layer uses a variety of technologies, which introduce several challenges and security risks. Figure 1 illustrates the different technologies, devices, and applications present within these layers. This section explores the most common and emerging attacks, threats, and vulnerabilities specific to each layer of the IoT architectural layer. Figure 2 depicts the potential attacks targeting these layers.

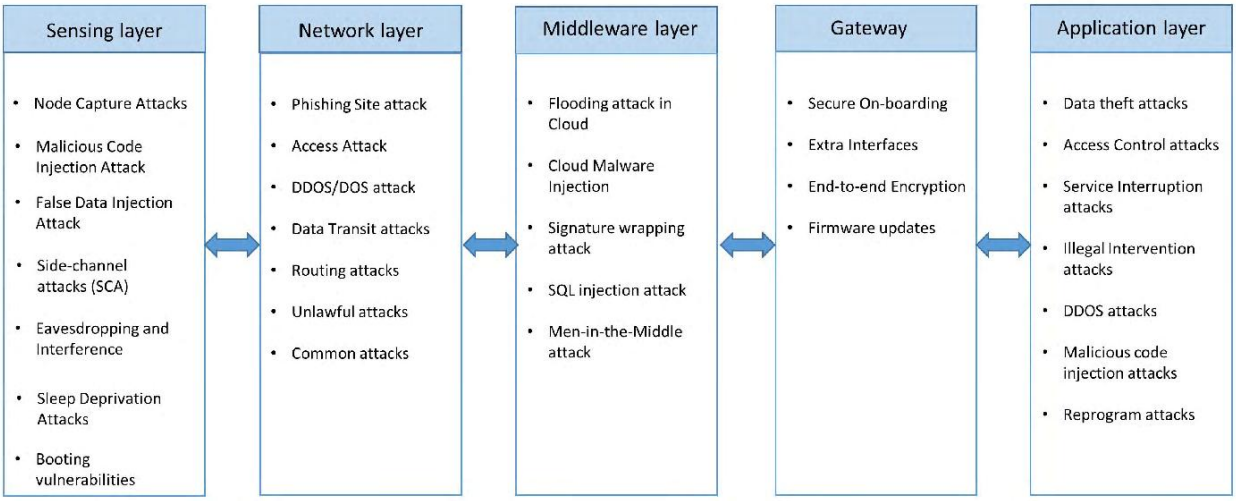


Figure 2 : Types of Attacks on IoT Based on the Different Layers

1.5.1 Perception or Physical Layer Attacks

The perception layer, also known as the physical or sensing layer, being the primary interface with the physical world, is particularly vulnerable to various attacks. This layer mainly deals with physical IoT sensors and actuators. Sensors sense the physical phenomenon happening around them. Actuators, on the other hand, perform a certain action on the physical environment based on the sensed data [347]. There are various kinds of sensors for sensing different kinds of data, e.g., ultrasonic sensors, camera sensors, smoke detection sensors, temperature and humidity sensors, etc. There can be mechanical, electrical, electronic, or chemical sensors used to sense the physical environment [347]. Various sensing layer technologies are used in different IoT applications like RFID, GPS, WSNs, RSNs, etc.

Major security threats that can be encountered at the perception layer are as follows:

- **Node Capture and Tampering**: IoT applications comprise of several low-power nodes, such as sensors and actuators. These nodes are vulnerable to a variety of attacks by adversaries. The attackers may try to capture or replace the node in the IoT system with a malicious node [347]. Attackers may physically access and manipulate IoT devices to extract sensitive information or alter their functionality [7] [9]. This can involve modifying the hardware or firmware, potentially compromising the entire IoT network. The risk is heightened by the often-unattended nature of IoT devices in remote or public locations. For

example, an attacker might tamper with a smart meter to manipulate energy consumption readings or extract cryptographic keys stored in the device.

- **Node Jamming Attacks**: These attacks overwhelm the communication channels with interference signals, preventing legitimate devices from transmitting data [7] [9]. This type of Denial of Service (DoS) attack is particularly effective against wireless sensor networks and can significantly disrupt IoT operations. Advanced jamming techniques can selectively target specific frequencies or protocols, making detection and mitigation more challenging [345].
- **Malicious Code Injection Attack**: The attack involves the attacker injecting some malicious code into the memory of the node. Generally, the firmware or software of IoT nodes is upgraded on the air, and this gives a gateway to the attackers to inject malicious code. Using such malicious code, the attackers may force the nodes to perform some unintended functions or may even try to access the complete IoT system [347].
- **False Data Injection Attack**: Malicious actors can introduce fake sensor readings or manipulate data at the source or a vulnerable node, leading to incorrect decision-making in the higher layers of the IoT system [7] [9]. Once a node is captured, the attacker may use it to inject erroneous data onto the IoT system. This may lead to false results and may result in malfunctioning of the IoT application [347]. This attack can have severe consequences, especially in critical infrastructure or healthcare applications. For instance, an attacker could manipulate temperature sensor readings in an industrial control system, potentially causing equipment damage or safety hazards.
- **Side-Channel Attacks (SCA)**: These sophisticated attacks exploit the physical implementation of IoT devices, such as power consumption patterns or electromagnetic emissions, to extract sensitive information like cryptographic keys [346, 347]. Apart from direct attacks on the nodes, various side-channel attacks may lead to the leakage of sensitive data [347]. Recent research has shown that even low-cost equipment can be used to perform power analysis attacks on IoT devices, highlighting the need for hardware-level security measures [347].
- **Sensor Spoofing**: This attack involves tricking sensors into reporting false data by manipulating their physical environment. For example, an attacker might use ultrasonic

signals to fool a smartphone's accelerometer or use fake GPS signals to mislead location-based services [346].

1.5.2 Network Layer Attacks

The network layer's key function is responsible for information transmission; it transmits the information received from the perception layer to the computational unit for processing. It faces several security challenges; some of the major attacks encountered in this layer are:

- **Man-in-the-Middle (MITM) Attacks**: Attackers intercept and potentially alter communication between IoT devices and servers [7] [9]. This can lead to data theft, manipulation of transmitted information, or unauthorized control of devices. The risk is particularly high in IoT systems that lack proper encryption and authentication mechanisms. Advanced MITM attacks may exploit vulnerabilities in protocols like Bluetooth Low Energy (BLE) to intercept and manipulate data transmitted between IoT devices and smartphones [345].
- **Routing Attacks**: In such attacks, malicious nodes in an IoT application may try to redirect the routing paths during data transit [9]. Various routing attacks, such as sinkhole attacks, wormhole attacks, and selective forwarding, can disrupt the network topology and data flow in IoT systems [7] [8]. These attacks can lead to data loss, increased latency, or complete network partition. In a sinkhole attack, a malicious node advertises fake routing information to attract traffic, potentially intercepting or dropping packets [4] [9]. Wormhole attacks create tunnels in the network to disrupt normal routing, while selective forwarding involves dropping specific packets to disrupt communication [347].
- **Sybil Attacks**: An attacker creates multiple fake identities to gain disproportionate influence in a network [7] [9]. This can compromise voting systems, routing mechanisms, and data aggregation in IoT networks, particularly in peer-to-peer or consensus-based systems. In the context of IoT, Sybil attacks can be particularly damaging in applications like smart grids or vehicular networks, where node reputation and consensus are critical [347].
- **DDoS/DoS Attacks**: By overwhelming IoT devices or servers with a flood of requests, attackers can render services unavailable [7] [8] [9]. The distributed nature of IoT makes

it particularly vulnerable to large-scale DDoS attacks, often facilitated by botnets comprised of compromised IoT devices [345]. Many IoT devices in IoT applications are not strongly configured and thus become easy gateways for attackers to launch DDoS attacks on the target servers. The Mirai botnet attack used this vulnerability and blocked various servers by constantly propagating requests to the weakly configured IoT devices [347].

- **Data Transit Attacks**: IoT applications deal with a lot of data storage and exchange. Data is valuable, and therefore it is always the target of hackers and other adversaries. Data that is stored in the local servers or the cloud has a security risk, but the data that is in transit or is moving from one location to another is even more vulnerable to cyber-attacks. In IoT applications, there is a lot of data movement between sensors, actuators, clouds, etc. Different connection technologies are used in such data movements, and therefore IoT applications are susceptible to data breaches [347].

1.5.3 Middleware Layer Attacks

The role of the middleware in IoT is to create an abstraction layer between the network layer and the application layer. Middleware can also provide powerful computing and storage capabilities. This layer provides APIs to fulfill the demands of the application layer. The middleware layer includes brokers, persistent data stores, queuing systems, machine learning, etc. [347]. Although the middleware layer is useful to provide a reliable and robust IoT application, it is also susceptible to various attacks. These attacks can take control of the entire IoT application by infecting the middleware.

Database security and cloud security are the other main security challenges in the middleware layer, where data processing and management are crucial. Various possible attacks in the middleware layer are discussed as follows:

- **Data Poisoning**: Attackers may inject malicious data into the system, compromising the integrity of data processing and decision-making algorithms [8] [9]. This is particularly concerning with the increasing use of machine learning in IoT systems. For example, an

attacker might inject false data to skew the results of a machine learning model used for predictive maintenance in an industrial IoT setting.

- **SQL Injection Attack**: Middleware is also vulnerable to SQL Injection (SQLi) attacks. In such attacks, attackers can embed malicious SQL statements in a program [347]. Then, the attackers can obtain private data of any user and can even alter records in the database [347]. Open Web Application Security Project (OWASP) has listed SQLi as a top threat to web security in their OWASP Top 10 2018 document [347].
- **Cloud Malware Injection**: In cloud malware injection, the attacker can obtain control, inject malicious code, or inject a virtual machine into the cloud. The attacker pretends to be a valid service by trying to create a virtual machine instance or a malicious service module [347]. In this way, the attacker can obtain access to service requests of the victim's service and can capture sensitive data, which can be modified as per the instance.
- **Man-in-the-Middle Attack**: The MQTT protocol uses a publish-subscribe model of communication between clients and subscribers using the MQTT broker, which effectively acts as a proxy. This helps in decoupling the publishing and the subscribing clients from each other, and messages can be sent without the knowledge of the destination [347]. If the attacker can control the broker and become a man-in-the-middle, then he/she can get complete control of all communication without any knowledge of the clients.

1.5.4 Application Layer Attacks

The application layer, being the user-facing component, is often targeted for various attacks. It deals directly with and provides services to end users. IoT applications like smart homes, smart meters, smart cities, smart grids, etc. lie in this layer. This layer has specific security issues that are not present in other layers, such as data theft and privacy issues [347]. The security issues in this layer are also specific to different applications. Many IoT applications also consist of a sub-layer between the network layer and application layer, usually termed an application support layer or middleware layer. The support layer supports various business services and helps in intelligent resource allocation and computation [347].

Major security issues encountered by the application layer are discussed below.

- **Malware Infections**: Attackers can exploit vulnerabilities to inject malicious software into IoT applications [9]. This can lead to data theft, unauthorized access, or device control. The limited resources of many IoT devices often make it challenging to implement robust anti-malware measures. Recent trends show an increase in IoT-specific malware, with some variants capable of spreading across multiple device architectures.
- **Data Thefts**: IoT applications deal with a lot of critical and private data. The data in transit is even more vulnerable to attacks than data at rest, and in IoT applications, there is a lot of data movement. The users will be reluctant to register their private data on IoT applications if these applications are vulnerable to data theft attacks [347]. This is particularly concerning given the personal nature of data collected by many IoT devices, such as health monitors or smart home systems. Recent high-profile breaches have highlighted the potential for IoT devices to leak sensitive personal information [348]. Data encryption, data isolation, user and network authentication, privacy management, etc. are some of the techniques and protocols being used to secure IoT applications against data thefts.
- **Social Engineering Attacks**: Users may be tricked into revealing sensitive information through fake interfaces or messages [7] [8] [9]. This human-centric vulnerability remains a significant challenge in IoT security. Phishing attacks targeting IoT device users, for example, might trick them into revealing login credentials for their smart home systems.
- **Firmware Attacks**: Exploitation of firmware vulnerabilities or the installation of malicious firmware updates can give attackers persistent control over IoT devices [9][346]. This highlights the importance of secure update mechanisms and firmware integrity checks. Advanced firmware attacks may exploit vulnerabilities in the boot process or update mechanisms to install rootkits or other persistent malware.
- **Command Injection**: This attack involves inserting malicious commands into an application, which are then executed with the privileges of the application. In an IoT context, this could allow an attacker to take control of a device or extract sensitive information [347]. For example, an attacker might exploit a vulnerable smart home hub to execute arbitrary commands on connected devices.

- **Access Control Attacks**: Access control is an authorization mechanism that allows only legitimate users or processes to access the data or account. Access control attack is a critical attack in IoT applications because once the access is compromised, then the complete IoT application becomes vulnerable to attacks [347].
- **Service Interruption Attacks**: These attacks are also referred to as illegal interruption attacks or DDoS attacks in existing literature. There have been various instances of such attacks on IoT applications. Such attacks deprive legitimate users from using the services of IoT applications by artificially making the servers or network too busy to respond [347].
- **Malicious Code Injection Attacks**: Attackers generally go for the easiest or simplest method they can use to break into a system or network. If the system is vulnerable to malicious scripts and misdirection due to insufficient code checks, then that would be the first entry point that an attacker would choose [347]. Generally, attackers use XSS (cross-site scripting) to inject some malicious script into an otherwise trusted website. A successful XSS attack can result in the hijacking of an IoT account and can paralyze the IoT system [347].
- **Sniffing Attacks**: The attackers may use sniffer applications to monitor the network traffic in IoT applications. This may allow the attacker to gain access to confidential user data if there are not enough security protocols implemented to prevent it [347].

Understanding these attacks and vulnerabilities is crucial for developing robust security measures across all layers of the IoT architecture. As the IoT ecosystem continues to expand and evolve, addressing these security challenges becomes increasingly critical to ensure the integrity, confidentiality, and availability of IoT systems and data.

1.6 Solutions and Mitigation Techniques Necessary for the Application of IoT

Personal computers (PCs) and smartphones are equipped with various security mechanisms, such as firewalls, anti-virus software, and address space randomization. These safety barriers are generally absent in numerous IoT devices that are already available on the market. IoT applications are currently encountering numerous security concerns. Currently, there is a lack of a well-defined structure and standard for a comprehensive IoT application. An IoT application is not

an independent application; it is a composite product that incorporates contributions from multiple individuals and industries. Each layer utilizes various items and technologies, from sensing to the application. The edge nodes are equipped with a multitude of sensors and actuators. Various communication standards exist, such as cellular networks, WiFi, IEEE 802.15.4, Insteon, Dash7, Bluetooth, and others. A handshake protocol is necessary to establish communication across all these standards. In addition, multiple networking technologies such as Zigbee, GLOWPAN, wireless HART, Z-Wave, ISA100, Bluetooth, NFC, RFID, etc. are utilized at various levels within the same IoT application. In addition, the generic HTTP protocol is not applicable to the application layer. HTTP is not appropriate for contexts with limited resources due to its heavyweight nature, which results in significant parsing overhead [347]. As a result, numerous alternative protocols have been implemented for IoT contexts, even at the application layer. Among the protocols used in the field of the Internet of Things (IoT) are MQTT, SMQTT, COAP, XMPP, AMQP, M3DA, and JavaScript.

The wide range of protocols, technologies, and devices in an IoT application leads to important considerations about cost effectiveness, security, dependability, privacy, coverage, and latency. Optimizing one metric may lead to the deterioration of other metrics. Excessive implementation of security checks and protocols for all data transactions in IoT applications may increase the program's cost and latency, making it unsuitable for customers.

An average Internet of Things (IoT) application consists of a complex network of interconnected devices, technologies, domains, and geographical locations. If any of the devices or technologies, or their combination, is compromised, it might pose a security risk for the entire program. The strength of the chain is determined by the strength of its weakest link. Recently, there has been a significant surge in the number of vulnerable connections in IoT applications. For instance, even simple IoT applications like smart lamps and smart door locks might serve as vulnerable points in a smart home IoT system, allowing attackers to obtain the user's WiFi password [352].

The increasing number of Internet of Things (IoT) devices worldwide, aimed at creating smart environments, results in a substantial volume of data pertaining to the environment and users. This data may reveal vast amounts of private information, posing a significant threat to both individuals and society [353]. Therefore, significant advancements and refinements in the existing

IoT application structure and framework are necessary to ensure its reliability, security, and resilience. Various sources have suggested multiple solutions and mitigation approaches in response to the security and attack risks described in section 1.5. This section will explore various strategies for improving the security of IoT systems at different levels.

1.6.1 Solutions for the Perception Layer

- To prevent node capture and manipulation, it is essential to incorporate tamper-resistant hardware designs and protective coatings. We can enhance this by using secure boot techniques, which prevent unwanted alterations to the firmware. Devices can utilize Physical Unclonable Functions (PUFs) for authentication, providing a unique hardware-based identity for each device. Furthermore, implementing measures to identify and counteract unwanted tampering, such as automatically deleting sensitive data upon detecting unauthorized access, can significantly enhance the device's security [7] [9][347].
- Advanced Anti-Jamming Techniques: To counteract jamming attacks, the use of frequency hopping spread spectrum (FHSS), or direct sequence spread spectrum (DSSS) can enhance the resistance of communications to interference. By implementing adaptive power regulation, devices can modify their signal intensity in response to jamming detection. Additionally, cognitive radio approaches enable the ability to switch to less-crowded frequency bands in a dynamic manner. References [7] [9][347] have discussed these concepts.
- To prevent the introduction of malicious code, it is necessary to have secure over-the-air (OTA) update protocols that include code signing and verification. Runtime application self-protection (RASP) approaches can identify and thwart code injection attempts in real-time. By utilizing sandboxing and containerization techniques, it is possible to isolate potentially harmful code, thereby minimizing its impact on the entire system [7] [9][347].
- Data validation and anomaly detection are crucial in countering false data injection attacks. It is imperative to employ data validation techniques at the sensor level to identify and eliminate fabricated or tampered data. Employing duplicate sensors and data fusion techniques allows for measurement validation, increasing the data's dependability. We can use machine learning techniques to detect anomalies in sensor data, providing an additional safeguard against data manipulation [7] [8][347].

- To prevent side-channel attacks, it is necessary to implement cryptographic algorithms that are resistant to side-channel vulnerabilities. Countermeasures such as random clock jittering and power consumption smoothing can enhance security against power analysis attacks by impeding the extraction of information by attackers. We can use electromagnetic shielding to reduce electromagnetic emissions, thereby offering additional protection against side-channel attacks [7] [9][347].

1.6.2 Solutions for the Network Layer

- DDoS Mitigation Strategies: To counteract DDoS attacks, it is crucial to employ traffic monitoring and filtering methods to identify and prevent unusual traffic patterns. Rate restriction and connection throttling can effectively prevent resource depletion, while cloud-based DDoS protection services provide scalable mitigation for bigger attacks [7] [9][347].
- Implementing secure routing protocols, such as secure RPL (SRPL), is necessary to prevent routing attacks in 6LoWPAN networks. Authentication and integrity checks for routing messages ensure the processing of only valid routing information. Intrusion detection systems (IDS) can be utilized to observe network activity and identify deviations in routing [7] [8][347].
- Implementing encryption and authentication measures to prevent man-in-the-middle (MITM) attacks: It is essential to employ robust encryption protocols such as TLS and DTLS for all network connections to safeguard against man-in-the-middle (MITM) attacks. Using certificate pinning to stop unauthorized certificate swapping works well and allowing devices and gateways to authenticate each other makes sure that both are who they say they are [7] [9][347].
- Countermeasures against Sybil attacks involve implementing resource testing to verify the uniqueness of devices. Trusted platform modules (TPMs) offer a safe means of authenticating hardware devices, enhancing the reliability of identity verification. We can utilize reputation systems to identify and separate bad nodes, thereby enhancing the overall security of the network [7] [9][347].
- Implementing network-based intrusion detection and prevention systems (IDS/IPS) can effectively identify and counteract a range of attacks, such as distributed denial of service

(DDoS), man-in-the-middle, and routing attacks [8]. Utilizing machine learning for anomaly detection can be highly efficient in detecting new attack patterns.

- Secure communication protocols, like Transport Layer Security (TLS) or Datagram Transport Layer Security (DTLS), can protect data during transmission [7]. These protocols ensure the verification, consistency, and secrecy of network communications.

1.6.3 Solutions for the Middleware Layer

- To mitigate the risk of SQL injection attacks, it is crucial to include input validation and parameterized queries. ORM frameworks provide a means to isolate database interactions, minimizing the potential for SQL injection problems. Implementing the least privileged rules for database access reduces the possible consequences of a successful attack [7] [9][347].
- To mitigate XML-based assaults, it is necessary to implement XML schema validation and input sanitization. This guarantees the processing of only valid and safe XML data. To prevent XML entity expansion attacks, it is advisable to utilize XML parsers that have security measures set, such as entity expansion limitations. Specifically at the application layer, Web Application Firewalls (WAFs) can identify and prevent XML-based attacks [7] [9][347].
- Cloud Malware Protection: To prevent the injection of malware into the cloud, it is necessary to establish robust access controls and authentication measures for cloud resources. We can use integrity verification approaches to ensure that the cloud environment executes only approved and unmodified code for virtual machine images and service modules. Behavior-based malware detection techniques can be utilized to detect and separate potentially harmful actions in cloud settings [7] [9][347].
- API Security Enhancement: To improve API security, deploy API gateways with robust authentication and permission procedures. Effective measures like rate limitation and throttling prevent API abuse by controlling and limiting usage. On the other hand, OAuth 2.0 and OpenID Connect are secure frameworks that establish a robust system for controlling access to APIs [7] [9][347].

1.6.4 Solutions for the Application Layer

- To prevent data theft, all sensitive data must be encrypted end-to-end. Data masking and tokenization techniques can be employed to safeguard sensitive information, even in the event of a security breach where the encrypted data is accessed by unauthorized individuals. Data loss prevention (DLP) solutions provide an additional layer of protection against data theft by overseeing and regulating data access and transfer [7] [9][347].
- To mitigate access control assaults, it is necessary to develop strong identity and access management (IAM) systems. For important tasks, it is advisable to employ multi-factor authentication. On the other hand, attribute-based access control (ABAC) can offer detailed permissions depending on user attributes and context [7] [9][347].
- To prevent service interruptions, application-level firewalls and intrusion prevention systems must be implemented. CDNs can be utilized to disperse traffic and counteract DDoS assaults. Auto-scaling and load balancing techniques are employed to manage traffic surges and ensure service availability [7] [9][347].
- Implementing secure coding techniques and executing regular code audits are crucial for preventing malicious code injection at the application layer and ensuring runtime protection. Web application firewalls (WAFs) are utilized to identify and prevent injection attempts, while runtime application self-protection (RASP) approaches offer an extra level of security against code injection assaults [7] [9][347].
- Implementing robust encryption for all network connections is necessary to effectively mitigate sniffing attempts. Virtual private networks (VPNs) can be used to establish secure remote access, while network segmentation reduces the risk of sniffing attacks by separating critical traffic [7] [9][347].

After addressing layer-specific security solutions, we must consider more comprehensive and adaptable methods to enhance the overall security of IoT systems. These intersecting solutions enhance layer-based techniques and offer a more all-encompassing security framework for IoT applications.

- An essential measure in ensuring the security of IoT devices is performing thorough penetration testing to assess the degree of risk involved in installing these devices in different applications. This procedure facilitates the creation of a prioritized list, ensuring

the appropriate deployment of devices based on their risk level [347]. Despite the extensive use of encryption techniques in various levels and protocols of the Internet of Things (IoT), the system's repeated cycles of encrypting, decrypting, and re-encrypting could potentially lead to vulnerabilities. Implementing end-to-end encryption greatly reduces the dangers by minimizing the number of possible points of attack.

- Authentication is a vital component of IoT security. "Authenticate-always" protocols ensure the confirmation of every device contact. This is achieved using digital certificates that are linked to cryptographic protocols, resulting in a smooth and safe authentication method [347]. Furthermore, it is imperative to subject any IoT security framework to comprehensive scalability testing to ascertain its efficacy when implemented on a large scale in public domains.
- IoT devices must have inherent secure data transfer capabilities, employing robust encryption algorithms like RSA, SHA256, or hash chains, to safeguard user and environmental data. This strategy not only ensures data protection but also fosters confidence among IoT application users, government agencies, and industries [347].
- With the rapid expansion of IoT applications, it is becoming more crucial to solve limitations in cost and capacity. The adoption of a decentralized method characterized by safe and autonomous communication between devices can successfully address these limitations. This decision also necessitates careful consideration of cloud-related risks, as numerous IoT applications rely on cloud services for data storage and retrieval. Encrypting data and adopting strict access controls can greatly improve data security in cloud environments by encrypting data and adopting strict access controls.
- Ensuring the data's dependability is another critical component of IoT security. Introducing means to authenticate data flow, particularly in distributed architectures, can aid in identifying and addressing problems associated with inaccurate data gathering or transmission by sensors [347]. Utilizing artificial intelligence and machine learning methods can improve the autonomous security features of IoT systems, thereby decreasing the reliance on human involvement and potentially reducing the burden of analysis and communication in the IoT environment [8].

By integrating these adaptable solutions with the layer-specific methods previously mentioned, enterprises may establish a strong and comprehensive security strategy for their IoT

implementations. This comprehensive strategy not only tackles existing security concerns but also offers a versatile framework that can adjust to the changing threat landscape in the fast-expanding IoT ecosystem.

1.7 Utilizing Federated Learning to Enhance IoT Security

Federated Learning (FL) presents a promising method for improving IoT security through decentralized and collaborative learning. In FL literature, devices utilize their data to train local models and only transmit model updates to a central server, thereby safeguarding data privacy [12]. This approach tackles various security challenges on the Internet of Things (IoT), including issues related to data privacy and scalability, by reducing the amount of data transferred and facilitating more robust data processing. Federated Learning (FL) improves the resilience of Internet of Things (IoT) security by consolidating information from various devices, hence enhancing the total capacity for detecting potential threats. It can assist in detecting intricate attack patterns that individual devices may overlook, enabling prompt upgrades to the global model using the most recent threat intelligence [12].

The FL process often encompasses multiple stages: Initially, the central server initiates a global model and disperses it across the participating IoT devices. Subsequently, every device proceeds to train the model using its own local data and transmits only the model's changes to the server. The server combines these updates to improve the global model, which it then distributes to the devices [12]. This recurrent process persists, enabling the model to acquire knowledge from many data sources without revealing confidential information.

When privacy concerns, regulatory limits, or bandwidth limitations prevent the centralization of data, FL is especially advantageous for IoT security [12]. It facilitates the creation of advanced intrusion detection systems, anomaly detection algorithms, and other security measures that can adjust to changing threats in various IoT devices and situations. Furthermore, Federated Learning (FL) enables the development of customized security models tailored to specific device types or user habits, leveraging the collective knowledge of the entire network [12].

Still, using Federated Learning (FL) for Internet of Things (IoT) security isn't easy. For example, you must make sure that model updates are real, protect against attacks while the learning process is going on, and make sure that IoT devices with limited resources can handle the computational load [12]. Although there are obstacles, FL's potential to completely transform IoT

security makes it a critical area for continuous research and development in the realm of IoT cybersecurity.

1.8 Conclusion

The Internet of Things (IoT) is a major technological advancement that links physical things and allows for automatic data exchange and activities. This chapter has examined the core elements of the Internet of Things (IoT), including its architecture, its applications, and the significant security challenges it encounters.

The standard IoT architecture comprises four layers: perception, network, middleware, and application. Each layer has distinct purposes and employs various technologies, ranging from sensors and actuators at the perception layer to sophisticated data processing and user interfaces at the application layer. The diverse applications of IoT, which include smart homes, healthcare, agriculture, and industrial automation, showcase its capacity to fundamentally transform multiple industries.

Nevertheless, the rapid growth and heterogeneous characteristics of IoT devices pose substantial security obstacles. The challenges stem from the diversity of devices, scaling difficulties, data privacy concerns, a lack of compatibility, and resource limitations. The chapter provides an in-depth analysis of the many attacks, threats, and vulnerabilities that are specific to each tier of the IoT architecture. It emphasizes the intricate security challenges that IoT systems encounter.

To tackle these security concerns, a comprehensive and multifaceted strategy is required. This involves the implementation of comprehensive security measures at every level, including the use of tamper-resistant hardware, secure communication protocols, and powerful encryption techniques. The chapter has delineated multiple strategies and techniques for mitigating risks, underscoring the importance of implementing a comprehensive security plan that includes all levels of the IoT architecture.

Moreover, the chapter presented the notion of Federated Learning (FL) as a viable method for improving IoT security. FL allows you to use the combined intelligence of multiple devices while maintaining data privacy. It tackles important issues in IoT security, such as data concentration and privacy concerns.

Chapter 2: Network Intrusion Detection Systems (NIDS) in the IoT

IDSs are defined as systems built to monitor and analyze network traffic and/or systems to detect anomalies, intrusions, or privacy violations. When an intrusion is detected, a NIDS is expected to (a) log the information related to the intrusion, (b) trigger alerts, and (c) take mitigation and corrective actions [1].

In short, an Intrusion Detection System (NIDS) is a security tool that watches over your computer or network for suspicious activity. Intrusion means someone unauthorized is trying to get in to steal information, mess with things, or block access. The detection system part is what finds this suspicious activity. NIDS constantly monitors traffic and if it sees something that breaks the security rules, it raises an alarm. There are different ways to set up a NIDS, like putting it on the network to watch all the traffic coming and going.

NIDS plays a critical role in safeguarding networks from unauthorized access and malicious activity. In the context of the ever-evolving IoT landscape, robust NIDS solutions are essential for protecting interconnected devices, applications, and the data they generate. This section explores some fundamentals of NIDS, exploring different types and detection techniques employed to secure the IoT ecosystem.

2.1 Fundamentals of NIDS

An intrusion detection system (NIDS) is a security mechanism that monitors network traffic or system activity for signs of malicious activity or policy violations. It analyzes network packets or system logs and generates alerts when it detects suspicious events that potentially indicate an ongoing or imminent attack. A NIDS tracks a network for unauthorized behavior or policy breaches. The NIDS logs the details related to the intrusion, initiates warnings and takes necessary mitigation or corrective measures in case an intrusion is detected [12]. Fig. 3 illustrates how a NIDS can be used in a corporate network.

Here are some of the key functionalities of a NIDS:

- **Traffic Monitoring**: A NIDS continuously monitors network traffic or system activity, searching for patterns or anomalies that deviate from established baselines.
- **Threat Detection**: The NIDS analyzes the collected data using various detection techniques to identify potential security threats, such as unauthorized access attempts, malware infections, or Denial-of-Service (DoS) attacks.
- **Alert Generation**: Upon detecting suspicious activity, the NIDS generates alerts that notify security personnel or automated defense mechanisms to take appropriate actions.
- **Logging and Reporting**: The NIDS logs detect events and generate reports that can be used for forensic analysis and security audits.

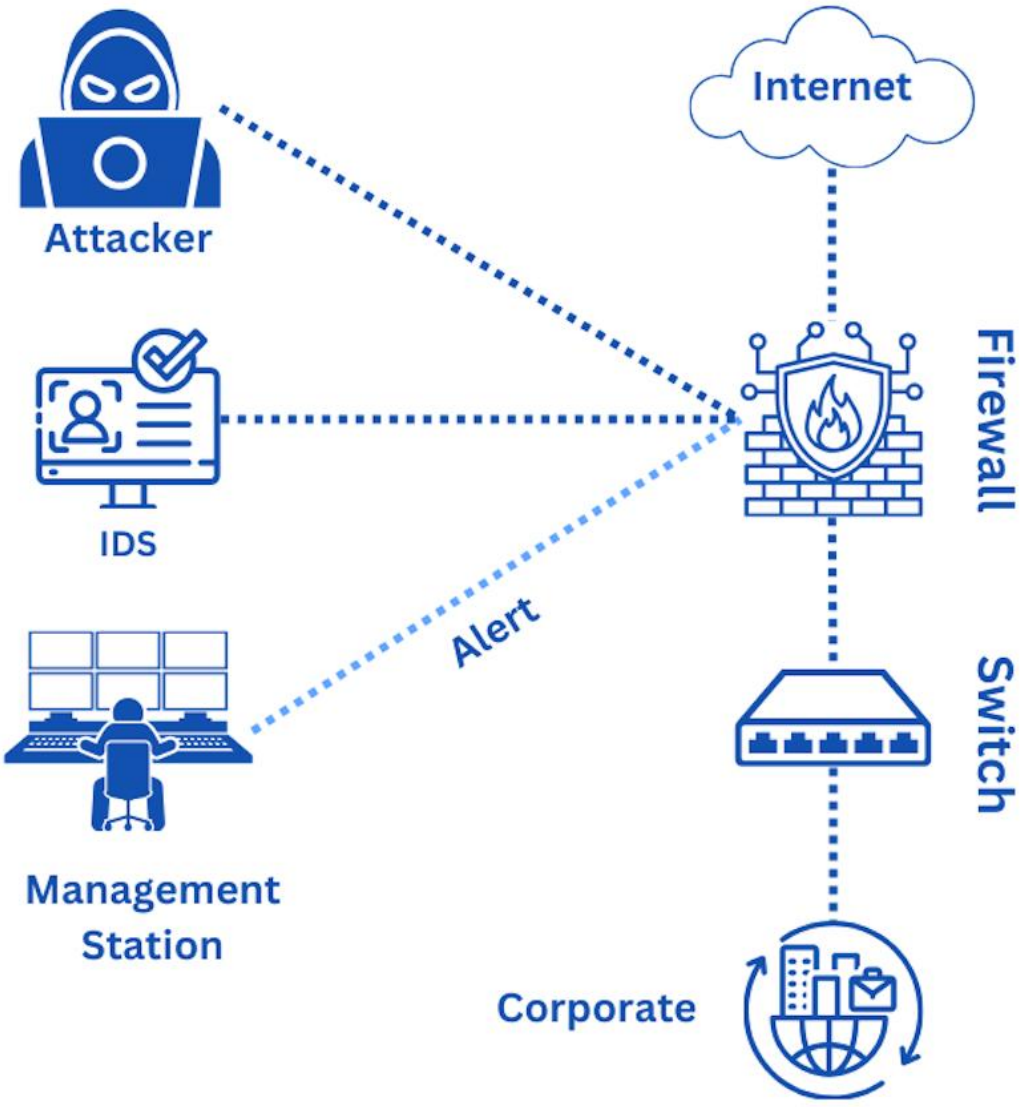


Figure 3 : Intrusion Detection System

2.1.1 Classification of NIDS

In this section, we provide an overview of NIDS classifications based on deployment methods and detection techniques. Figure. 4.

2.1.1.1 Deployment Method-Based Classification

A NIDS can be classified as either a host-based NIDS (HIDS) or a network-based NIDS (NIDS) based on its deployment strategy. HIDS resides on a single host, monitoring its activities and searching for suspicious behavior or policy violations. However, deploying HIDS on every machine requiring protection increases processing overhead and reduces overall NIDS performance. In contrast, NIDS sits on the network, safeguarding all devices and the entire network from intrusions. NIDS continuously monitors network traffic for potential security breaches [13]. This work focuses on various methods employed in NIDS.

2.1.1.2 Detection Method-Based Classification

From a detection perspective, NIDS falls into two categories: signature-based intrusion detection systems (SIDS) and anomaly detection-based intrusion detection systems (AIDS). SIDS, also known as misuse- or knowledge-based NIDS, relies on defining attack pattern signatures stored in a database for data pattern matching. SIDS offers high detection efficiency for known attacks with available signatures but struggles to identify novel attacks due to the lack of corresponding patterns. Additionally, maintaining a vast signature database and comparing it with data packets for potential intrusions makes SIDS resource intensive.

AIDS, also known as behavior-based NIDS, establishes a profile for normal activity, flagging any deviations as anomalies. The benefits of AIDS include the ability to detect unknown attacks and the adaptability of normal activity profiles for various networks and applications. However, the high false alarm rate (FAR) due to the difficulty of defining a clear boundary between normal and abnormal profiles poses a challenge for AIDS.

The rise of the Internet of Things (IoT) has led to a surge in IoT devices. Wireless Sensor Networks (WSNs), a crucial technology in IoT networks, consist of sensor nodes collecting information. The vast amount of critical data collected by these sensor nodes and shared over the internet, coupled with the complex structure of resource-constrained sensor nodes in WSNs, creates security challenges for IoT networks. NIDS is considered a vital security mechanism for the IoT and WSNs [13].

2.1.2 Host-Based Intrusion Detection System (HIDS) and Network-Based Intrusion Detection Systems (NIDS)

Host Intrusion Detection System (HIDS) and Network Intrusion Detection System (NIDS) are two types of NIDS [14]. HIDS is responsible for monitoring a system internally, including connections to log files, user triggers or actions, etc. NIDS, on the other hand, analyzes both incoming and outgoing traffic between network nodes.

- Host-based Intrusion Detection System (HIDS): A host-based intrusion detection system (HIDS) monitors the activity on a specific host system, such as a server, workstation, or IoT device. It analyzes system logs, files, and running processes to detect suspicious activities within the host itself. HIDS are effective in detecting malware infections, unauthorized modifications, and privilege escalation attempts. However, a host-based NIDS is not an ideal solution on its own. It suffers from severe disadvantages, such as a high consumption of resources that harms the host's performance. Such attacks may not be exposed unless they penetrate the host [12].
- Network Intrusion Detection System (NIDS): A Network Intrusion Detection System (NIDS) monitors network traffic for malicious activity. It typically resides at a predetermined location throughout the network and analyzes all packets flowing through it. NIDS are well-suited for detecting network-based attacks such as port scans, unauthorized access attempts, and DoS attacks. Similarly to HIDS, [12] outlines that with NIDS, there are issues with scrutinizing every packet during heavy traffic and failing to detect an attack at that time, as well as problems with the identification of virtual private networks (VPN) and understanding encrypted data, which add to the list of challenges.

2.1.3 Signature-Based Detection (SIDS) and Anomaly-Based Detection (AIDS)

SIDS and AIDS are two main approaches used by NIDS to detect suspicious activity:

- Signature-based Detection: This technique relies on a predefined database of signatures, which are unique patterns or characteristics associated with known attacks. The NIDS compares incoming network traffic or system activity against the signature database. If a

match is found, the NIDS raises an alert, indicating a potential attack. [12] highlights a major challenge faced by this technique: if an intruder uses a new attack that is not already existing in the library, the method will fail to identify the attack. This type of attack is also known as a zero-day attack.

- Anomaly-based Detection: Anomaly-based intrusion detection overcomes the challenges posed by signature-based intrusion detection. The detection of intrusion is based on the behavior of the user [12]. This technique analyzes network traffic or system activity for deviations from established baselines. The NIDS learns the normal behavior patterns of the system or network and identifies significant deviations as potential threats. Anomaly-based detection can be effective in identifying novel attacks but may generate false positives due to legitimate activities that deviate from the norm. Another limitation faced by this technique seen in [12] is its inability to recognize encrypted packets, which thus leaves an opportunity for attack.

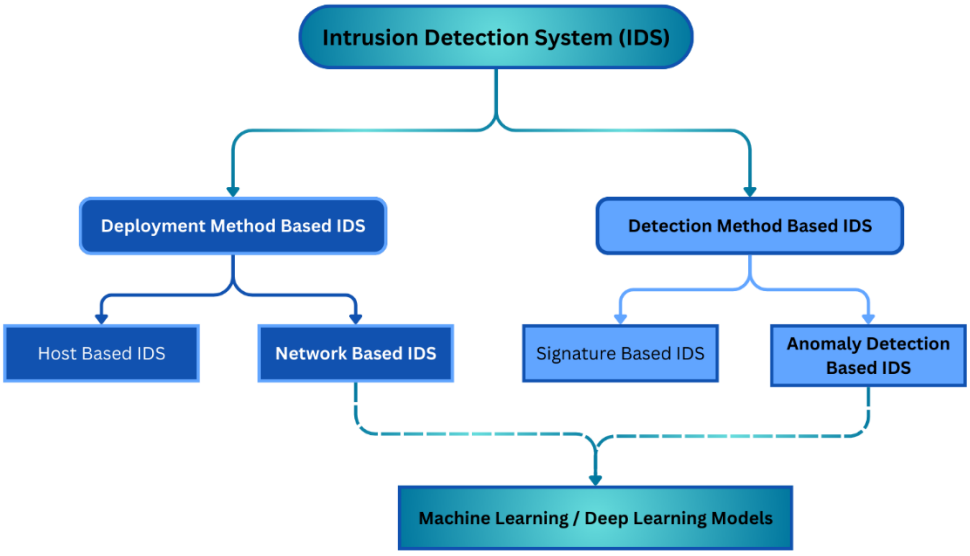


Figure 4 : Classification of Intrusion Detection Systems

2.2 Challenges of NIDS in the IoT Environment

While intrusion detection systems (NIDS) offer a valuable layer of security in traditional networks, their deployment in the realm of IoT presents unique challenges. The dynamic and resource-constrained nature of IoT devices and networks necessitates adaptations to traditional NIDS approaches. Here, we explore the specific hurdles encountered when implementing NIDS in the IoT environment.

2.2.1 Challenges of Traditional NIDS in IoT Networks

- **Heterogeneity of Devices**: The IoT landscape is characterized by a vast array of devices with varying capabilities, operating systems, and communication protocols. Traditional NIDS solutions may not be able to effectively monitor and analyze traffic or activity from such a diverse range of devices, potentially leaving vulnerabilities undetected.
- **Resource Limitations**: Many IoT devices are resource-constrained, possessing limited processing power, memory, and battery life. Deploying traditional NIDS software on such devices can be impractical due to the computational overhead it may incur, potentially impacting device performance and battery life.
- **Real-time Processing Demands**: The IoT ecosystem often involves real-time data collection and analysis. Traditional NIDS solutions might struggle to keep pace with the high-velocity data streams generated by numerous devices, potentially delaying threat detection and response times.
- **Lack of Standardization**: The absence of standardized communication protocols and security mechanisms in the IoT environment can further complicate NIDS deployment. Traditional NIDS solutions may not be able to interpret or analyze data from devices utilizing non-standard protocols, creating visibility gaps.
- **Data Privacy Concerns**: Traditional NIDS solutions often rely on deep packet inspection to analyze network traffic, potentially raising privacy concerns. In the IoT realm, where sensor data may contain sensitive personal information, intrusive data analysis methods by NIDS can be a point of contention.

2.2.2 Limitations of Centralized NIDS Architectures

Traditional NIDS architectures often rely on a centralized approach, where data from network devices is forwarded to a central server for analysis. While this approach can be effective in controlled environments, it presents limitations in the context of large-scale IoT deployments.

- **Scalability Issues**: The vast number of devices in an IoT network can overwhelm a centralized NIDS server, leading to scalability bottlenecks and potential delays in threat detection. Managing and updating security configurations on millions of devices can be a complex and resource-intensive task for a centralized system.
- **Privacy Concerns**: The centralized storage of data collected from all devices raises privacy issues. Data breaches or unauthorized access to the central server could compromise sensitive user information collected by IoT devices.
- **Single Point of Failure**: A centralized NIDS architecture creates a single point of failure. If the central server malfunctions or is compromised, the entire network's intrusion detection capabilities are crippled.

These challenges necessitate the development of novel NIDS solutions specifically designed for the unique characteristics of the IoT landscape. Decentralized architectures, lightweight anomaly detection techniques, and privacy-preserving data analysis methods are promising areas of research for securing the ever-evolving realm of the Internet of Things.

2.3 Recent Advancements in IoT NIDS

The ever-changing nature of the Internet of Things (IoT) landscape necessitates innovative security solutions. Traditional intrusion detection systems (NIDS) struggle to adapt to the vast array of devices (heterogeneity), limited resources, and need for real-time processing within the IoT. To address these challenges, researchers have developed advanced NIDS solutions specifically designed for the unique security needs of IoT environments. These solutions leverage various methodologies and technologies, each offering distinct advantages and limitations.

This section explores some recent advancements in IoT NIDS, categorized by the techniques employed.

- **Lambda Architecture**: This method combines batch and real-time data processing to efficiently handle massive data streams. This architecture benefits NIDS by enabling timely threat detection and response in large-scale IoT deployments.
- **Machine Learning-Based NIDS**: Machine learning and the use of machine learning techniques and algorithms have shown promising results in analyzing network traffic to identify malicious activities. Techniques like Random Forests (RF) and Support Vector Machines (SVMs) have achieved high accuracy in classifying traffic as normal or malicious [15]. A comprehensive survey by compared machine learning algorithms for NIDS based on datasets, data reduction approaches, classifier types, and achieved outcomes [15].
- **Deep Learning-Based NIDS**: Deep learning techniques, such as deep neural networks (DNNs), recurrent neural networks (RNNs), and convolutional neural networks (CNNs), are used to detect complex attack patterns that traditional methods might miss. These models can effectively handle large volumes of IoT data but require substantial computational resources and risk the integrity and confidentiality of the data, potentially limiting their application in resource constrained IoT environments.

While this section provides a brief overview of these advancements, a future chapter will explore and provide a deeper overview of specific machine learning and deep learning solutions for IoT NIDS.

2.4 Challenges and Future Directions in IoT NIDS

Securing the vast and diverse ecosystem of IoT devices presents unique challenges for intrusion detection systems (NIDS). Heterogeneity, resource constraints, and data privacy concerns are just some of the hurdles. NIDS frameworks must also contend with high false positive/negative rates, real-time processing demands, and the ever-evolving threat landscape.

Future research in IoT NIDS will likely focus on machine learning enhancements tailored to the distributed nature and limited processing power of IoT devices, as well as decentralized and edge-based NIDS solutions that operate closer to data sources, enabling faster response times and

dynamic adaptation to threats. These solutions should seamlessly integrate with existing and future IoT technologies while adhering to data privacy regulations.

This is where our research project comes in. We propose a federated learning-based NIDS solution that leverages its decentralized and privacy-preserving nature to address the challenges and contribute to a more secure IoT ecosystem.

The dynamic field of IoT NIDS research remains crucial for safeguarding the ever-growing number of connected devices from a constantly evolving cyber threat landscape.

2.5 Conclusion

Network Intrusion Detection Systems (NIDS) are essential for detecting and mitigating security threats in IoT environments. However, traditional NIDS face several challenges, including resource limitations and the increasing complexity of attacks specific to IoT devices. This chapter explored different approaches to NIDS, particularly highlighting anomaly-based detection methods that offer better capabilities in identifying unknown threats. The findings stress that existing NIDS are inadequate for securing IoT due to their limitations in scalability and privacy handling. Consequently, there is a need for decentralized, lightweight, and adaptive NIDS capable of real-time detection and integration with IoT's constrained environments.

Chapter 3: Machine Learning and Deep Learning for NIDS

The dynamic nature of cyber threats necessitates adaptable and intelligent security solutions. Machine learning (ML) and Deep Learning (DL) techniques are increasingly employed in the development of intrusion detection systems (NIDS) for identifying and responding to security threats, offering the ability to learn from data and identify sophisticated attacks.

As a part of this chapter, we will review some of the AI methodologies, algorithms, achievements, and related works of ML/DL in solving recent real-world intrusion detection problems in various applications. We also dive into the role of machine learning and deep learning in NIDS, as well as exploring their advantages and limitations.

3.1 AI Techniques for Network Intrusion Detection Systems (NIDS)

This section explores the general methodology behind AI-based NIDS and delves into the most common machine learning (ML) and deep learning (DL) algorithms used for designing effective NIDS solutions. Both ML and DL can be categorized into supervised and unsupervised learning approaches [16]. Supervised learning algorithms leverage labeled data to extract valuable information, whereas unsupervised algorithms rely on unlabeled data to identify useful features and patterns [17].

3.1.1 Categories of AI-Based Learning Approaches

- **Supervised Learning**: This approach involves training a machine learning model using labeled data sets. The data sets contain network traffic samples or system activity logs that are pre-classified as normal or malicious. The model learns to identify patterns and relationships within the data, enabling it to classify new, unseen data points as normal or anomalous (a potential attack). Common supervised learning algorithms used in NIDS include support vector machines (SVMs), decision trees, and random forests. These models are trained to recognize patterns corresponding to attacks or normal behavior, thus enabling them to detect known types of attacks effectively.

- **Unsupervised Learning**: In contrast, unsupervised learning algorithms are used when the labels are unknown. This approach involves training a machine learning model on unlabeled data sets. The model identifies inherent patterns and structures within the data without prior classification. In the context of NIDS, unsupervised learning can be used to establish baselines for normal network traffic or system activity. Deviations from these baselines can then be flagged as potential anomalies or suspicious behavior. They are particularly useful for detecting new or unknown types of attacks (zero-day attacks), as they don't rely on predefined attack signatures.

3.1.2 General Methodology for AI-based NIDS

Network intrusion detection systems built with ML and DL techniques typically follow a three-step process illustrated in Figure 5:

- **Data Preprocessing**: The raw data undergoes preprocessing to transform it into a format suitable for the chosen algorithm. This stage often involves encoding and normalization techniques. Additionally, data cleaning might be necessary to remove entries with missing information or duplicates.
- **Training Phase**: The preprocessed data is then randomly divided into two sets: training and testing data. The training set, typically comprising around 80% of the data, is used to train the ML or DL model. The training time can vary depending on the data size and model complexity, with DL models generally requiring more training time due to their intricate architecture.
- **Testing Phase**: Once trained, the model is evaluated using the testing set (around 20% of the data). In the context of NIDS, the model predicts whether network traffic is benign (normal) or malicious. This evaluation helps assess the model's effectiveness in identifying potential threats.

The following section provides a comprehensive overview of widely used ML and DL algorithms for NIDS. Additionally, Table 1 presents a taxonomy of recent ML- and DL-based techniques employed for NIDS.

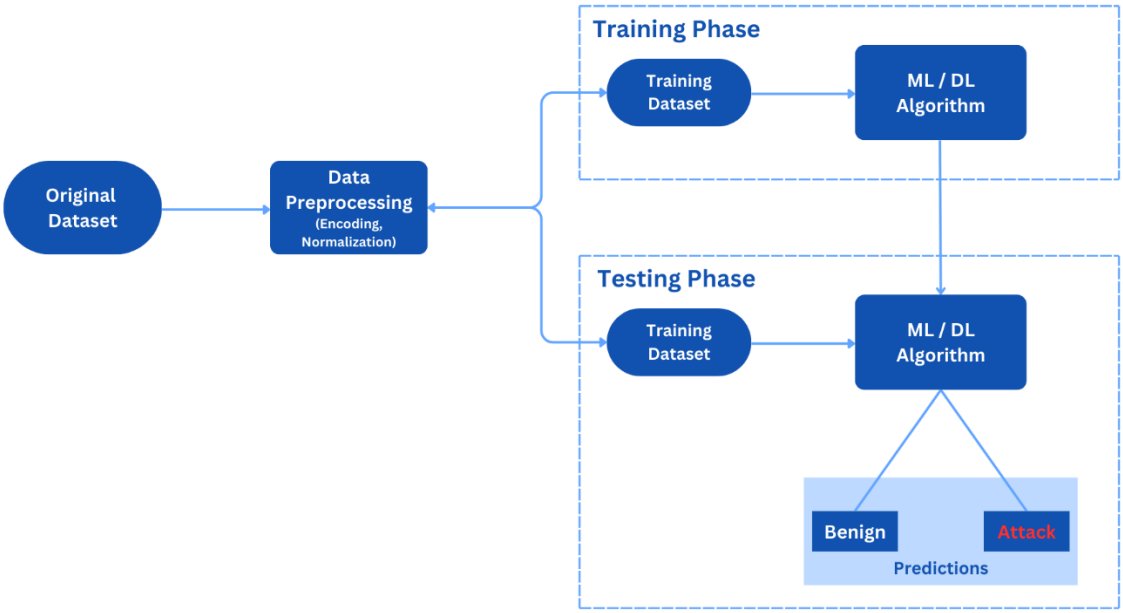


Figure 5 : Generalized Machine Learning-/Deep Learning-Based Network-Based Intrusion Detection System Methodology

3.2 Machine Learning in NIDS

Machine learning covers a wide range of techniques that can be applied to intrusion detection. Here, we explore some commonly used algorithms, the benefits, and limitations of using machine learning in NIDS.

3.2.1 Common Machine Learning Algorithms for Network Intrusion Detection Systems (NIDS)

This section explores various machine learning (ML) algorithms commonly used in designing effective NIDS solutions.

3.2.1.1 Decision Tree (DT)

A fundamental supervised ML algorithm, DT, is used for both classification and regression. It builds a tree-like structure with nodes, branches, and leaves [18], where each node represents a

feature, branches represent decisions, and leaves represent class labels (e.g., normal or attack) [19]. DT automatically selects features for building the tree and prunes irrelevant branches to prevent overfitting. Common DT models include CART, C4.5, and ID3 [20]. Advanced learning algorithms like Random Forest (RF) and XGBoost are built using multiple decision trees [21] [22].

3.2.1.2 K-Nearest Neighbor (KNN)

KNN is a straightforward supervised ML algorithm that uses "feature similarity" to predict the class of a data sample. It identifies a sample's class based on its closest neighbors, determined by a distance calculation. The value of K (the number of neighbors) significantly impacts performance. A small K-value can lead to overfitting, while a large K-value might misclassify instances [23] [24]. Karatas et al. [25]. addressed dataset imbalance by applying SMOTE (Synthetic Minority Oversampling Technique [26]) to improve detection rates for minority class attacks when using KNN with the CSE-CIC-IDS2018 benchmark dataset.

3.2.1.3 Support Vector Machine (SVM)

SVM is a supervised ML algorithm that employs the concept of "max-margin separation" in a high-dimensional feature space. It can handle both linear and non-linear problems using kernel functions, which map low-dimensional input vectors into a higher-dimensional space where an optimal separation hyperplane can be established [27] [28]. SVMs can enhance NIDS efficiency and accuracy by effectively classifying normal and malicious traffic [29] [30].

3.2.1.4 K-means clustering

K-means clustering is a popular unsupervised ML algorithm that groups similar data into clusters. It iteratively assigns data points to clusters based on their distance from the cluster's centroid (center). The main objective is to minimize the total distance between data points and their respective centroids within a cluster [31] [32] [33]. Yao et al. [34]. proposed a multilevel semi-supervised ML framework (MSML) for NIDS that combines clustering with a random forest model. This approach forwards unlabeled attacks to subsequent modules for detection if they are not identified in the initial stages. The model achieved promising results using the KDD Cup'99 dataset.

3.2.1.5 Artificial Neural Networks (ANNs)

Inspired by the human brain's structure, ANNs are supervised ML algorithms composed of interconnected processing elements (neurons) organized in layers. Backpropagation is used as a learning technique. ANNs excel at non-linear modeling with large datasets [35] but can be time-consuming to train due to their complexity [36], which makes reaching a suboptimal solution take time [37].

To overcome the limitations of ANN, Huang et al. [38] proposed a new ANN called an extreme learning machine (ELM). The ELM is a single hidden layer feed-forward neural network that randomly uses the input weights and hidden layer bias without tuning and determines the output weights in an analytical way [39]. Based on the idea of ELM, Li et al. [40] proposed a Fast-Learning Network (FLN). FLN is based on connecting a multilayer feed-forward neural network and a single-layer feed-forward neural network in parallel. FLN showed reasonable performance and stability using a smaller number of hidden nodes and utilizing less time.

Ali et al. [41] addressed the NIDS problem by proposing a model based on FLN, particle swarm optimization [42], and PSO-FLN and testing the model using the KDD Cup'99 dataset. The model was tested by comparing the FLN with different optimization algorithms. Results showed that PSO-FLN outperforms the other FLN models with different optimization algorithms, such as genetic algorithms, money search optimization, and accelerated teaching and learning-based optimization. They also demonstrated that increasing the number of neurons in the hidden layer increases accuracy. The main drawback is the lower detection rate accuracy for lower attack classes.

3.2.1.6 Ensemble Methods

Ensemble methods leverage the strengths of multiple classifiers to create a more robust model. Since individual classifiers have varying strengths and weaknesses for different attack types, combining them through voting algorithms can lead to superior performance. Shen et al. [43] proposed an ensemble method using ELM (Extreme Learning Machine) as a base classifier and a BAT optimization algorithm for pruning. This ensemble approach achieved better results than individual ELMs on datasets like KDDCup'99, NSL-KDD, and Kyoto. Similarly, Gao et al.

[44] presented an adaptive ensemble model that combines various base classifiers (DT, RF, KNN, and Deep Neural Network) and employs an adaptive voting algorithm for selection. While demonstrating improved performance compared to other models on the NSL-KDD dataset, this approach also faced challenges with weaker attack classes.

3.2.2 Machine Learning works for IoT Network Intrusion Detection Systems

This subsection explores how machine learning (ML) models are utilized to address intrusion detection challenges in various real-world applications. We will examine several studies that leverage ML techniques and algorithms to safeguard critical systems from cyberattacks. These studies demonstrate the effectiveness of ML in identifying anomalies and malicious activities within different network environments.

In a recent effort to bolster security in Internet of Things (IoT) networks, [45] proposed an anomaly-based intrusion detection system (NIDS). This system tackles the challenge of diverse cyberattacks within IoT environments. The proposed method utilizes Kernel Principal Component Analysis (KPCA) to preprocess data by reducing its dimensionality, allowing for more efficient anomaly detection. To classify traffic flow as benign or malicious, the system employs the Kernel Extreme Learning Machine (KELM) for binary classification. Furthermore, KELM is used for multiclass classification to identify the specific type of attack within malicious traffic. The effectiveness of the proposed NIDS is assessed through its evaluation of two modern datasets. The results demonstrate a significant improvement in detection efficiency and various performance metrics, including accuracy, specificity, sensitivity, F1-score, and Area Under the Curve (AUC).

In [46], Xu, H., Sun, Z., Cao, Y., et al. propose a data-driven approach that leverages machine learning with data pre-processing. To improve the training data, they employ a technique called SMOTE and mutual information. Additionally, they utilize automated machine learning to find the best hyperparameters for the algorithm. This not only reduces computation costs but also achieves high accuracy, exceeding 99.7% in multi-class attack classification, significantly outperforming current methods.

In a recent study, Khan, M.M., and Alkhathami, M., proposed machine learning models for anomaly detection in healthcare IoT data using the CIC IoT dataset [47]. This dataset includes

33 attack types across 7 categories. The researchers applied various supervised learning algorithms, including random forest, logistic regression, and deep neural networks. To optimize the models, they performed pre-processing, balanced class representation, and feature reduction techniques. Their findings indicate that Random Forest achieved the best performance (around 99.55% accuracy) in both binary and multi-class attack classifications. This approach also reduced training time, which is crucial for real-time anomaly detection in healthcare IoT applications.

Similarly, in research proposes a novel hybrid intrusion detection system (NIDS) for securing medical devices in IoMT networks [48]. It tackles cyberattacks by combining two detectors: logistic regression analyzes network traffic features to identify suspicious activity, and gradient-boosted trees analyze medical sensor data to detect anomalies. This approach achieves high accuracy (95.4%) using only 11 features, surpassing existing methods that require 40 features. Additionally, combining the detectors improves the attack detection rate (from 111 to 305 attacks identified), demonstrating the effectiveness of this hybrid system. Overall, the study offers a new approach for securing IoMT devices across healthcare networks.

A recent study reviewed 145 papers to assess how bio-inspired algorithms can improve intrusion detection systems (NIDS) for IoT networks [49]. Machine learning (ML) and deep learning (DL) are common tools in NIDS, but achieving high accuracy with low false alarms can be difficult. Bio-inspired algorithms are seen as a potential solution. The researchers analyzed 25 relevant studies and found that combining bio-inspired techniques with ML and DL approaches is effective in enhancing NIDS performance. This suggests a promise for strengthening security in IoT networks. The review also offers a comparison of the studies and identifies future directions for integrating these techniques.

A new study proposes a machine learning (ML)-based intrusion detection system for large and imbalanced network data [50]. To address challenges like data imbalance and high dimensionality, the system uses a combination of techniques: Random oversampling (RO) tackles imbalanced data by replicating samples from minority classes. Stacking with feature embedding leverages clustering results for improved feature representation. Principal Component Analysis (PCA) reduces data dimensionality for efficient ML model training. The researchers evaluated their model on three benchmark datasets, achieving high accuracy (up to 99.99%). These results

outperform existing methods, demonstrating the effectiveness of their approach for network intrusion detection.

B. Gao, et al. [51] The authors propose a two-layer intrusion detection system for train-ground communication. The first layer uses machine learning algorithms like Random Forest and Gradient Boosted Decision Tree to identify anomalies. The second layer uses a state observer to detect physical state abnormalities.

C.-C. Sun et al. [52] This work proposes a two-stage system to protect Advanced Metering Infrastructure (AMI) from cyberattacks. Support Vector Machines (SVM) are used in the first stage to identify suspicious meter behavior. The second stage uses a Temporal Failure Propagation Graph (TFPG) to identify possible attack routes.

Venturi et al. [53] The authors used gradient boosting for feature selection along with decision-tree-based algorithms to classify various attacks and normal events in Smart Grid SCADA systems.

3.2.3 Benefits of Machine Learning in NIDS

Machine learning (ML) significantly enhances Network Intrusion Detection Systems (NIDS) by enabling adaptive learning capabilities, allowing systems to evolve as they encounter new data or threats. This adaptability is crucial given the constantly changing nature of cyber threats [261]. ML models can automate the detection process, reducing the need for human intervention and allowing for real-time threat detection and response [262].

- **Adaptability**: ML models can continuously learn and adapt to evolving threats. As new attack vectors emerge, the model can be retrained on data containing these new threats, improving its detection capabilities over time [263]. For instance, Kaloudi and Li demonstrated an adaptive ML-based NIDS that could effectively detect emerging threats in IoT networks [264].
- **Pattern Recognition**: ML algorithms excel at identifying complex patterns within datasets. This allows them to detect subtle anomalies that might be missed by traditional signature-based NIDS approaches [265]. A study by Sommer and Paxson showed that ML

techniques could identify complex attack patterns in network traffic with high accuracy [267].

- **Generalization**: ML models can learn from a variety of data sources and generalize their knowledge to identify unseen attacks. This is crucial in the context of IoT security, where new attack vectors are constantly being developed [267]. Research by Debar et al. demonstrated that ML-based NIDS could effectively detect novel attacks in IoT networks with a false-positive rate below 1% [268].

3.2.4 Limitations of Machine Learning for NIDS

Despite their benefits, machine learning models also face several challenges in the context of NIDS:

- **Labeled Data Dependency**: Supervised learning algorithms require large datasets of labeled network traffic or system activity for effective training. Obtaining high-quality labeled data, particularly for novel attacks, can be challenging [269]. A comprehensive study by Ring et al. highlighted the scarcity of up-to-date, publicly available datasets for training ML-based NIDS [270].
- **Bias and Overfitting**: ML models can inherit biases present within the training data. This can lead to the model overlooking certain types of attacks or generating false positives for specific scenarios [271]. Overfitting is another risk, where a model might perform well on training data but poorly on unseen data. Soekhoe et al. demonstrated that biased training data could significantly impact the performance of ML-based intrusion detection systems [272].
- **Computational Overhead**: Training and running complex ML models can be computationally expensive. This can be a limiting factor for deployment on resource-constrained devices within an IoT network [258]. However, recent work by Zhao et al. has shown promise in developing lightweight ML models that can run efficiently on edge devices while maintaining high detection accuracy [273].

While ML offers significant advantages for NIDS, it's crucial to acknowledge its limitations. Addressing data quality issues, mitigating bias, and exploring lightweight ML models are ongoing areas of research to enhance the effectiveness of ML in intrusion detection [274].

3.3 Deep Learning for Enhanced NIDS Capabilities

Building upon existing machine learning works in IoT network intrusion detection, the integration of deep learning offers convincing advantages. Machine learning has provably improved intrusion detection, but deep learning architectures offer even greater potential.

NIDS based on machine learning (ML) algorithms have gained the attention of the scientific community due to their possibility of high detection performance and their ability to automate the intrusion detection process [54]. ML algorithms for NIDS are promising, but they do not represent a universal solution. Moreover, they are vulnerable to adversarial attacks, in which adaptive adversaries modify their malicious communication patterns to evade detection. As cybersecurity intrinsically involves active attackers, this issue requires improving the robustness of ML-based NIDS [54].

Generally, machine learning is a subset of artificial intelligence, and deep learning is a subset of machine learning techniques. Deep learning networks are constructed from several connected layers, in which the first one is called the input layer, the last one is denoted as the output layer, and all layers between are called hidden layers. Each of the hidden layers consists of several neurons, and the signal strength of a neuron depends on factors such as bias, weight, and the activation function [55].

Deep learning's ability to automatically extract complex features from raw data enhances detection accuracy. Its adaptability to complex and evolving attack patterns ensures robust security. Moreover, deep learning models excel at handling large-scale datasets, which is vital in IoT environments. By leveraging deep learning, IoT ecosystems can achieve real-time threat detection with minimal human intervention. This approach empowers proactive defense mechanisms, safeguarding sensitive IoT networks effectively. Ultimately, embracing deep

learning for intrusion detection in IoT ecosystems represents a forward-thinking strategy to enhance cybersecurity resilience.

This section explores how deep learning architectures can enhance NIDS capabilities, particularly in feature extraction and pattern recognition for network traffic analysis.

3.3.1 Common Deep Learning Models for Network Intrusion Detection Systems (NIDS)

Deep learning models, particularly Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Auto-Encoders (AE), Deep Neural Network (DNNs) and Deep Belief Network excel at extracting features and identifying patterns in complex data sets [13]. This makes them well-suited for analyzing network traffic data, which often contains intricate temporal and spatial relationships.

This section explores deep learning (DL) models or algorithms that are increasingly being adopted for NIDS due to their ability to learn complex features from data.

3.3.1.1 Recurrent Neural Networks (RNNs)

RNNs extend the capabilities of traditional feed-forward neural networks by effectively modeling sequential data. They are built with input, hidden, and output layers, where hidden units act as memory elements, considering past inputs when making decisions. RNNs are widely used in various fields, like speech processing and activity recognition [56] [57] [58] [59]. In NIDS, RNNs can be used for supervised classification and feature extraction. However, they struggle with long sequences due to short-term memory limitations [60]. Variants like Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) address this issue [61] [62] [63].

Yin et al. [64] proposed an RNN-based NIDS for binary and multi-class classification on the NSL-KDD dataset. Their model achieved good accuracy with optimal learning rates and hidden nodes, outperforming traditional ML algorithms and a reduced-size RNN model [65]. However, the model suffers from high training times and lower detection rates for specific attack

classes (R2L and U2R) due to increased computational demands. Additionally, a comparison with other DL methodologies is missing.

In another study, Xu et al. [66] proposed an NIDS using RNN with GRU as the main memory unit, along with a multi-layer perceptron and a SoftMax classifier. Their approach demonstrated good detection rates on the KDD Cup'99 and NSL-KDD datasets but exhibited lower performance for minority attack classes (U2R and R2L).

Similarly, Naseer et al. [67] conducted a comparative analysis of DL and ML algorithms for NIDS on a GPU-based platform. Their findings showed LSTM and Deep CNN achieving superior accuracy compared to other models on the NSL-KDD benchmark dataset.

3.3.1.2 Autoencoders (AEs)

AEs are unsupervised neural networks [68] that learn data representations by attempting to reconstruct the input at the output layer. They consist of input and output layers with the same dimensionality and hidden layers with typically smaller dimensions. AEs operate in an encoder-decoder fashion, with variants including stacked AEs, sparse AEs, and variational AEs [69].

Shone et al. [70] proposed a NIDS that combines deep AEs with random forest (RF) for improved computational efficiency and training time. They utilized only the encoder part of the AE in a non-symmetric manner with two stacked deep AEs, each with three hidden layers. Experiments using the KDD Cup'99 and NSL-KDD datasets demonstrated efficiency compared to Deep Belief Networks (DBNs) in terms of accuracy and training time [71]. However, the model showed limitations in detecting R2L and U2R attacks due to insufficient training data.

Yan et al. [72] proposed a NIDS using a stacked sparse autoencoder (SSAE) and SVM. The SSAE served as a feature extraction method, while the SVM acted as a classifier. This approach achieved superior performance compared to other feature selection, ML, and DL methods on the NSL-KDD dataset for both binary and multi-class classification problems. While achieving reasonable detection rates for U2R and R2L attacks, they were still lower compared to other attack classes.

Several studies adopted similar approaches based on sparse AEs and SVMs, demonstrating improved overall performance compared to other DL and ML models on the NSL-KDD dataset (e.g., A-Qatf et al. [73]). However, the performance in R2L and U2R classes is often not discussed.

Papmartizivanous et al. [74] proposed an autonomous misuse detection system [75] that combined sparse AEs for unsupervised learning with the MAPE-K framework [76]. Experiments using the KDD Cup'99 and NSL-KDD datasets revealed a lack of detection accuracy for U2R and R2L attack classes.

Khan et al. [77] introduced an efficient two-stage model based on deep-stacked AEs. The first stage classified the data into attack and normal classes with probabilities. These probabilities were then used as additional features for the final stage, classifying normal and multi-class attacks. This approach was tested on the KDD Cup'99 and UNSWNB15 datasets, with class imbalance addressed through down-sampling (KDD Cup'99) and SMOTE up-sampling (UNSWNB15) techniques to improve attack class detection efficiency.

Malaiya et al. [78] explored various NIDS models based on fully connected networks, variational AEs, and sequence-to-sequence (Seq2Seq) structures, respectively. These models were examined for different datasets: NSL-KDD, Kyoto Honeypot, UNSW-NB15, IDS2017, and MAWILab traces. [79] Results showed that the Seq2Seq model constructed using two RNNs performed the best compared to other models in terms of detection accuracy across all the datasets.

Yang et al. [80] also propose a NIDS model that leverages autoencoders (AEs). Their approach utilizes a specific type of AE called a Supervised Adversarial Variational Autoencoder (SAVAER) with regularization, combined with a Deep Neural Network (DNN). This method is particularly effective at detecting low-frequency and novel attacks, as demonstrated on benchmark datasets like NSL-KDD and UNSW-NB15.

In contrast, Andresini et al. [81] take a multi-stage approach that incorporates AEs alongside a 1D convolutional layer and fully connected layers. During the unsupervised stage, they train two separate AEs, one using normal data and another using attack data, to reconstruct the samples. Then, in the supervised stage, these reconstructed samples are combined to create an augmented dataset. This augmented dataset is then fed into a 1D CNN for classification. Their

model achieves superior performance compared to other DL models on datasets like KDDCup'99, UNSWNB15, and CICIDS2017. However, there are limitations to their approach. They do not evaluate how the model performs on minority classes, and their method does not provide information on the specific characteristics of the attacks it detects.

3.3.1.3 Deep Neural Networks (DNNs)

DNNs are fundamental DL architectures that enable models to learn complex patterns through multiple hidden layers. They consist of an input layer, an output layer, and several hidden layers in between. DNNs are adept at modeling non-linear relationships within data. A greater number of hidden layers allows for increased abstraction levels, enhancing the model's capability [82].

Jia et al. [83] proposed a network NIDS using a DNN with four hidden layers for classifying the KDD Cup'99 and NSL-KDD datasets. The output layer comprised a fully connected layer and a SoftMax classifier. The rectified linear unit (ReLU) served as the activation function for the hidden layers [84]. Their model achieved high detection rates for most attack classes except U2R due to the limited number of instances in the dataset. However, increasing the number of nodes and layers can lead to a more complex structure, which in turn increases computation time and resource consumption. Optimization algorithms and automatic tuning can potentially address these issues.

Wang et al. [85] investigated the vulnerability of DNN-based NIDS to adversarial examples using the NSL-KDD dataset. They evaluated the role of individual features in generating adversarial samples, which were produced using techniques like FGSM, JSMA, DeepFool, and CW attacks [86] [87] [88] [89]. Their findings revealed that commonly used attributes are more susceptible to manipulation by attackers, highlighting the need for additional safeguards.

Vinayakumar et al. [90] introduced Scale-hybrid-NIDS-AlertNet, a scalable hybrid DNN framework for intrusion detection at both network and host levels. This scalable platform was implemented using the Apache Spark cluster co{mputing platform [91]. For NIDS, the model was tested on publicly available datasets like KDDCup’99, NSL-KDD, Kyoto, UNSW-NB15, WSN-

DS, and CICIDS 2017. The results demonstrated the superiority of the proposed model compared to various ML algorithms.

3.3.1.4 Deep Belief Networks (DBNs)

DBNs are DL models constructed by stacking multiple restricted Boltzmann machines (RBMs) layer-by-layer, followed by a final SoftMax classification layer [92]. An RBM is a two-layer (input and hidden) model with bidirectional data flow. Within a DBN, connections exist between each node in a layer and all nodes in the preceding and subsequent layers, but not between nodes within the same layer. DBNs are pre-trained using unsupervised greedy layer-wise learning, followed by supervised fine-tuning to learn valuable features [93]. In the context of NIDS, DBNs are employed for feature extraction and classification tasks.

Marir et al. [94] proposed a distributed model based on DBN and multi-layer ensemble SVM for large-scale network ID using Apache Spark. DBN was used for feature extraction, with features then forwarded to the ensemble SVM. Finally, predictions were made using a voting mechanism. This method's efficiency was tested on the KDDCUP'99, NSL-KDD, UNSW-NB15, and CICIDS2017 datasets, demonstrating promising results for distributed detection of abnormal network behavior.

To improve NIDS detection accuracy, Wei et al. [95] proposed a DL-based model that combines DBN with particle swarms, fish swarms, and genetic algorithms for optimization. The model was evaluated using the NSL-KDD dataset and achieved significant improvement in detection rates for U2R and R2L classes. However, the complex structure of this model leads to increased training time.

3.3.1.5 Convolutional Neural Networks (CNNs)

CNNs are a type of DL architecture particularly well-suited for data stored in arrays. They typically consist of an input layer, followed by a stack of convolutional and pooling layers for feature extraction. Finally, they have fully connected layers and a SoftMax classifier in the output layer. CNNs have achieved remarkable success in computer vision applications [96]. In NIDS, they are used for supervised feature extraction and classification tasks.

Xiao et al. [97] proposed an efficient NIDS based on CNNs. Their approach involved performing feature extraction using principal component analysis (PCA) and AE first. The one-dimensional feature vector was then transformed into a two-dimensional matrix suitable for input into the CNN. Experiments on the KDD Cup'99 dataset demonstrated the effectiveness of their model in terms of training and testing time. However, it exhibited lower detection rates for the U2R and R2L classes compared to other attack classes.

Zhang et al [98] built a complex, multi-layered NIDS model that combines Convolutional Neural Networks (CNNs) and GCForest. To prepare the data for CNN, they introduced a novel P-Zigzag algorithm that converts raw data into two-dimensional grayscale images. Their model utilizes a two-stage approach. First, an improved CNN model (GoogLeNetNP) in a coarse grain layer performs initial detection. Then, a gcForest (caXGBoost) in the fine-grained layer further classifies the abnormal classes into more specific categories. This model was trained on a dataset combining the UNSW-NB15 and CIC-IDS2017 datasets and achieved significant improvements in accuracy, detection rate, and false alarm reduction compared to using either algorithm alone.

Jiang et al. [99] proposed an efficient NIDS system that combines CNNs and Bidirectional Long Short-Term Memory (BiLSTM) networks in a deep architecture. To address the class imbalance problem where some attack types are less frequent, they employed SMOTE to increase the number of minority samples, allowing the model to learn their features more effectively. The CNN component extracts spatial features from the data, while BiLSTM focuses on temporal features. This approach, tested on the NSL-KDD and UNSWNB15 datasets, achieved higher accuracy and detection rates overall. However, while the detection rate for minority classes improved slightly, it remained lower compared to other attack types. Additionally, the complex structure of the model leads to longer training times.

Yu et al. [100] introduced a novel approach based on few-shot learning (FSL) for NIDS. FSL [101] allows training with a small amount of balanced, labeled data. Their model utilizes DNNs and CNNs as embedding functions to extract essential features and reduce data dimensionality. Experiments on the NSL-KDD and UNSW-NB15 datasets demonstrated the model's efficiency in achieving reasonable detection rates for minority attack classes. Notably, this

model required less than 2% of the data for training, highlighting its effectiveness with limited data.

By leveraging these deep learning architectures, NIDS can achieve:

- **Improved Feature Extraction**: Deep learning models can automatically discover and extract informative features from network traffic data, potentially uncovering hidden patterns that might be missed by traditional feature engineering techniques.
- **Enhanced Pattern Recognition**: Deep learning models can learn complex patterns and relationships within network traffic data, enabling them to identify subtle anomalies indicative of malicious activity, such as zero-day attacks or novel attack vectors.
- **Adaptability to Evolving Threats**: Deep learning models can continuously learn and adapt to new attack patterns as they emerge. This is crucial in the dynamic threat landscape of cybersecurity.

3.3.2 Case Studies: Deep Learning for IoT NIDS

Having discussed the various common deep learning algorithms and models for NIDS with several studies using these models, this section discusses some other works that use DL for intrusion detection in the IoT. The emergence of smart and complex IoT devices with various IoT ecosystems requires new NIDS models to be resilient, adaptive, and closer to the edge. Edge nodes already provide computing, storage, communication, and control services. Similarly, they can host security services. Attacks on IoT systems range from gaining access to the local system to DoS, et cetera. Most of the existing work in IoT environments considers centralized solutions for DL-based intrusion detection systems, with a few new studies introducing distributed architectures for network intrusion detection.

Alom et al. [251] developed the first IDS based on Deep Belief Networks (DBN). Their system was trained using the NSL-KDD dataset, which was normalized through a numerical encoding procedure. For the classification phase, their IDS achieved a detection accuracy of 97.5% after just fifty iterations, marking a significant improvement over existing systems, which typically operated with much lower accuracy on similar datasets. The system is notable for not

only detecting attacks but also classifying them into five groups based on limited, incomplete, and nonlinear data sources.

Kim et al. [252] focused on detecting distributed denial of service (DDoS) attacks using long short-term memory (LSTM) networks, a type of recurrent neural network (RNN). Using the KDD Cup 1999 dataset, their approach successfully identified and predicted complex patterns in time-series data by detecting anomalies in temporary patterns of unreliable symptoms. Their system achieved an accuracy of 98.8%, demonstrating the efficacy of deep learning approaches in IDS applications.

Kwon et al. [253] chose a Multi-Layer Perceptron (MLP) model for their IDS evaluation. They employed the NSL-KDD dataset, preprocessing it by normalizing numerical values and encoding categorical values as numerical ones. The network was trained with various hyper-parameter configurations, including the number of units, hidden layers, epochs, and learning rates. The outputs were produced using a SoftMax layer, and the system achieved a promising accuracy of 90%.

Ferrag et al. [254] conducted a comprehensive comparative study of different deep learning models for intrusion detection systems. They evaluated Recurrent Neural Networks (RNN), Deep Neural Networks (DNN), Restricted Boltzmann Machines (RBM), Convolutional Neural Networks (CNN), Deep Boltzmann Machines (DBM), and Deep Autoencoders (DAE). Using two datasets, CSE-CIC-IDS2018 and the Koroniotis version of BoT-IoT, which included 5% of the total dataset for training and testing, they found that CNN and DAE achieved the highest accuracy, at 97% and 98%, respectively, across both datasets.

Yadav and Subramanian [255] proposed a system for detecting DDoS attacks at the application layer through traffic classification using stacked encoders (SAEs). Their dataset was constructed from features extracted from web server logs, including request flooding, session flooding, and asymmetric attacks. The system's features were transformed into numerical form, learned through the SAE, and classified using a logistic regression classifier. Their method improved the detection rate to 98.99% with an average false-positive rate of 1.27%.

Martin et al. [256] introduced a system based on conditional variational autoencoders (CVAE), which not only detected intrusions but also performed feature reconstruction, recovering missing features from incomplete datasets. Using a deviation-based approach and a discriminative framework for traffic samples and classification, the system labeled traffic samples with intrusions that had the least reconstruction error. Tested on the NSL-KDD dataset, the system could recover missing categorical features with accuracies of 99%, 92%, and 71% for features with 3, 11, and 70 values, respectively. Luo and Nagarajan [257] focused on anomaly detection in wireless sensor networks (WSN) using autoencoders (AE), addressing the high computational resource consumption of deep learning in WSNs. Their two-part algorithm detected anomalies in a fully distributed manner without sensor-to-sensor communication, with training offloaded to the cloud. Their dataset was collected over four consecutive months in a real WSN indoor testbed, achieving high accuracy and a low false-positive rate (AUC > 0.8 in most cases).

Diro and Chilamkurti [258] proposed a distributed architecture for IoT networks, where fog nodes handled model training and attack detection while master nodes conducted collaborative parameter sharing and optimization. Their deep learning model, tested with the NSL-KDD dataset, used three layers of neurons (150, 120, and 50) and a SoftMax layer. The distributed model outperformed the centralized one, improving the overall detection accuracy from 96% to 99%.

Finally, Roopak et al. [259] analyzed four different classification deep learning models: MLP, CNN, LSTM, and a combination of CNN and LSTM. They found that the CNN+LSTM model achieved the highest accuracy of 97.16%. Using the CICIDS dataset and balancing the DDoS attack dataset by duplicating data, their system improved the training effectiveness of the deep learning methods.

These case studies highlight the promising capabilities of deep learning for enhancing network intrusion detection. As research and development progress, deep learning models are poised to play an increasingly significant role in safeguarding networks against sophisticated cyber threats.

3.3.4 Benefits of Deep Learning for IoT NIDS

Deep learning offers several advantages over traditional machine learning approaches for NIDS, particularly in handling complex, high-dimensional data typical of modern network traffic. These benefits make deep learning models increasingly attractive for cybersecurity applications [258].

- **Feature Learning**: Deep learning models can automatically learn relevant features from raw data, reducing the need for manual feature engineering [85]. This is particularly valuable in NIDS, where the relevance of different traffic characteristics may not be immediately apparent [70].
- **Scalability**: Deep learning architectures can efficiently handle large-scale datasets, making them suitable for processing the vast amounts of network traffic data generated in modern IoT environments [90].
- **Temporal Modeling**: Certain deep learning models, such as RNNs and LSTMs, are particularly adept at capturing temporal dependencies in network traffic patterns, which is crucial for detecting complex, time-dependent attack sequences [64].
- **Hierarchical Representation**: Deep neural networks can learn hierarchical representations of data, allowing them to capture both low-level features and high-level abstractions of network behavior. This multi-level understanding can lead to more nuanced and accurate threat detection [253].
- **Robust to Noise**: Deep learning models, especially when properly regularized, can be more robust to noise and variations in input data [69]. This resilience is valuable in NIDS, where network traffic patterns can be highly variable [260].

3.3.5 Limitations and Challenges of Deep Learning for NIDS

While deep learning offers significant advantages for NIDS, there are challenges to consider for practical implementation:

- **Data Integrity and Privacy**: Deep learning models often require vast amounts of labeled data for effective training. Obtaining high-quality labeled data, especially for novel attacks,

can be challenging [266]. A study by Ring et al. highlighted the scarcity of up-to-date, publicly available datasets for training deep learning-based NIDS [270]. Additionally, collecting and storing network traffic data, particularly in user environments, raises privacy concerns. Shokri et al. demonstrated that deep learning models can potentially leak sensitive information about the training data [275].

- **Computational Overhead**: Training and running complex deep learning models can be computationally expensive. This can be a limiting factor for deployment on resource-constrained devices within an IoT network or in scenarios with limited processing power [258]. However, recent work by Wang et al. has shown promise in developing lightweight deep learning models that can run efficiently on edge devices while maintaining high detection accuracy [276].
- **Explainability and Transparency**: Deep learning models can be complex "black boxes," making it difficult to understand how they arrive at specific decisions. This lack of explainability can be problematic in security-critical applications where understanding the reasoning behind an alert is essential [277]. Marino et al. proposed a method for improving the interpretability of deep learning-based NIDS but noted that significant challenges remain [278].
- **Security of the Learning Process**: The training process itself can be vulnerable to attacks. Malicious actors could potentially manipulate training data to poison the model and hinder its effectiveness in detecting real threats [279]. A study by Jagielski et al. demonstrated the vulnerability of deep learning-based NIDS to data poisoning attacks and proposed potential mitigation strategies [280].
- **Limited Applicability in Sensitive Environments**: The intrusive nature of deep learning, often requiring extensive data collection and analysis, might not be suitable for highly sensitive environments like medical institutions. Strict data privacy regulations and the need to minimize disruption to critical systems can limit the applicability of deep learning-based NIDS in such settings [281]. Berman et al. discussed the challenges of implementing deep learning-based security solutions in healthcare environments, highlighting the need for privacy-preserving techniques [282].

These limitations pave the way for exploring alternative approaches like federated learning, which can address data privacy concerns and potentially enable deep learning in resource-constrained environments [117]. McMahan et al. demonstrated the potential of federated learning for training deep learning models without centralizing the training data, which could be particularly useful for NIDS in sensitive or distributed environments [116].

Study	Algorithm Type	Methodology
Yin et al. [64]	DL	Recurrent Neural Network
Shen et al. [43]	ML	Ensemble Method with Optimization using BAT algorithm
Shone et al. [70]	ML + DL	Non-Symmetric Deep Auto Encoder with Random Forest
Ali et al. [41]	ML	Fast Learning Network and Particle Swarm Algorithm
Jia et al. [83]	DL	Deep Neural Network
Wang et al. [85]	DL	Deep Neural Network
Yan et al. [72]	ML + DL	Sparse Auto Encoder with SVM
Naseer et al. [67]	ML + DL	Comparison between different ML and DL-based NIDS Models
Xu et al. [66]	DL	Neural Network using gated Recurrent Units as memory with Multiple Layer Perception
Al-Qatf et al. [73]	ML + DL	Self-Taught Learning Model based on Sparse Auto Encoder and SVM
Marir et al. [94]	ML + DL	Deep Belief Network and SVM
Papmartizivanous et al. [74]	DL	Self-Taught Learning based on Sparse Auto Encoder
Khan et al. [77]	DL	Two-Stage Model using Stacked Auto Encoder

Xiao et al. [97]	DL	Convolutional Neural Network with Principal component analysis (PCA) and Auto Encoder for dimension reduction
Yao et al. [46]	ML	A Multilevel Model based on K-Means Clustering and Random Forest
Vinayakumar et al. [90]	DL	Deep Neural Network
Gao et al. [44]	ML	Ensemble Machine Learning methods with Voting algorithm
Wei et al. [95]	DL	Deep Belief Network along with optimization algorithms Particle Swarm, Fish Swarm and Genetic Algorithms
Zhang et al. [98]	DL	Multi-Layer Convolutional Neural Network
Malaiya et al. [78]	DL	Model based on Fully Connected Networks (FCNs), Variational Autoencoder (VAE), and Sequence-to-Sequence (Seq2Seq) structures
Karatas et al. [25]	ML	Performance Comparison of different ML algorithm by first reducing the dataset imbalance ratio using SMOTE
Jiang et al. [99]	DL	Deep Hierarchical Network based on CNN and BiLSTM
Yang et al. [80]	DL	Supervised adversarial Variational Auto Encoder with regularization (SAVAER) and Deep Neural Network (DNN)
Yu et al. [100]	DL	Few Shots Learning (FSL) using DNN and CNN
Andresini et al. [81]	DL	Multistage Auto Encoder and CNN

Table 1 : Recent machine learning (ML) and deep learning (DL)-based network intrusion detection studies and their methodologies as shown in [13]

Study	Strengths	Weaknesses
Yin et al. [64]	A NIDS based on RNN model is proposed that achieves higher performance in terms of detection accuracy using ML algorithms	The model has increased complexity and requires more time for training. Also, it uses an older dataset NSL-KDD for model evaluation. Results demonstrate lower detection rates for minority attack classes like R2L and U2R.
Shen et al. [43]	An idea of using a BAT optimization algorithm together with the Ensemble method based on ELM as the base classifier. Many ELMs combined in an Ensemble way outperform individual ELM performance.	The use of old datasets KDD Cup'99, NSL-KDD, and Kyoto for model evaluation. Moreover, the model achieves lower detection accuracy for the U2R attack class.
Shone et al. [70]	Proposed a NIDS model using a non-symmetric deep AE and RF classifier. The complexity of the model is reduced by using non-symmetric deep AE for efficient feature selection.	The model is evaluated using old datasets KDD Cup'99 and NSL-KDD. The model performance for minority classes R2L and U2R is on the lower side.
Ali et al. [41]	Use of FLN and particle swarm optimization algorithm for proposing a NIDS model. The proposed model showed better performance than other models based on FLN with different other optimization algorithms.	The model is tested using a very old dataset KDD Cup'99. Also, the detection rate for the attack class with less training data (R2L and U2R) is low.
Jia et al. [83]	Proposed a NIDS based on DNN with four hidden layers that have shown	For validating the model, older datasets KDD cup'99 and NSLKDD are used. Also, lower

	superior performance than NIDS based on ML methods.	detection accuracy for the U2R attack class is recorded.
Wang et al. [85]	Studied the effect of DNN-based NIDS against adversaries and evaluated the state-of-the-art attack algorithms against DNN based intrusion detection on the NSL-KDD dataset.	Use of older dataset NSLKDD for validation purposes.
Yan et al. [72]	Efficient use of stacked sparse autoencoder (SSAE) for feature extraction and SVM as a classifier to propose a NIDS.	An older dataset NSL-KDD is used for evaluating the model. The model achieves reasonable detection rates for U2R and R2L attacks, but it is still lower comparing the other attack classes of the dataset.
Naseer et al [67]	Comparison of ML- and DL-based NIDS algorithms by implementing on GPU-integrated testbed.	Use of an older dataset NSL-KDD for evaluation.
Xu et al. [66]	Use of GRU as the main memory for RNN together with the multilayer perceptron and a SoftMax classifier to propose a NIDS solution.	The model is evaluated using old datasets KDD Cup'99 and NSL-KDD. Also, lower detection rates for R2L and U2R attack classes are recorded.
Al-Qatf et al. [73]	The proposal of a NIDS by an efficient idea of self-taught learning based on Sparse AE and SVM.	The model is tested using an older dataset NSL-KDD. Also, no results are provided for the performance of the model against minority attack classes (e.g. R2L and U2R).

Marir et al [94]	Use of DBN and ensemble SVM to detect abnormal behavior in a distributed way. DBN is used for feature extraction, which is then forwarded to the ensemble SVM and finally, the voting mechanism is used for prediction.	The model is complex, and the training time increases slightly for deeper layers.
Papmartizivanous et al. [74]	Propose an autonomous misuse detection system by combining the self-taught learning and MAPE-K frameworks. For learning useful features, a sparse AE is used.	The proposed model is validated using old datasets KDD Cup'99 and NSLKDD. Also, it shows the incapability of detecting R2L and U2R attack classes
Khan et al [77]	Propose an efficient two-stage NIDS model based on deep stacked AE. The result of the first stage along with the obtained probability scores are input the final decision stage as an additional feature to improve for normal and multiclass attack classification. The use of different methodologies (down-sampling/SMOTE for different datasets) during the preprocessing of the dataset dramatically improves the detection rate efficiency of minority attack classes.	For evaluation purposes, KDD Cup'99 and UNSW-NB15 datasets are used. The detection accuracy for KDD Cup'99 is 99.996% while for the UNSW-NB15 dataset detection accuracy is 89.134% which suggests that although the model performed well for an older dataset, however for the relatively newer dataset, the detection accuracy is decreased by almost 10%. Further, it is not shown how the model performs in detecting minority attack classes.
Xiao et al. [97]	An efficient NIDS is proposed based on the DL algorithm CNN. For dimensionality reduction, feature	The performance of the proposed model is evaluated using an older dataset KDD Cup'99. Moreover, the model achieves lower

	extraction is performed considering PCA and AE.	detection rates for the U2R and R2L classes.
Yao et al. [46]	A clustering concept is used along with RF to propose a multilevel intrusion detection model. The model exhibits superior performance in detecting the attacks even with low instances in the dataset.	Use of a very old dataset KDD Cup'99 for model validation purposes.
Vinayakumar et al. [90]	A hybrid scalable DNN framework called scale-hybrid-NIDS-AlertNet is proposed for effective real-time monitoring of network traffic to detect intrusions. The model is tested using different new and old datasets.	A complex model which gives a lower detection rate performance for minority attack class.
Gao et al. [44]	Use of an adaptive ensemble model by using several base classifiers as decision tree, RF, K-Nearest Neighbor, DNN, and choosing the best classifier using an adaptive voting algorithm.	The model is tested using an older dataset NSL-KDD which exhibits unsatisfactory results for weaker attack classes.
Wei et al. [95]	Proposed a DL-based model DBN and is optimized by combining the different optimization algorithms as Particle Swarm, Fish swarm, and Genetic Algorithm.	The proposed model is complex and requires more training time. Furthermore, it is evaluated using an older dataset NSL-KDD.
Zhang et al. [98]	A multilayer NIDS model based on CNN and gcForest is proposed. Also, for converting raw data into a two-dimensional greyscale image, a novel P-Zigzag algorithm is proposed. The model is evaluated using by	The detection accuracy for the attack classes with less training data is low.

	combining new datasets UNSW-NB15 and CIC-IDS2017.	
Malaiya et al. [78]	Different models for ID detection are proposed using fully connected networks, Variational AE, and Seq2Seq structures, respectively. Both new and older datasets are used for model evaluation purposes.	The Seq2Seq model performs best among other models at the cost of greater training overhead than others. Moreover, it is not shown which model performs best in the detection of the minority attack classes.
Karatas et al. [25]	The analysis of six ML-based NIDSs is presented that addresses dataset imbalance by reducing the imbalance ratio using SMOTE. This eventually improves the detection rate for the minority attack class. Also, an up-to-date dataset CSECIC-IDS2018 is used for evaluation.	Adaboost algorithm is shown to achieve higher detection accuracy, but it is done at the cost of higher execution time.
Jiang et al. [99]	A NIDS is proposed using CNN and bi-directional long short-term memory (BiLSTM) in a deep hierarchy. The class imbalance problem is addressed by using SMOTE to increase the minority samples. Evaluation is performed using both older (NSL-KDD) and newer (UNSW-NB15) datasets.	The proposed model is complex, and the detection rate of minority data classes improved slightly but still, it is very low comparing other attack classes.
Yang et al. [80]	A NIDS is proposed based on the supervised adversarial variational AE with regularization and DNN. Evaluation is performed using both	For the NSL-KDD dataset, the model achieves reasonable detection rates for U2R and R2L attacks, but it is still lower compared to the other attack

	older (NSL-KDD) and newer (UNSW-NB15) datasets.	classes of the dataset. For the UNSW-NB15 dataset, the model performance for the detection of minority class attacks is not shown.
Yu et al. [100]	An efficient NIDS model is proposed based on the novel DL idea of Few-shot Learning (FSL). DL algorithms as DNN and CNN are adopted as embedding functions in the model for extracting the essential feature and reducing the dimension. The model is evaluated using NSL-KDD and UNSW-NB15 datasets.	For the NSL-KDD dataset, the model achieves reasonable detection rates for U2R and R2L attacks, but it is still lower compared to the other attack classes of the dataset. Also, FSL requires labeled data for learning which makes its application limited in scenarios, where the model is trained frequently using the unlabeled network traffic to help it learn more patterns for effective detection.
Andresini et al. [81]	Use of AE to propose a multi-stage model involving the intrusion detection convolutional layer and two stacked fully connected layers. Also, the model's performance is tested using both new and older datasets.	The model's effectiveness in detecting the minority attack class is not given. Also, the model is unable to provide details about the structure and characteristics of the attack.

Table 2 : Strengths and weaknesses of the proposed methodologies as shown in [13]

3.4 Challenges and Motivation to Adapt Federated Learning (FL) for IoT Network Intrusion Detection Systems (NIDS)

Traditional machine learning (ML) and deep learning (DL) models have made significant contributions to network intrusion detection systems (NIDS). However, these models face several challenges that motivate the exploration of alternative approaches like federated learning.

3.4.1 Key Challenges in Traditional ML/DL NIDS Implementations

- **Data Privacy and Centralized Storage**: Traditional ML/DL models often rely on centralized servers for data storage and training, raising privacy and security concerns [117]. Users must upload their private data to a central server, creating a privacy risk if the server is compromised. This is particularly concerning for IoT devices in sensitive environments like smart homes or medical institutions [281].
- **Scalability and Real-Time Analysis**: Centralized processing becomes a bottleneck as networks scale, creating a single point of failure that can disrupt service [258]. This centralization also introduces latency, hindering NIDS's ability to provide real-time analysis for critical threats [283].
- **Data Collection Complexity and Diversity**: Modern networks, especially 5G/6G, handle diverse data types (text, audio, video, AR/VR), making centralized data collection burdensome and expensive [284]. This complexity is compounded in IoT environments, where sensor nodes collect vast amounts of critical data with limited resources [285].
- **Dataset Challenges**:
    - **Lack of Up-to-Date Datasets**: Many proposed methodologies struggle to detect zero-day attacks due to training on outdated datasets [270]. Creating and maintaining up-to-date datasets that reflect new attack types is resource-intensive and requires expert knowledge [286].
    - **Imbalanced Datasets**: Most datasets are imbalanced, leading to lower detection accuracy for minority attack classes [26]. While techniques like SMOTE have been used to address this, there's still room for improvement [26][287].

- **Performance in Real-World Environments**: Most proposed methodologies are tested using public datasets in controlled environments. Their performance in real-world scenarios, especially with modern network traffic, remains unclear [266].
- **Resource Constraints**:
    - **Complex Models**: Many proposed IDS methodologies, especially DL-based ones, are computationally intensive and resource-hungry [276]. While high-performance GPUs can speed up computation, they are expensive and not always feasible, especially in IoT environments [288].
    - **Lightweight IDS for IoT**: IoT sensor nodes have limited computational power, storage capacity, and battery life. Designing efficient, lightweight IDS models for these resource-constrained environments is challenging [289].
- **Feature Selection and Optimization**: Efficient feature selection algorithms are needed to reduce computational overhead and improve processing speed [290]. While various optimization algorithms have been explored, there's still scope for improvement in this area [291].

These challenges motivate the exploration of alternative approaches like federated learning, which can potentially address issues of data privacy, scalability, and resource constraints while enabling more effective intrusion detection in diverse network environments.

3.4.2 Federated Learning: A Promising Solution

Federated learning (FL) [126] emerges as a promising approach to address the challenges faced by traditional machine learning and deep learning implementations in network intrusion detection systems. FL introduces a novel collaborative training paradigm where devices train models on their local data and share only model updates with a central server, preserving the privacy of raw data. This decentralized approach offers numerous advantages for intrusion detection systems. By keeping raw data on user devices, FL significantly enhances data privacy and minimizes associated risks. The distributed nature of FL improves scalability, allowing for efficient handling of large-scale networks by spreading processing across multiple devices. Local processing in FL reduces communication overhead, leading to reduced latency and enabling faster, more responsive intrusion detection. The absence of centralized data storage in FL also increases

overall security by reducing the potential attack surface for data breaches. Additionally, FL eliminates the need for cumbersome and expensive centralized data collection in complex networks, resulting in more efficient data management. By addressing these critical challenges, federated learning paves the way for more efficient and secure applications of machine learning and deep learning algorithms in intrusion detection systems, promising significant advancements in the field of network security.

3.5 Conclusion

Machine learning (ML) and deep learning (DL) methods have become integral to developing sophisticated intrusion detection systems (IDS), especially in handling complex and previously unseen attacks. These technologies can identify attack patterns by learning from vast amounts of data, outperforming traditional rule-based systems. However, their application in IoT introduces challenges such as computational overhead, energy consumption, and data privacy concerns. This chapter revealed that combining ML/DL with Federated Learning could significantly enhance the detection capabilities of IDS in IoT ecosystems. The reviewed literature further underscores the importance of resource-efficient algorithms and the need for continuous improvements in both model performance and privacy preservation.

Chapter 4: Federated Learning for Secure and Privacy-Preserving IoT

In this chapter, we dive into the application of Federated Learning (FL) within network intrusion detection systems (NIDS) in the context of the Internet of Things (IoT). We commence with a comprehensive review of the current state-of-the-art in Federated Learning, particularly its relevance to NIDS and the inherent challenges it aims to address. Through this review, we explain the classical processes of machine learning (ML) and deep learning (DL) model training, highlighting their limitations and the need for alternative approaches. Subsequently, we explore the core principles of Federated Learning, shedding light on its mechanisms for preserving privacy in collaborative machine learning scenarios.

Moreover, we discuss the distinct advantages of Federated Learning in decentralized environments such as the IoT, highlighting its potential in mitigating privacy concerns and enhancing security measures. Drawing from existing research efforts, we analyze the application of Federated Learning in constructing secure and privacy-preserving NIDS within IoT networks. We specifically investigate how Federated Learning facilitates collaborative threat detection while ensuring the confidentiality of device data.

Furthermore, we identify key challenges and research gaps in the deployment of Federated Learning for NIDS in IoT environments, notably focusing on issues related to communication efficiency and model aggregation techniques. Finally, we delve into practical implementation considerations, evaluating the viability of the Federated Learning architecture for NIDS implementation. This includes an examination of privacy measures, communication protocols, dataset heterogeneity, and the efficiency of Federated Learning for resource-constrained IoT devices, emphasizing the importance of scalability and low-power consumption in real-world applications.

4.1 Federated Learning: A Paradigm Shift in Collaborative and Distributed Machine Learning

With the rapid evolution of artificial intelligence technologies, machine learning (ML) and deep learning (DL) are becoming increasingly common in various IT applications, such as healthcare, engineering, the Internet of Things (IoT), marketing, and business analytics. While this technological progress represents a significant stride towards the 4th Industrial Revolution, the effectiveness of these algorithms relies heavily on extensive data for model training and testing. While data collection is straightforward in some instances, privacy concerns arise when user data is required for model training. To address these concerns, regulations like the Algerian National Data Protection Authority (ANPDP), established under Law No. 18-07 [102], the General Data Protection Regulation (GDPR) [103], and the Health Insurance Portability and Accountability Act (HIPAA) [104] are famous regulations that have been implemented to safeguard user privacy.

Federated learning (FL) has emerged as an innovative approach to machine learning, facilitating collaborative training on decentralized datasets. This section explores the fundamental

principles of FL, emphasizing its advantages over traditional centralized learning, particularly in the realm of network intrusion detection systems (NIDS) for the Internet of Things (IoT).

4.1.1 Classic Machine Learning and Deep Learning

Before we begin to discuss federated learning in more detail, it is essential to revisit the conventional workings of the most common and classic machine learning and deep learning processes today.

In the realm of machine learning and deep learning, we have a model, and we have data. This model may take the form of a neural network (as illustrated here) or other structures, like classical linear regression.



Figure 6 : Model and Data.

The model is trained using the data to execute specific tasks effectively. These tasks could involve simple predictions or classification problems, object detection in images, transcribing audio recordings, or even mastering strategic games like Go.

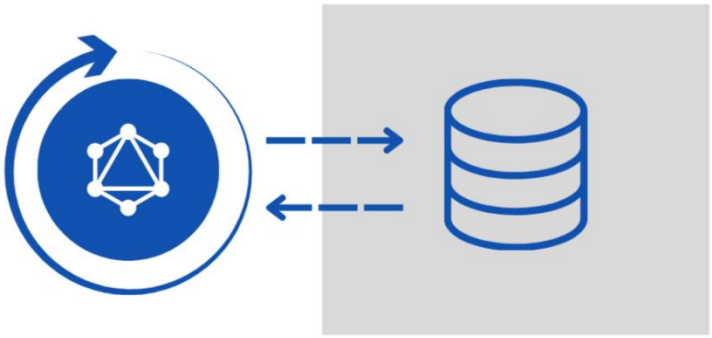


Figure 7 : Training the Model with Data.

In practical scenarios, the training data utilized does not necessarily originate from the same machine where the model is being trained. Instead, it is generated elsewhere.

This data could originate from various sources, such as user interactions with smartphone applications, sensor data collected by vehicles, input received through laptops, or voice commands captured by smart speakers.



Figure 8 : Model and Data Sourced from Smartphone.

It is noteworthy that this data is not limited to a singular source but is often dispersed across multiple locations. These sources could cut across numerous devices running the same applications or distinct organizations contributing data for a common objective.

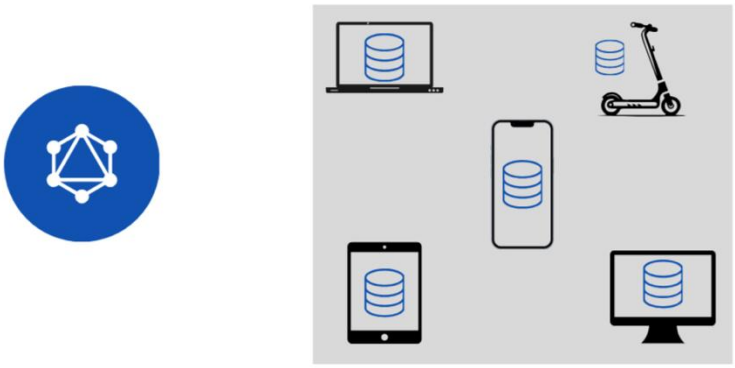


Figure 9 : Fig. 9: Distributed Nature of Data.

Historically, the common approach for machine learning and data analysis involved consolidating all data onto a central server. This server could be situated within a data center or in the cloud.

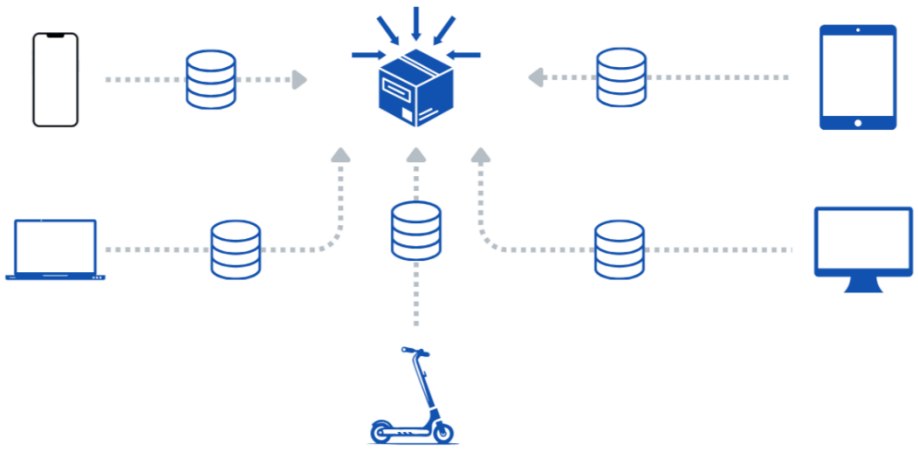


Figure 10 : Centralized Data Aggregation.

Once the data is centralized, machine learning and deep learning algorithms are employed to train models effectively. This centralized method of model training has been the bedrock of traditional machine learning practices.

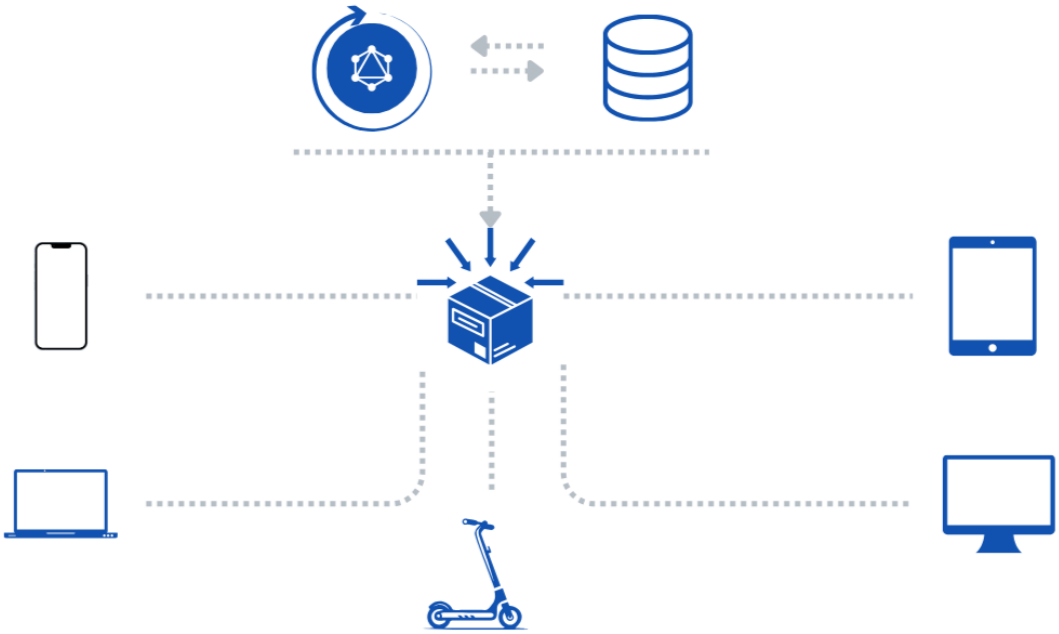


Figure 11 : Centralized Model Training with Aggregated Data.

4.1.1.1 Challenges of Classical Centralized Machine Learning and Deep Learning

While the centralized approach to machine learning and deep learning has its merits, it encounters limitations in various scenarios. For instance, it may be suitable for tasks like categorizing holiday photos or analyzing web traffic where data is readily available on a centralized server.



Figure 12 : Classical Centralized ML/DL Advantages or Possibilities.

However, this centralized approach falls short in numerous scenarios where data is inaccessible on a central server or where the data available on a single server is insufficient for robust model training.

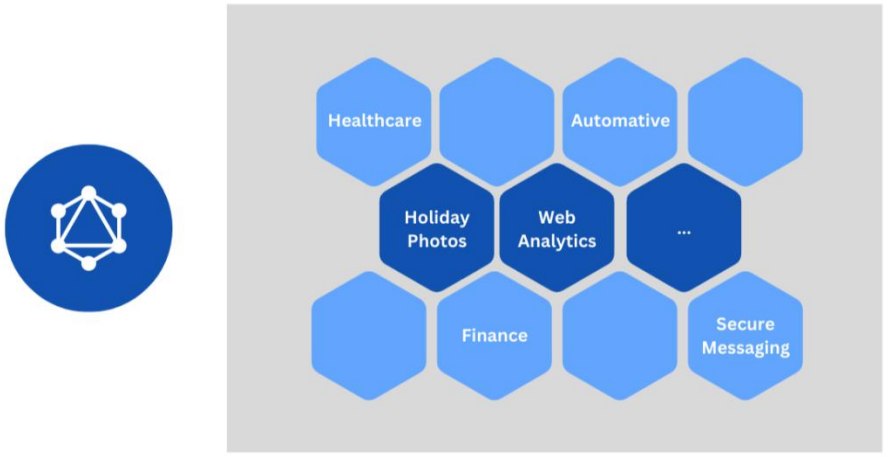


Figure 13 : Classical Centralized ML/DL Challenges.

Numerous factors contribute to the ineffectiveness of the classic centralized machine learning methodology in addressing crucial real-world use cases. These factors include:

- **Regulatory Constraints**: Stringent data protection regulations such as ANPDP (Algeria) [102], GDPR (Europe) [103], HIPPA (US) [104], CCPA (California) [105], PIPEDA (Canada) [106], and other global mandates safeguard sensitive data from cross-border transfers. These regulations often hinder organizations from amalgamating user data for AI training due to diverse data governance frameworks across regions.
- **User Privacy Concerns**: In addition to regulatory barriers, user preferences play a pivotal role. With the increased user awareness of privacy concerns and the associated laws, users are concerned about the security of their data and their privacy and are often reluctant to share their data [107]. Users entrusting sensitive information like passwords and financial details to their devices expect that data will remain within the confines of their devices.
- **Data Volume Challenges**: Certain sensors, like cameras, generate copious amounts of data that make centralized collection impractical or economically unfeasible. The accuracy of

the machine learning models is then negatively affected due to the lack of data availability. The centralized training of ML and DL models requires huge amounts of data as well as powerful computational resources. Arranging enough energy and power for the training and testing of these models is another hindrance to the deployment of these algorithms. Also, the data collected from various sources is in different formats, which requires a lot of pre-processing and cleaning before it is used to train a model. For instance, the extensive data output from security cameras at numerous train stations across a country necessitates substantial infrastructure for processing and storage, posing logistical and financial challenges.

Other instances and situations where centralized machine learning falls short include:

- Utilizing sensitive healthcare records from multiple hospitals to train cancer detection models.
- Aggregating financial data from diverse organizations to identify financial irregularities.
- Harnessing location data from electric vehicles for enhanced range predictions.
- Analyzing end-to-end encrypted messages to refine auto-complete functionalities.

Due to the above-mentioned reasons, classic ML and DL are restricted from reaching their full potential for privacy-constrained applications, IoT ecosystems, etc. Federated Learning (FL), or Federated Machine Learning (FedML), has emerged as a front-runner solution to all the above-mentioned problems.

4.1.2 Federated Learning: Concepts, History, and Intuitive Explanation

4.1.2.1 Concepts and Literature Overview

Federated learning flips the traditional approach by allowing machine learning on decentralized data, shifting the training process to the data itself rather than transferring the data for training. Federated Learning (FL) is a distributed learning approach to machine learning that enables collaborative learning on large datasets without sharing the raw data with a central server or other participants [108]. Instead, the training process occurs locally on individual devices or edge nodes, with only the model updates shared with a central server or coordinator. FL is

particularly valuable when data privacy and security are critical, ensuring that sensitive user data remains decentralized and protected [109].

Here's the single-sentence explanation:

- Central machine learning: move the data to the computation
- Federated (machine) learning: move the computation to the data

This innovative method opens possibilities for utilizing machine learning in new areas that were previously unattainable. For instance, it facilitates collaboration among hospitals to develop advanced medical AI models, aids in combating financial fraud through AI model training with data from various financial institutions and enables the creation of privacy-enhancing applications with superior AI capabilities compared to non-privacy-focused alternatives. These are just a few examples of the transformative potential of federated learning, as it grants access to previously inaccessible data, sparking innovation in various fields. As federated learning is implemented, new opportunities arise for reinventing sectors that now have access to extensive data resources.

FL has gained significant attention in applications involving distributed data sources, including autonomous vehicles, healthcare systems, mobile devices [111], and, specifically, in the context of Internet of Things (IoT) networks [112].

4.1.2.2 History and Pioneering Works

While the core idea of collaborative learning can be traced back to the 1990s, federated learning gained significant traction in the late 2010s.

The idea was first presented by Google in 2016 as a machine learning approach that allows models to be trained collaboratively across multiple devices while keeping the data decentralized and private [113]. It was later implemented on GBoard for Android in 2017 to enhance virtual keyboard search suggestions, demonstrating improvements in quality and user privacy without direct access to user data, showcasing the potential of this decentralized approach in improving user experiences while safeguarding privacy. [4][115].

Since then, FL has garnered significant interest from researchers and industry players due to its potential to address privacy issues and enable collaborative learning in distributed environments. Pioneering works by McMahan et al. (2017) [116] introduced the concept of federated learning in the context of mobile devices, where training data resides on individual devices due to privacy concerns and computational limitations. Qiang Yang et al. (2019) [117] provided a comprehensive overview of Federated Learning, including its principles, architectures, and applications, while discussing the challenges and potential solutions for implementing Federated Learning in various domains. Keith Bonawitz et al. (2019) [118] in this work, also from Google AI researchers, described the system design considerations and challenges for implementing Federated Learning on a large scale. It discussed aspects such as secure aggregation, differential privacy, and communication efficiency.

4.1.2.3 Intuitive Explanation of Federated Learning

Federated learning operates on the fundamental principle of training a global model collaboratively without directly sharing private data from individual devices. On this basis, every FL model follows the below-mentioned steps:

- The server initializes and shares the global model with a set of clients or peers.
- The clients train the model locally with their devices or center's data.
- Each client sends back the encrypted weights or gradient updates to the server.
- The server aggregates the gradients from all clients.
- The global model is updated and shared with the clients in the next iteration.

The following overview provides an intuitive explanation of how federated learning works in five (5) steps.

Initialization of the Global Model

We begin by setting up the central model on the server, like traditional centralized learning.

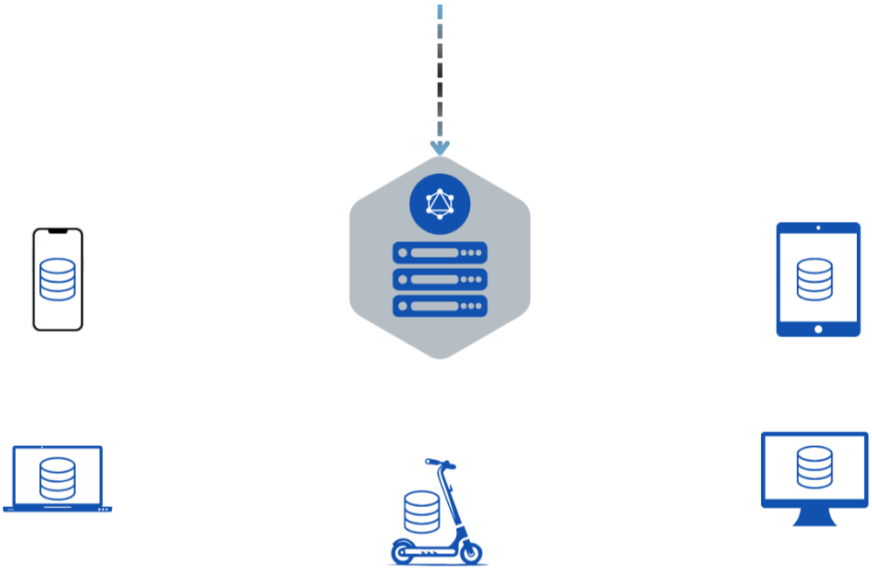


Figure 14 : Initialize and Setup Central Federated Server.

Distribution of Global Model to Client Nodes

Next, we transfer the global model parameters to selected client nodes (edge devices like smartphones or servers belonging to organizations) to ensure uniform starting points for training.

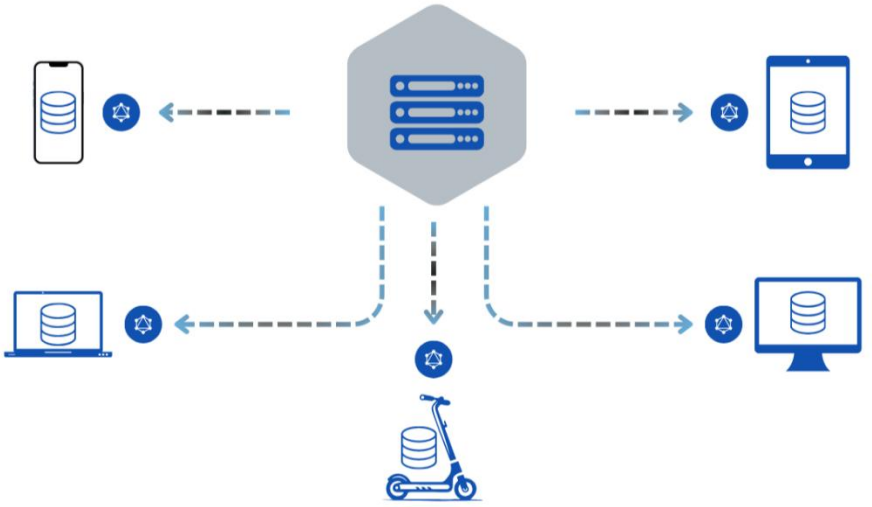


Figure 15 : Distribution of Global Model to Client Nodes.

Local Model Training on Client Devices

Selected client nodes now train their models on their local data for a brief period or in mini batches, using the global model parameters that were originally passed to them.



Figure 16 : Local Model Training on Client Devices.

Return of Model Updates by Clients

After the training is done locally on each participating client, client nodes share their model updates, which the server aggregates using techniques like Federated Averaging to create a new global model.

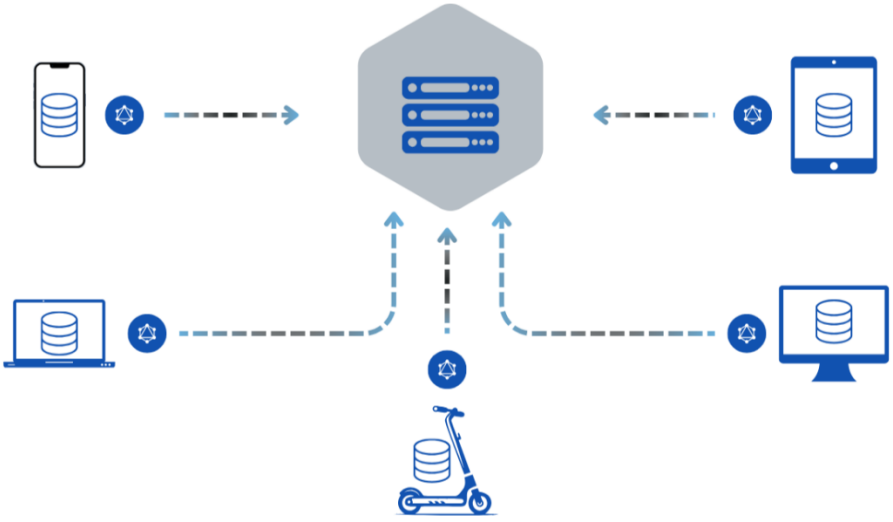


Figure 17 : Return of Model Updates by Clients.

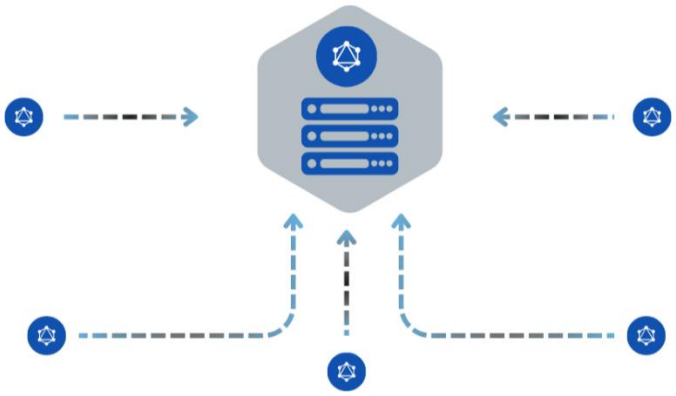


Figure 18 : Aggregation of Client Model Updates and Global Model Update by FL Server.

Iterative Process

The above steps combined are referred to as a "single round of federated learning," so steps 1 to 4 get repeated until the model converges, continually enhancing the global model with insights from each client's data.

4.1.3 Categorization of Federated Learning

Federated Learning (FL) can be categorized based on three main aspects: data distribution, learning model, and architecture [107]. This categorization provides a comprehensive framework for understanding and implementing FL in various domains, considering the nature of the data, the learning task, and the system design.

- **Data Distribution** [107]: a. **Cross-Device**: This approach involves training models on many edge devices, each with a small amount of local data. It is often used in scenarios like mobile keyboard prediction or voice assistants. Google's GBoard project is an example of cross-device horizontal FL working in a centralized fashion [4][127]. **b. Cross-Silos**: This involves training local models at data centers or data silos with a smaller number of participants, each having a substantial amount of data. The local models are then aggregated into a global model at a centralized point. Examples include collaborations between hospitals for medical research or banks for fraud detection.
- **Learning Model** [107]: a. **Horizontal Federated Learning**: Participants have datasets with the same feature space but different sample spaces. For instance, two banks in different regions may have the same type of customer data but for different individuals. [128] b. **Vertical Federated Learning (Feature Space FL)**: Participants have datasets with different feature sets, and the sample space may or may not be the same. Entity alignment and data modeling techniques are required. The FedAI Technology Enabler FATE project is an example of vertical FL [129, 130]. c. Federated Transfer Learning: Used when datasets have neither the same feature space nor the same sample space. It involves using a pre-trained model on a similar dataset to solve a different problem. For example, making movie recommendations based on a user's past browsing behavior [116, 132]. The FedHealth Project uses Federated Transfer Learning.

- **Architecture** [107]: a. **Centralized**: The traditional FL setup has a central server coordinating communication with various data sources. The server is responsible for client selection, model aggregation, and communication of the global model. b. **Decentralized**: i. **Blockchain-based**: utilizes blockchain technology for secure, transparent, and tamper-resistant model updates and aggregation [133]. ii. **Peer-to-Peer**: Clients communicate and coordinate directly without a central server, enhancing privacy and reducing single points of failure.

This categorization highlights the versatility of FL in adapting to various data scenarios and architectural constraints and provides a comprehensive framework for understanding and implementing FL in various domains, considering the nature of the data, the learning task, and the system architecture. It allows researchers and practitioners to choose the most appropriate FL strategy based on their specific requirements for privacy, data characteristics, and system design.

4.2 Federated Learning for IoT Systems (FL + IoT)

Federated Learning (FL) is an innovative distributed machine learning paradigm that facilitates collaborative training of a global model across multiple clients or devices while ensuring data privacy [116]. This approach is particularly advantageous for Internet of Things (IoT) systems, where vast amounts of data are generated by distributed devices. Traditional centralized machine learning methods require aggregating all data to a central server, raising significant privacy and security concerns due to the high scalability of modern IoT networks along with increased bandwidth usage [134]. FL addresses these issues by enabling the training of models locally on edge devices, subsequently aggregating only the learned parameters (model updates) rather than the raw data. [134, 135].

The FL training process consists of two major phases: local training and global aggregation. Initially, the clients download an initial global model from the central server. Each client then trains an updated model based on their local data. These selected clients send their model updates back to the central server, which aggregates the updates (typically by averaging) and generates a new global model. This process is repeated for multiple iterations until the desired performance is achieved [135].

4.2.1 Key Advantages of Federated Learning for IoT Systems

With its innovative operational concept, FL can offer some important benefits for IoT applications, as follows:

- **Data Privacy Enhancement [134]**: Federated Learning (FL) does not require raw data for training on the central FL server, thus minimizing the risk of sensitive user information leakage to external parties and providing a level of data privacy. Considering strict data privacy regulations like the Algerian National Data Protection Authority (ANPDP), established under Law No. 18-07 [102], the General Data Protection Regulation (GDPR) [103], and the Health Insurance Portability and Accountability Act of 1996 (HIPAA) [104], FL's privacy features make it a prime choice for developing secure and intelligent IoT systems.
- **Low-latency Network Communication [134]**: FL eliminates the need to transmit IoT data to the server since the data always remains on the local devices, thereby reducing communication delays due to data offloading and conserving network resources such as spectrum and transmission power during data training.
- **Enhanced Learning Quality [134]**: FL can potentially improve the convergence rate of the training process and achieve higher learning accuracy [136] by leveraging numerous computing resources and diverse datasets from a network of IoT devices. This enhancement may not be possible with centralized AI methods that suffer from limited data and computational power. Consequently, FL also bolsters the scalability of intelligent networks through its distributed learning approach.

Given its unique benefits, Federated Learning (FL) is suggested for a range of Internet of Things (IoT) applications, including smart healthcare, smart finance, smart transportation, smart agriculture, military operations, and unmanned aerial vehicles (UAVs). The subsequent sub-section will delve into various applications of FL in IoT, along with an examination of the numerous studies and research conducted in this field.

4.2.2 Applications of Federated Learning in the IoT

Federated Learning (FL) has emerged as a promising solution for enabling privacy-preserving machine learning in IoT systems across various sectors. The decentralized nature of FL, coupled with its ability to leverage distributed data sources while maintaining data privacy, has opened numerous applications in diverse fields.

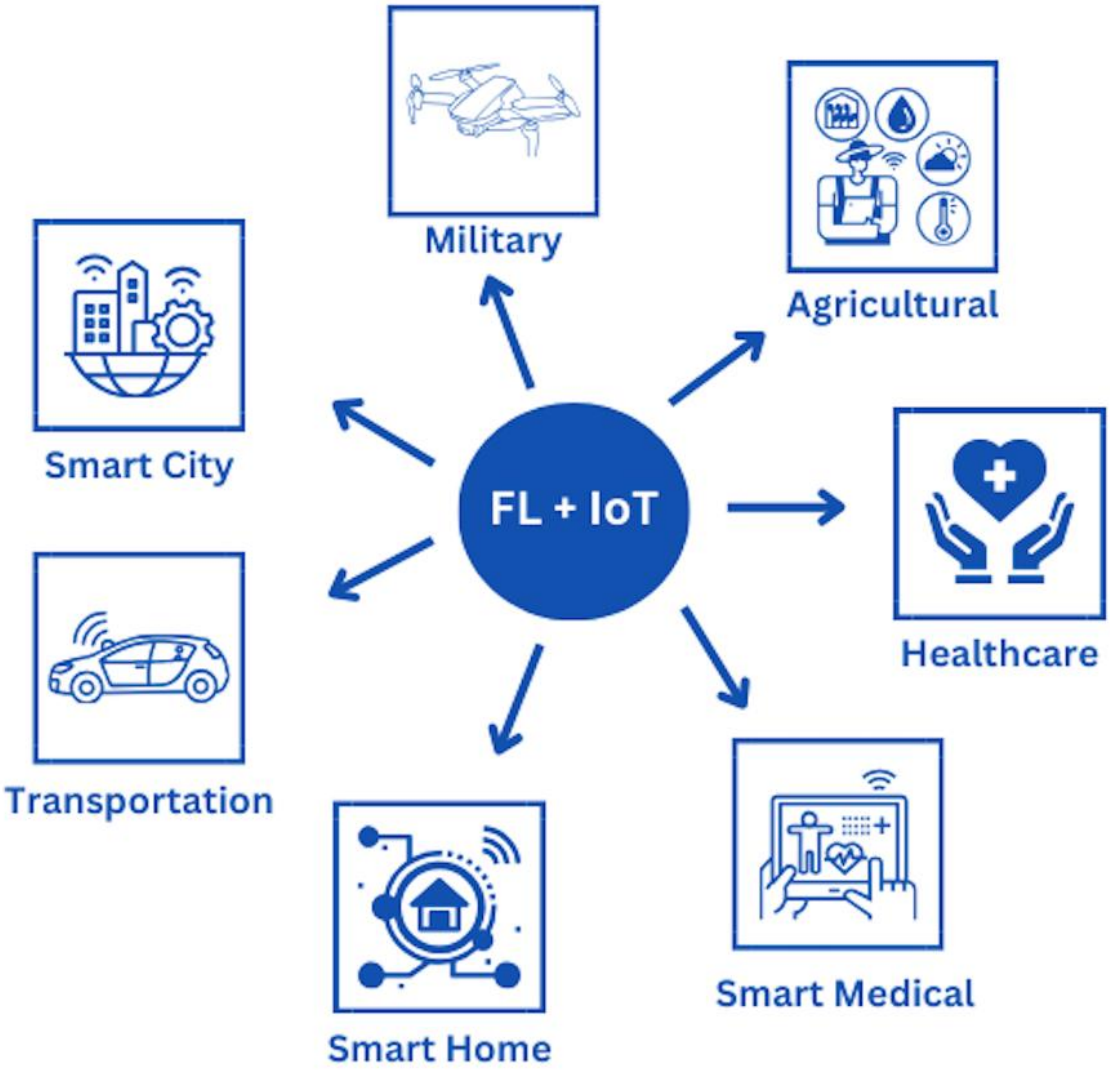


Figure 19 Applications of IoT.

- **Healthcare**: Federated Learning (FL) has gained significant traction in the healthcare domain, where data privacy and security are of crucial importance. There are research articles that encourage the use of FL in healthcare and medical informatics due to its privacy-preserving nature. Data-driven medicines and treatments require patients' data that is confidential in nature. Thus, FL is the best solution for such cases [137]. Teo et al. [138] provide a comprehensive systematic review of federated learning (FL) in health; they identify relevant articles in scientific, engineering, and medical journals in English and include 612 articles; similarly, Rahman et al. [139] explore integrating federated learning (FL), artificial intelligence (AI), and explainable AI (XAI) in healthcare for improved data privacy, accuracy, and transparency in AI decision-making. Zhang et al. [140] in this review focus on methodological advances in FL for healthcare applications between 2015 and 2023, emphasizing the complexity and unique challenges healthcare data pose. Some organizations, like AI Sweden, are using FL to predict the survival rate of emergency care patients in their project "Federated Mortality Prediction" [141]. Another project, "FedHealth" [142], uses Federated Transfer Learning to train the model for personalized prediction on the data collected from wearable health care devices and is tested for Parkinson's disease. Li et al. [143] implemented and evaluated FL for brain tumor segmentation on the BraTS data set and applied differential privacy to guarantee the alliance with the data privacy regulations. Fed-BioMed [144] is an open-source front-end framework for FL in medical science. By leveraging FL, healthcare organizations can collaboratively train machine learning models without sharing sensitive patient data, enabling improved diagnostic accuracy and treatment recommendations.
- **Smart Cities**: In the context of smart cities, FL has been applied to various aspects, including traffic management [145], urban planning, and environmental monitoring [146]. FL allows the integration of data from multiple sources, such as sensor networks, vehicles, and citizen-generated data, while preserving privacy. Zheng et al. and Pandya et al. [147, 148] discuss the applications of federated learning (FL) in smart cities, emphasizing its role in addressing data privacy and protection issues arising from big data and artificial intelligence evolution. This facilitates the development of intelligent systems for optimizing city operations and improving the quality of life for citizens.

- **Industrial IoT**: FL has found applications in industrial settings, where data from various sources, such as machines, sensors, and production lines, can be leveraged for predictive maintenance, process optimization, and quality control [149]. By keeping sensitive industrial data on-premises, FL enables organizations to benefit from collaborative machine learning while maintaining data confidentiality and compliance with regulations. Bahar et al. [150] explore the integration of industrial Internet of Things (IIoT) and artificial intelligence (AI) in smart factories within Industry 4.0. It highlights challenges like data fragmentation, privacy concerns, and inefficient AI models due to a lack of high-quality data. The document proposes a decentralized architecture utilizing federated learning to enable cross-company collaborations and address fragmented data landscapes. Similarly, Rashid et al. [151] proposed a federated learning (FL) method for intrusion detection to protect Internet of Things (IoT) networks. It highlighted the data privacy risks of centralized machine learning for growing IoT devices. The FL approach allows local IoT devices to collaboratively train an intrusion detection model without sharing raw data. Local models are aggregated at a central server to improve the global detection algorithm iteratively. Experiments on a new Edge-IIoT dataset showed the proposed FL method achieved high accuracy comparable to centralized approaches.
- **Transportation**: With the increase of IoT sensors in vehicular networks, it has become crucial to collect data and train machine learning (ML) models for vehicle and traffic management. Since the data in such scenarios is distributed across various vehicles, federated learning (FL) can be applied. One particular use case of interest is predicting the energy demand of electric vehicles (EVs). EVs have emerged as one of the most sustainable solutions in transportation, as they help reduce oil consumption, high energy usage, and gas emissions. Saputra et al. [152] proposed some ML approaches that aimed to improve the efficiency and accuracy of energy demand prediction while reducing communication overhead in EV networks. Similarly, S. Zhang et al. [153] conducted a comprehensive survey on applying federated learning (FL) to intelligent transportation systems (ITS). It highlighted the challenges in ITS, like vehicle behavior decisions and privacy and security risks. It discussed motivations for FL in ITS, like privacy protection and scalability. It reviewed FL deployments in ITS scenarios, analyzed challenges like uneven data and limited resources, and explored potential solutions. Finally, it discussed open challenges

and proposed future research directions for FL in ITS. In a recent work, Bhavsar et al. [154] present an FL-NIDS system for enhancing vehicular network security of connected and autonomous vehicles (CAVs) IoT edge devices. It employs local learning to protect data privacy and utilizes ML and DL classifiers like LR and CNN. The system, implemented on the Raspberry Pi and Jetson Xavier and subsequently deployed on two testbeds with 2 and 4 clients, respectively, outperforms centralized learning in accuracy and loss metrics. Evaluation of the NSL-KDD and Car-Hacking datasets shows significant performance improvements.

- **Agriculture**: FL has been explored for applications in precision agriculture, such as crop yield prediction [158], soil health monitoring, and pest and disease detection [161]. By leveraging data from distributed sources, such as IoT sensor networks, drones, and farm machinery, FL enables the development of data-driven agricultural solutions while preserving the privacy of farm data. [155] Zalik et al. comprehensively reviewed federated learning (FL) applications in agriculture, analyzing different aspects such as data partitioning methods, FL types (horizontal, vertical, hybrid), architectures (centralized, decentralized), federation levels (cross-device, cross-silo), and aggregation algorithms employed across various agricultural use cases, providing an insightful overview of FL techniques and their progress in this domain. Similarly, in [156], Kondaveeti et al. explored the applications of federated learning (FL) in smart agriculture, focusing on disease diagnosis and severity assessment, the Internet of Agricultural Things (IoAT), and yield forecasting, while also highlighting the challenges and opportunities associated with leveraging FL for smart agricultural practices and demonstrating its potential to revolutionize the sector. Le et al. [157] With the aim of bridging the research gap in federated learning for agriculture in ASEAN countries, this work compiled studies on machine learning, deep learning, and federated learning techniques in agriculture from both industrialized nations and regions with similar economic constraints, emphasizing the combination of federated learning with IoT devices as a promising approach for developing smart agriculture. M.T. et al. [158]. To address agricultural production risks and crop yield prediction challenges, this study proposed a federated learning method for training yield prediction models on horizontally distributed datasets, demonstrating that federated averaging using the ResNet-16 regression model with the Adam optimizer yielded optimal

results compared to centralized learning models, paving the way for efficient deployment in federated settings. G. Idoje et al. [159] explored the application of federated learning to smart farming, employing the federated averaging model for crop classification based on climatic parameters, and observed that decentralized models achieved faster convergence and higher accuracy than centralized models, showcasing the potential of federated learning in enabling efficient and accurate crop classification in smart farm decentralized networks. A. Durrant et al. [160], recognizing the challenges of data sharing in the agri-food sector, propose a technical solution based on federated learning that leverages decentralized data to develop a cross-silo machine learning model for soybean yield prediction, demonstrating its ability to facilitate data sharing across supply chains without raw data exchange while enabling the adoption of emerging machine learning technologies to boost productivity.

- **Smart Homes and Buildings**: In the sector of smart homes and buildings, FL has been explored for energy management [162], anomaly detection [163], home automation, and more. By leveraging data from distributed IoT devices and sensors, FL enables the development of personalized and energy-efficient systems while preserving the privacy of household data from IoT devices. Z. Xingsen et al. [164] introduced the framework of federated learning and its current applications in the smart home domain, analyzing potential challenges and providing insights to facilitate the in-depth application of federated learning for smarter and more convenient smart homes and user experiences while preserving data privacy and promoting collaboration among smart devices. M. Angan et al. [165] proposed a federation framework to orchestrate distributed learners across smart buildings, demonstrating that leveraging inter-learner knowledge transfer through federated learning enables achieving personalized lower generalization loss for regression problems like time series forecasting and image classification, paving the way for secure and sustainable ambient intelligence via edge devices. Z. Boqi et al. [166] addressed the challenges of data privacy and single points of failure in smart homes. This work introduced a blockchain-based federated learning approach that leverages edge nodes to maintain a decentralized blockchain for model training, ensuring efficient learning while preserving data privacy from local IoT devices, and proposed a clustering method to tackle

non-homogeneous data distribution, significantly improving model accuracy and generalization.

- **Autonomous Vehicles/Driving**: Along with the advancement of the vehicular IoT, autonomous driving technology is making its way into everyday cars. A reliable self-driving system needs frequent real-time communication in a multi-access communication environment [167]. Additionally, the dynamic spatial and temporal aspects of the vehicular environment necessitate an intelligent approach capable of evolving with environmental changes. The conventional centralized-over-the-cloud method requires the driving system to send vast amounts of raw data to the server, risking potential privacy breaches. The substantial communication overhead from large data transfers and restricted network bandwidth could prevent the driving system from accurately responding to real-time spatial changes. Therefore, integrating federated learning into vehicular edge computing for autonomous driving is emerging as a viable solution to these issues. With federated learning, vehicles need only send minimal data to the cloud, enabling more sensitive adaptation to real-time local variations. Z. Hongyi et al. [168] introduced an approach for end-to-end on-device machine learning by utilizing federated learning for autonomous driving vehicles, specifically for wheel steering angle prediction. Results showed that federated learning can significantly improve local edge model quality, achieve the same accuracy as centralized learning without its drawbacks, accelerate model training speed, and reduce communication overhead, demonstrating its strengths for deploying ML/DL in real-world embedded systems. Similarly, A. Nguyen et al. [169] propose that to address privacy concerns in autonomous driving, a peer-to-peer deep federated learning (DFL) approach be used to train deep architectures in a fully decentralized manner without central orchestration. The proposed Federated Autonomous Driving Network (FADNet) improved model stability, ensured convergence, and handled imbalanced data distribution, achieving superior accuracy compared to recent methods while maintaining user privacy. W. Shuai et al. [170] presented networking and training frameworks for federated deep learning-assisted connected autonomous vehicle (FLCAV) perception. They proposed multi-layer graph resource allocation and vehicle-road contrastive sensor placement techniques to address network management and sensor deployment problems, respectively. Experiments on the CarlaFLCAV platform confirmed the superiority of the proposed techniques over

various benchmarks. Shiva et al. [171]. To enable privacy-aware and efficient vehicular communication networking, this paper developed a mathematical framework for an autonomous blockchain-based federated learning (BFL) design where local on-vehicle machine learning model updates are exchanged and verified in a distributed fashion. The framework captured the impact of controllable network and BFL parameters on system-level performance and quantified the end-to-end delay with BFL, providing insights into deriving optimal block arrival rates.

- **Finance**: In the financial sector, FL has been explored for applications such as fraud detection [172], credit risk assessment [173], and anomaly detection [174]. By leveraging data from multiple financial institutions while preserving data privacy, FL enables the development of more robust and accurate models for financial risk management and security. S. Yueyue et al. [175] present in their survey a comprehensive taxonomy of responsible and effective federated learning (RE-FL) approaches that address stakeholder concerns in FL-based financial services. It identifies six major dimensions: accountability, controllability, fairness, privacy, security, and effectiveness. The paper aims to guide readers in understanding this interdisciplinary field and proposes potential directions for future research, being the first literature review conducted on RE-FL in the financial sector. Similarly, C. Pushpita et al. [176] recognize the vulnerabilities, threats, and risks in financial services. Their paper explores the use of federated learning and blockchain to enhance security and address challenges like cost-effectiveness, automation, and security control. It discusses existing issues, potential solutions using blockchain-enabled federated learning methods, and outlines future research directions for leveraging these technologies in the financial sector.

Other applications of federated learning in the IoT include military and defense settings [180], recommender systems [177], insurance [178], communication enhancement [116, 179], metaverse and virtual reality [167], industry 4.0 (smart manufacturing) [181], robotics [181] et cetera. These applications highlight the versatility of FL in addressing privacy concerns and leveraging distributed data sources across various IoT domains. As research in this field continues to advance, we can expect to see further adoption of FL in numerous other sectors where data privacy and distributed and decentralized learning are crucial.

4.3 Federated Learning for IoT Network Intrusion Detection Systems (NIDS)

4.3.1 Overview of FL in IoT NIDS

Network Intrusion Detection Systems (NIDS) are crucial for maintaining the security and integrity of IoT networked systems by monitoring network traffic and identifying suspicious activities that may indicate security breaches. The heterogeneity of IoT devices and the necessity for frequent training to maintain optimal performance make it a tough task to configure an ML-based anomaly detection method [119], [120]. This, in turn, makes security issues in IoT a major issue for both developers and end users. Over the years, many research proposals have been made to address the open vulnerabilities of IoT devices [121], [122]. Among those, ML-based solutions [123] and [124] for detecting network intrusions have surged.

However, ML-based solutions and traditional NIDS are the least preferred options because they rely on centralized data collection and analysis, which presents several challenges: they raise privacy concerns by aggregating sensitive data from various sources, can face scalability issues, especially in large-scale networks with high data volumes, and represent a single point of failure, making them vulnerable to attacks that can compromise the entire network.

Federated Learning (FL) [112] offers a promising alternative to centralized ML-based solutions and traditional NIDS by enabling the decentralized training of machine learning models directly on devices or edge nodes where the data resides [125], [126]. This approach provides several benefits: it enhances privacy by keeping data localized and only sharing model weights, significantly reducing privacy risks [127]; it improves scalability by leveraging the computational resources of multiple devices; and it enhances robustness by reducing the risk of a single point of failure, making FL the preferred approach over non-FL solutions.

In the context of IoT NIDS, FL allows collaborative training of intrusion detection models using data from multiple network nodes without sharing the raw data. This method ensures the models are up to date with the latest threat patterns while preserving data privacy.

4.3.2 Key and Practical Considerations for FL-NIDS Implementation

Implementing federated learning for network intrusion detection in IoT systems or environments necessitates careful attention to various practical aspects. Properly addressing these considerations will facilitate efficient communication with clients participating in model training and ensure maximum benefits from the effectively trained model. Key aspects encompass techniques for client selection to guarantee optimal responses from end devices, the selection of the most appropriate aggregation algorithms, and framework choice tailored to the specific task [107]. Other considerations include managing data heterogeneity to enhance model learning and performance, ensuring communication efficiency between the FL server and clients, addressing resource constraints inherent in IoT architectures or systems, and maintaining security and privacy to ensure the integrity of model weights transferred to the aggregator, among others.

4.3.2.1 Client Selection Techniques

Implementing Federated Learning (FL) for Network Intrusion Detection Systems (NIDS) in IoT environments requires an effective strategy for selecting clients. The right client selection techniques can significantly impact the efficiency, accuracy, and robustness of the model training process [182]. The following synthesizes key ideas from multiple research papers to offer a comprehensive overview of diverse client selection techniques in Federated Learning (FL) NIDS.

Client selection is a crucial aspect of model training. According to Khan et al. (2023), some techniques proposed in the literature include random selection, which selects clients randomly from a larger pool; FedCS, also known as greedy selection, which selects clients based on their resources, such as computational power and wireless channel conditions; and the use of incentive mechanisms based on contract theory to encourage mobile devices with high-quality data to participate in model training by rewarding their contributions. Additionally, the multi-armed bandit (MAB) approach is highlighted as a useful method when there is no prior information about the clients' computational and communication resources, as it balances exploration of unknown clients with exploitation of those with larger resources. The Pulling Reduction with Local Compensation (PRLC) technique is also mentioned, where clients pull the global model updates only when there is new or updated data at the data source. Each of these methods has its pros and

cons, with some being faster but lacking specific criteria for client selection, while others consider various factors like client availability and resources to improve the system's overall efficiency [107].

Fu et al. (2022) emphasize that client selection algorithms in FL systems prioritize clients based on statistical and system utilities, which optimize performance. They also discuss the importance of balancing exploration and exploitation in scheduling clients and the use of tools such as TensorFlow Federated and FedScale for experimentation. However, challenges such as biased client selection and device availability are noted as significant issues in this context [183].

Hayyolalam and Özkasap (2023) focus on communication efficiency in client selection, categorizing existing methods into loss value-based, reputation-based, threshold-based, and multi-criteria-based strategies. They highlight that FL improves communication efficiency by distributing model training to client devices, reducing latency, energy consumption, and network bandwidth usage. The paper also addresses future trends, such as evaluating network performance in terms of communication efficiency, addressing data privacy and energy consumption challenges, and optimizing client selection in dynamic environments [184].

Gouissem et al. (2023) provide a comprehensive survey on client selection in Florida, underscoring the importance of selecting clients that are representative of the entire population and have the necessary resources for training. They discuss conventional selection techniques, performance-aware selections, resource-aware selections, and methods for enhancing model security. The survey also identifies challenges related to resource constraints, security, heterogeneity, dynamicity, and performance trade-offs, emphasizing the need to balance multiple objectives in client selection to ensure the success of FL systems [185].

These techniques offer diverse approaches to client selection, each with its own strengths and limitations, making it crucial to choose the most appropriate method based on the specific needs and constraints of the IoT ecosystem and NIDS application, as the goal of client selection is to select the proper set of participating clients in a way that mitigates the negative effects of that heterogeneity in order to increase the performance of FL [182].

4.3.2.2 Aggregation Algorithms

Federated Learning (FL) for Network Intrusion Detection Systems (NIDS) in IoT environments relies on aggregation algorithms to combine local model updates from multiple clients into a global model, ensuring privacy and security by keeping raw data on devices. The following synthesis of key aggregation techniques from several research papers provides an overview of these algorithms and their principles. Table 3 presents a summary of significant averaging methods in Federated Learning (FL).

S. No	Averaging Algorithm	Description
1	Federated Averaging – FedAvg [116, 181]	A few communication rounds are enough to train high-quality models using FedAvg. Demonstrated on various model architectures like multi-layer perceptron, CNNs, and LSTMs. Concepts like Differential Privacy and Secure Multiparty Computations can be easily applied.
2	Personalized Federated Learning by Pruning (Sub-FedAvg) [181]	Introduces Hybrid Pruning with averaging on the intersection of clients' subnetworks, improving communication efficiency, resource constraints, and personalized model accuracies.
3	Dynamic Aggregation - Inverse Distance Aggregation (IDA) [181]	Uses the distance of model parameters as a strategy to minimize the effect of outliers and improve the model's convergence rate.
4	Hybrid Federated Dual Coordinate Ascent (HyFDCA) [181]	Solves convex problems in the hybrid FL setting, extending CoCoA for cases where both samples and features are partitioned across clients, with a convergence rate matching or exceeding existing methods.
5	Federated ViT using Dynamic	Uses Vision Transformer models to enhance FL accuracy over large and diverse data, with a reconstructive strategy

	Aggregation (FED-REV) [181]	reducing computation costs of edge devices while preserving accuracy.
6	Federated Stochastic Gradient Descent – FedSGD [181, 191]	Based on selective stochastic gradient descent for Neural Networks, providing privacy and control over learning objectives using differential privacy.
7	Federated Personalization – FedPer [192]	Addresses personalization using a base + personalization layer approach for deep feedforward neural networks, dealing with statistical heterogeneity.
8	Federated Match Averaging – FedMA [193]	Layer-wise matching and averaging of hidden elements with similar feature extraction signatures for constructing the global model. Works well for CNNs and LSTMs, resolving data bias issues.
9	Federated Learning with Dynamic Regularization – FedDyn [181, 194]	Chooses a subset of devices randomly for each round. A dynamic regularizer aligns device level and global solutions, different in regularizing loss and attempting to parallelize gradient computations.
10	Federated Distribution – FedDist [195]	Modifies model architecture by identifying dissimilarities between specific neurons amongst clients. Uses Euclidean distance dissimilarity measurement, performing well on non-IID data with good generalization and personalization.
11	Bayesian Nonparametric Federated Learning of Neural Networks [196]	Uses a Bayesian nonparametric approach to handle heterogeneous data distributions and model uncertainties, improving flexibility and robustness in FL.
12	Similarity-based Federated Learning – SimFL [197, 198]	Enhances model accuracy by selecting clients based on the similarity of their data distributions to the global data distribution, reducing bias and improving generalization.
13	Overlap-FedAvg [199]	Extension of FedAvg that addresses overlapping data distributions among clients, improving model convergence and performance in non-IID environments.

14	FedProx [186]	Modifies FedAvg objective by adding a proximal term to tackle data heterogeneity, ensuring more stable and accurate convergence in FL.
15	LEGATO [186]	Dynamically adjusts learning rates based on the quality of received updates, improving model accuracy and convergence speed.
16	FedAW [188]	Introduces a weighted aggregation mechanism sensitive to client dataset quality, enhancing model performance and convergence. Evaluates local data quality using biased models trained with Generalized Cross Entropy Loss.
17	SAFA [186]	Uses scalable and adaptive federated averaging to improve model performance and address issues of model scalability and efficiency in large-scale FL systems.
18	FedWorse [190]	A robust aggregation strategy modifying model architecture based on client losses, showing improved convergence bounds and trade-offs between accuracy and speed.
19	FedSoftWorse [190]	Like FedWorse, this strategy focuses on soft modifications to client models, balancing convergence and accuracy in FL tasks.
20	FedBetter [190]	Enhances model training by focusing on client updates that improve the global model, demonstrating effectiveness in classification tasks.
21	FedSoftBetter [190]	Combines soft updates from clients to enhance the global model's performance, ensuring stability and efficiency in FL systems.
22	FedSage [190]	A novel strategy combining traditional and new aggregation approaches to improve robustness and efficiency in federated learning environments.
23	Think Tank Aggregation [189]	A two-layer aggregation method selecting valuable models containing rare class data for global aggregation, improving accuracy and resilience to Byzantine attacks.

Table 3 : Averaging Algorithms for Federated Learning.

Khan et al. (2023) discusses various averaging techniques used in FL, such as FedSGD, which was the first technique introduced by Google, and its upgraded version, FedAvg. FedAvg is based on model averaging rather than gradient averaging and has a lower communication cost, though both techniques have been shown to occasionally result in poor performance. More advanced methods, such as FedMA by IBM and FedPer by Adobe, address specific issues like model bias and personalization. Other noteworthy techniques include Bayesian Nonparametric Federated Learning of Neural Networks, Similarity-Based Federated Learning (SimFL), and Overlap-FedAvg, which have been proposed to enhance model performance in diverse scenarios [107]. In addition to the previously discussed aggregation algorithms, in [181] several techniques were mentioned, including FedSGD, FedDyn, Sub-FedAvg, IDA, HyFDCA, and FED-REV. These algorithms address various challenges in federated learning, such as non-IID data, communication efficiency, and local computation requirements, showcasing the diversity and complexity of approaches to improving federated learning systems.

Moshawrab et al. (2023) provide a comprehensive review of FL aggregation algorithms, highlighting strategies to address model staleness and enhance FL efficiency. They discuss fault-tolerant mechanisms that improve the robustness and reliability of FL systems. Learning quality enhancement algorithms like FedProx and LEGATO focus on improving model accuracy, while scalability, personalization, and generalization are key aspects addressed by algorithms such as SAFA and FedDist [186].

Ek et al. (2021) introduce FedDist, an aggregation algorithm designed for pervasive computing environments. FedDist aims to balance generalization and personalization, and it is evaluated using the REALWORLD dataset for human activity recognition on smartphones. The study demonstrates that FedDist outperforms other FL algorithms by achieving a balance between these two metrics, addressing challenges related to heterogeneous devices, asynchronous learning, and model evolution [187].

Tang (2024) presents FedAW, a novel FL algorithm that uses weighted aggregation sensitive to client dataset quality to improve model performance and convergence. FedAW evaluates local data quality using biased models trained with generalized cross-entropy loss,

adapting reweighing to prevent performance degradation due to biased datasets. Empirical assessments on the colored MNIST dataset show FedAW's effectiveness in handling non-IID datasets with biased features, significantly improving convergence rates and overall performance [188].

Jiang et al. (2023) propose a secure two-layer aggregation method for FL to tackle the challenges of unbalanced data distribution and Byzantine node attacks. This method employs a "think tank" concept to select valuable models containing rare-class data information for global aggregation. The experiments demonstrate that this approach outperforms existing algorithms in accuracy and resilience to attacks, effectively learning from rare class data and protecting against malicious models [189].

Mansour et al. (2022) introduce new robust aggregation algorithms for FL, such as FedWorse, FedSoftWorse, FedBetter, and FedSoftBetter, which modify model architecture based on client losses. Mathematical analysis shows the trade-offs between accuracy and speed, and experimental results confirm the effectiveness of these strategies in classification tasks using synthetic datasets. Hybrid strategies combining traditional and novel approaches are also discussed, showing promise for improving FL efficiency and stability [190].

These aggregation algorithms cover a range of approaches to enhance federated learning's efficiency, accuracy, robustness, and scalability, each addressing specific challenges encountered in decentralized model training environments.

4.3.2.3 Federated Learning Frameworks

Federated learning (FL) frameworks provide the necessary infrastructure and tools for implementing FL systems and for Network Intrusion Detection Systems (NIDS) in IoT environments, ensuring data privacy and efficient model training across distributed networks. The following synthesis of key frameworks from several research papers provides an overview of these frameworks and their features.

S. No	Name	Developer	Description
1	IBM Federated Learning [204]	IBM Watson Project	Provides infrastructure for federated learning jobs, supporting Deep Learning and Machine Learning algorithms, with components for both the aggregator and the clients.
2	TensorFlow Federated [205]	Google	Enables experimentation with existing and novel FL algorithms, offering a high-level interface for training and evaluation, and a lower-level core for developing new FL algorithms.
3	PySyft + PyGrid by PyTorch [206, 207]	OpenMind	Provides secure peer-to-peer data management, allowing owners to manage data and access clusters, and is built as an extension of PyTorch, TensorFlow, and Keras.
4	Flower [208, 209]	Joint project by Oxford, UCL, and Cambridge	Supports large scale FL experiments on heterogeneous devices, focusing on scalability, support for heterogeneous devices, realistic results, and privacy preservation.
5	OpenFL [210]	Intel Corp	Collaborative framework for FL that supports various ML and DL models, with multiple aggregation algorithms like FedAvg, FedProx, and FedOpt.
6	Fed-BioMed [211, 212]	French Computer Institute-INRIA	Focuses on biomedical research, providing an FL analysis framework that is user-friendly and easy to deploy.
7	XayNet [213]	Xayn-Berlin	A masked cross-device FL framework that supports horizontal and transfer FL in both cross-device and cross-silos settings.
8	FATE [214, 215]	Webank	Provides secure computing frameworks enabling collaborative model training without data sharing.

9	NVIDIA CLARA [216]	NVIDIA	Focuses on healthcare applications, offering tools for medical imaging and genomics.
10	Substra [217]	Substra Foundation	Designed for collaborative machine learning without centralizing data.
11	OpenFed [218]	OpenFed Consortium	Supports diverse use-cases with a focus on privacy and scalability.
12	UniFed [201]	UniFed Project	Standardizes open-source FL frameworks, simplifying distributed experimentation and deployment.
13	FedLab [202]	NS	Lightweight open-source framework for FL simulation, focusing on communication efficiency and algorithm effectiveness.
14	FederatedScope [203]	NS	Flexible FL platform addressing heterogeneity challenges, supporting synchronous/asynchronous training and customization.

Table 4 : Federated Learning Frameworks.

Khan et al. (2023) discusses various FL frameworks, emphasizing their role in ensuring security and supporting diverse machine learning models. Notable frameworks include FATE, which provides secure computing frameworks enabling collaborative model training without data sharing; NVIDIA CLARA, which focuses on healthcare applications offering tools for medical imaging and genomics; Substra, designed for collaborative machine learning without centralizing data; and OpenFed, which supports diverse use cases with a focus on privacy and scalability. These frameworks cater to different use cases, highlighting the versatility and security features of FL systems [107].

Kholod et al. (2020) provide a comparative review of open-source FL frameworks for IoT systems, including TensorFlow Federated (TFF), which enables developers to try existing FL algorithms and experiment with novel ones; Federated AI Technology Enabler (FATE), offering promising features for FL in IoT but with constraints in documentation and support; Paddle

Federated Learning (PFL), tailored for industrial applications but requiring further development to be suitable for IoT environments; and PySyft, facilitating secure peer-to-peer data management and access, built as an extension of PyTorch, TensorFlow, and Keras. These frameworks address data privacy, security, and communication schemes but face challenges in resource requirements for IoT environments [200].

Liu et al. (2022) introduce UniFed, a unified platform that standardizes open-source FL frameworks. Key features include task specification addressing workflow variations, distributed execution management simplifying distributed experimentation and deployment, and logging and data analysis assisting in evaluating and comparing FL frameworks. UniFed empowers practitioners to select the appropriate FL framework for specific use cases, providing insights and recommendations [201].

Zeng et al. (2021) present FedLab, a lightweight open-source framework for FL simulation, focusing on communication efficiency by providing customizable interfaces and modules for communication and algorithm effectiveness by supporting various deployment scenarios, including standalone, cross-process, and hierarchical setups. FedLab aims to simplify the implementation of FL algorithms and relieve researchers from repetitive tasks, encouraging standard optimization implementation [202].

Xie et al. (2022) introduce FederatedScope, a flexible FL platform designed to handle heterogeneity in data, resources, and learning goals. Key features include an event-driven architecture supporting synchronous or asynchronous training strategies, plug-ins for privacy protection, attack simulation, and auto-tuning, and customization options allowing users to tailor training processes for various federated learning scenarios. FederatedScope addresses the challenges of heterogeneity, providing flexibility, usability, and extensibility [203].

4.3.2.4 Data Heterogeneity

In federated learning, data is often non-identically distributed. For example, in IoT settings, devices gather data in various formats and types for analysis, making local data unrepresentative of the overall distribution. Additionally, the volume of data can significantly differ across devices based on the scenario. The multitude of devices and the communication challenges between clients

and servers are also critical factors to consider in the development of federated learning applications. Therefore, to implement Federated Learning (FL) for Network Intrusion Detection Systems (NIDS) in IoT environments, we must handle data heterogeneity arising from diverse data sources and non-IID data. This section synthesizes insights from several research papers addressing these challenges.

Wikipedia contributors (2024) provide a detailed overview of non-IID data challenges in FL, discussing various types of distribution shifts such as covariate shift, prior probability shift, concept drift, and concept shift. The entry explains how these shifts impact the training process and performance, emphasizing the need for sophisticated data normalization methods to mitigate accuracy loss. The description is based on an analysis of the joint probability between features and labels for each node, categorizing non-IID data into distinct types to better understand their effects on FL systems [181].

Agrawal et al. (2022) discuss the use of FL for intrusion detection systems (NIDS), highlighting the challenges of data heterogeneity and the benefits of using FL to address these challenges. The paper explores how FL can help manage insufficient and unlabeled data in NIDS, reducing false alarms and improving precision. It reviews various FL methods for handling non-IID data in NIDS, such as semi-supervised learning with disagreement-based labeling and multi-task deep neural networks. The study shows that FL can outperform traditional centralized architectures in terms of prediction accuracy and efficiency, despite the challenges posed by data heterogeneity [12].

Chai et al. (2019) presents a comprehensive study on the impact of resource and data heterogeneity on FL, highlighting issues such as stragglers and drop-out parties. The paper discusses the challenges posed by heterogeneity in resources and data within FL systems. Experimental studies reveal that stragglers significantly affect training time, and the paper proposes research problems to mitigate these issues. Questions regarding the classification of parties based on response time, seamless data integration without stragglers, and dynamic identification of drop-out parties are explored. The research underscores the limitations of existing FL approaches and advocates for developing straggler-aware techniques to enhance FL model efficiency and effectiveness [219].

Luca et al. (2022) explore the use of data augmentation to address non-IIDness and improve out-of-domain generalization in FL. The study shows that applying appropriate transformations related to the data-generating process can improve out-of-domain accuracy and communication efficiency. Experiments on datasets like Rotated MNIST, PACS, and OfficeHome demonstrate that data augmentation can mitigate data heterogeneity, reduce gradient norms, and close the performance gap between centralized and federated training. The study connects concepts from domain generalization, federated learning, and causal modeling to enhance understanding of data augmentation's impact on FL [220].

Dai et al. (2022) introduce FedNH, a method for addressing data heterogeneity and class imbalance in Florida. The study shows that FedNH has the smallest standard deviation in testing accuracy across clients, indicating fair performance. The algorithm effectively addresses data heterogeneity and class imbalance, outperforming state-of-the-art methods in terms of accuracy and fairness. FedNH significantly improves performance across all clients equally, surpassing baseline methods across multiple datasets [221].

By addressing the complexities of data heterogeneity, these studies contribute to the development of more robust and efficient FL systems for Network Intrusion Detection Systems (NIDS) in IoT environments that can handle diverse data sources and non-IID data effectively.

4.3.2.5 Other Considerations

Implementing federated learning (FL) for network intrusion detection systems (NIDS) in IoT environments involves addressing several key and practical considerations. Ensuring communication efficiency is crucial, which can be achieved by developing efficient communication protocols to minimize the overhead of transmitting model updates and employing model compression techniques like pruning and quantization to reduce update sizes. The resource limitations of IoT devices, such as limited computational and storage capacities, necessitate the use of lightweight FL algorithms and optimizations to reduce energy consumption, especially for battery-powered devices. Security and privacy are paramount, requiring secure aggregation techniques to encrypt model updates and differential privacy methods to obscure individual data points. Lastly, system scalability must be considered, with architectures designed to accommodate

a growing number of IoT devices and the capability for dynamic participation, allowing devices to join and leave the FL network without disrupting the overall training process.

4.3.3 Related Works in FL for IoT NIDS

As a technology that typically necessitates remote device connections to a central location, IoT is well-suited to the Federated Learning (FL) paradigm. Likewise, NIDSs are usually organized as a distributed network, which further justifies the application of FL in IoT network intrusion detection. Indeed, there has been a growing interest in FL for IoT NIDSs in recent times. The research in this field encompasses various facets of this emerging technology. This section will explore the existing literature on IoT network intrusion detection system implementations with Federated Learning.

Mothukuri et al. [181] propose a decentralized federated learning (FL) approach to anomaly detection in IoT networks, aiming to enhance security while maintaining user data privacy. The methodology involves using gated recurrent units (GRUs) trained on decentralized on-device data, where only the learned model weights are shared with a central server. The study utilizes the PySyft deep learning framework to create virtual instances replicating IoT devices and a central server, facilitating asynchronous training. The model is evaluated using the Modbus network dataset, achieving an average detection accuracy of 90.25%. The architecture that yields the best results includes two hidden layers with 256 neurons and a dropout rate of 0.01. An ensemble learning technique, specifically a random forest decision tree classifier, aggregates updates from multiple sources to optimize the global model's accuracy.

Belarbi et al. [135] proposed a federated deep learning-based IDS for IoT networks using the TON_IoT dataset to design an experiment representative of the real world and evaluate the performance of an FL-based IDS. Their study explores pre-training and investigates various aggregation methods to mitigate the impact of data heterogeneity. They compared their approach to a centralized solution. The comparison shows that the heterogeneous nature of the data has a considerable negative impact on the model's performance when trained in a distributed manner. However, in the case of a pre-trained initial global FL model, their approach demonstrated a performance improvement of over 20% (F1-score) compared to a randomly initiated global model.

They demonstrated that pre-training the global model in FL can address data heterogeneity to some extent. In addition, they demonstrated that FedProx and FedYogi increase detection performance since they are designed to address highly non-IID datasets.

Mauricio et al. [114] employed a federated learning approach using the IoT-23 dataset in their study of anomaly detection in IoT environments. This study addresses the critical issue of securing IoT networks from malware attacks, a concern heightened by the exponential increase in IoT device deployments. They utilized a Federated Learning model, which operates by training local models on individual IoT devices and only sharing model updates with a central server, thus preserving data privacy and reducing the risk of data breaches. The model architecture was based on a multi-layer perceptron (MLP) for both the central server and the federated members, optimized using gradient descent with Nesterov momentum to manage the large volume of data and ensure efficient model convergence. The study's evaluation highlighted that the Federated Learning approach could achieve 100% accuracy on some devices. However, the overall system performance was capped at 70% accuracy due to the non-IID (non-independent and identically distributed) nature of the data and the unbalanced data distribution across different devices. This work is significant as it demonstrates both the potential and the challenges of applying Federated Learning in a practical IoT security context, providing valuable insights for further research in this domain.

Huang et al. [131] proposed a federated learning strategy for improving the training of non-IID data by allowing mutually trusted nodes to self-aggregate into clusters of trust that will participate in federated learning as peers, and results show that using this strategy is considerably more accurate for federated learning, comparable to proposals that envisage centrally sharing a portion of node data and comparable to centralized machine learning accuracy. They evaluated their strategy using the Aposemat IoT-23 (IoT-23 Dataset for short).

Nguyen et al. [222] introduce DÏoT, an autonomous, self-learning distributed system for detecting compromised IoT devices using a federated learning approach. The system utilizes device-type-specific communication profiles to detect anomalous deviations in device behavior, achieving a 95.6% detection rate with no false positives in real-world smart home deployments. DÏoT employs GRU-based models trained with custom-generated network traffic to identify

Mirai-infected devices. The system's architecture includes a local security gateway and a global IoT security service. The security gateway hosts the anomaly detection component, while the global service maintains a repository of device-type-specific models, aggregating locally learned models. Limitations noted include potential conflicts with the learned model due to device behavior evolution and the risk of poisoned models by infected devices.

Li et al. [223] propose DeepFed, a federated deep learning scheme for detecting cyber threats in industrial cyber-physical systems (CPSs). The system comprises a trust authority, a cloud server, and multiple industrial agents. The trust authority manages secure communication keys, while the cloud server builds and shares the intrusion detection (ID) model using the FedAVG algorithm. Each industrial agent trains a local model on data collected from CPS devices and updates its parameters via interactions with the cloud server. The ID model, based on a CNN-GRU architecture, achieves a minimum F1-score of 97.34% on a real industrial CPS dataset. The model's architecture includes three convolutional blocks, two GRU layers, and a multilayer perceptron block, ensuring robust detection performance.

Chen et al. [224] introduce FedAGRU, an intrusion detection system for wireless edge networks (WENs) that combines gated recurrent unit (GRU) models with a support vector machine (SVM) under a custom federated learning algorithm. FedAGRU employs an attention mechanism to prioritize the importance of model updates, significantly reducing communication overhead and ensuring learning convergence. The system is evaluated using the KDD-CUP99, CICIDS-2017, and WSN-DS datasets, achieving a 97.12% F1-score in non-IID scenarios and a 70% reduction in communication costs compared to standard FedAVG. The attention mechanism helps filter out less important updates, thus improving overall efficiency and robustness against poisoning attacks.

Al-Marri et al. [225] propose a novel federated mimic learning approach to create a distributed intrusion detection system (NIDS) while preserving user privacy. This method combines the advantages of federated learning (FL) and mimic learning, using teacher models trained on private user data to label a public dataset, which is then used to train student models. The trained student models are aggregated using the FedAVG algorithm. The NIDS achieves a 98.11% detection accuracy on the NSL-KDD dataset. The system uses feedforward multilayer perceptron (MLPs) with two hidden layers of 256 units each, employing ReLu activation, a

dropout rate of 0.4, and the ADAM optimizer. Both Federated Teacher Mimic Learning (FTML) and Federated Student Mimic Learning (FSML) methods produce similar results, demonstrating the system's effectiveness.

Oliveira et al. [226] propose F-NIDS, a federated learning-based network intrusion detection system (NIDS) designed to enhance scalability and security in IoT networks. The system employs asynchronous communication techniques between system entities to achieve horizontal scalability and utilizes differential privacy techniques to ensure data confidentiality. F-NIDS is adaptable to cloud- and fog-based environments, making it versatile for various IoT scenarios. The study evaluates the robustness and performance of the detection model by introducing Gaussian noise and conducting membership inference black-box attacks and model inversion attacks. Experimental results show that F-NIDS effectively predicts and determines the nature of attacks, with the optimal Gaussian noise level and sample sizes determined to be 21 and ranging from 10K to 25K, respectively. The model achieves high multi-class accuracy, binary accuracy, precision, and recall, demonstrating its efficacy in maintaining a balance between data privacy and classification performance.

Bhavsar et al. [227] introduce FL-NIDS, a federated learning-based intrusion detection system designed to enhance the security of vehicular networks within IoT environments. FL-NIDS leverages edge devices for local learning, sharing only model updates with a central server to protect data privacy. The system integrates machine learning (logistic regression) and deep learning (convolutional neural networks) classifiers to detect cyber threats. The FL-NIDS was evaluated using the NSL-KDD and Car-Hacking datasets and deployed on two testbeds with different client configurations (2 clients and 4 clients). Results indicate that FL-NIDS outperforms traditional centralized learning methods, achieving accuracy rates of 94% and 99% and reducing losses to 0.28 and 0.009, respectively. This study highlights the potential of federated learning to enhance the security and privacy of connected and autonomous vehicles (CAVs) in transportation IoT systems.

Rahman et al. [228] explore the effectiveness of a federated learning-based intrusion detection system (NIDS) for IoT devices, comparing it to centralized and on-device approaches. The federated learning scheme allows local training on IoT devices while sharing model updates

with a central server, thereby preserving data privacy and reducing the risk of a single point of failure. The study uses the NSL-KDD dataset to evaluate the proposed approach, considering three scenarios with different data distributions. Results show that the federated learning-based NIDS achieves accuracy comparable to centralized methods and surpasses distributed, unaggregated on-device models. The study underscores the advantages of federated learning in terms of scalability, privacy preservation, and robustness against cyber threats.

Popoola et al. [229] present a federated deep learning (FDL) approach for detecting zero-day botnet attacks on IoT-edge devices. The method employs a deep neural network (DNN) architecture and the federated averaging (FedAvg) algorithm to aggregate local model updates from multiple IoT-edge devices. This approach ensures data privacy and security by keeping training data on local devices. The FDL model was evaluated using Bot-IoT and N-BaIoT datasets, simulating zero-day attack scenarios. Results demonstrate that the FDL model achieves high classification performance, guarantees data privacy, has low communication overhead, requires minimal memory for training data storage, and maintains low network latency. The FDL method outperforms centralized, localized, and distributed learning methods, proving its efficacy in detecting zero-day botnet attacks in IoT environments.

Huong et al. [230] propose LocKedge, a low-complexity multi-attack detection mechanism designed for IoT edge computing environments. LocKedge aims to provide quick and accurate attack detection at the edge layer, near the source of attacks, thereby reducing the workload on cloud servers and enabling prompt responses. The system is implemented in both centralized and federated learning modes to evaluate its performance from different perspectives. Using the BoT-IoT dataset, LocKedge demonstrated superior performance in terms of accuracy and complexity compared to other machine learning and deep learning methods such as neural networks (NN), convolutional neural networks (CNN), recurrent neural networks (RNN), k-nearest neighbors (KNN), support vector machines (SVM), random forests (RF), and decision trees. LocKedge combines a deep neural network (DNN) model with principal component analysis (PCA) for feature engineering, effectively enhancing the security of IoT networks while maintaining low computational complexity.

Sarhan et al. [231] present a federated learning-based framework for sharing cyber threat intelligence across multiple organizations without exchanging raw data. The approach utilizes a common data format to extract meaningful patterns and a federated averaging algorithm to aggregate model updates, ensuring data privacy. The system is tested using the NF-UNSW-NB15-v2 and NF-BoT-IoT-v2 datasets. The federated model demonstrates effectiveness comparable to centralized and localized models, successfully classifying various traffic types while maintaining data privacy. The study highlights the challenges of maintaining a common feature set across diverse IoT environments.

Zhao et al. (2020) [232] propose an intelligent intrusion detection system using a federated learning (FL) framework with long short-term memory (LSTM) networks. This approach addresses privacy concerns and central server vulnerabilities by training models locally and aggregating updates. Evaluated on the SEA command dataset, the FL-LSTM system achieves a 99.21% F1-score, outperforming centralized LSTM and CNN models. Despite the positive results, the use of an outdated dataset raises concerns about its applicability to modern command sets.

Zhao et al. (2021). [233] introduce a federated learning framework to detect false data injection (FDI) attacks in solar photovoltaic (PV) systems. The decentralized approach leverages LSTM networks for local training, enhancing data privacy and efficiency. Extensive experiments on synthetic data demonstrate the framework's effectiveness in identifying FDI attacks. The study marks a pioneering application of federated learning in power electronics, aligning with data privacy regulations and improving detection capabilities over centralized methods.

Zhang et al. [234] developed FedIoT, a federated learning platform for anomaly detection in IoT devices. The FedDetect algorithm utilizes a local adaptive optimizer and a cross-round learning rate scheduler to enhance performance. Tested on Raspberry Pi devices with the N-BaIoT and LANDER datasets, FedDetect achieves efficient anomaly detection with manageable training times and memory costs, making it suitable for resource-constrained IoT environments. The study underscores federated learning's potential to improve IoT security while preserving data privacy.

Chatterjee and Hanawal [235] propose a hybrid ensemble model named PHEC, adapted to a federated learning framework for intrusion detection in IoT networks. The model combines local training with parameter aggregation, addresses label noise through noise-tolerant techniques, and leverages the robustness of KNN classifiers. Experimental results on four benchmark datasets show high true positive rates and low false positive rates, even with noisy data. The study highlights the model's effectiveness in federated settings, achieving performance close to that of centralized methods.

Lazzarini et al. [236] explore the application of federated learning (FL) as a decentralized approach to intrusion detection systems (NIDS) in IoT environments. They emphasize the privacy-preserving aspect of FL compared to centralized models, where data remains on local devices. The study employs a shallow artificial neural network (ANN) architecture and evaluates it with the ToN_IoT and CICIDS2017 datasets, focusing on binary and multiclass classification tasks. FedAvg is utilized as the aggregation algorithm to merge locally trained models without sharing raw data. Results demonstrate that FL achieves comparable accuracy, precision, recall, and F1-score metrics to centralized approaches while maintaining data privacy. Additionally, alternative aggregation algorithms such as FedAvgM, FedAdam, and FedAdagrad are tested using the Flower FL framework, with FedAvg and FedAvgM showing superior performance in the evaluated scenarios.

Saha et al. [237] introduce FogFL, a federated learning framework enhanced with fog nodes to optimize training in resource-constrained IoT environments. They highlight the challenges of communication overhead and computational requirements in traditional FL setups and propose integrating geospatially distributed fog nodes as local aggregators. These fog nodes help reduce latency and energy consumption by locally aggregating model updates before transmitting them to a central server or cloud. A novel greedy heuristic approach is devised to select the optimal fog node as a global aggregator dynamically based on round-by-round communication between edge devices and the cloud. Experimental results validate FogFL's effectiveness, showing significant reductions of up to 92% in energy consumption and 85% in communication latency compared to conventional centralized FL frameworks. The study underscores FogFL's potential to enhance reliability and efficiency in training federated models for IoT applications.

Zhang et al. [238] propose an anomaly-based intrusion detection system (NIDS) tailored for industrial Internet-of-Things (IIoT) networks using federated learning (FL) with transfer learning techniques. They address the challenge of non-IID data by leveraging instance-based transfer learning at the local level to improve model performance. The study introduces a rank aggregation algorithm with a weighted voting approach to enhance the NIDS's detection capabilities across various attack scenarios. Experimental evaluations on real-world IIoT datasets demonstrate superior accuracy (95.97% with AdaBoost) compared to traditional centralized approaches, highlighting FL's ability to maintain data privacy while achieving robust intrusion detection. The research contributes to advancing privacy-preserving machine learning solutions for secure IIoT environments.

Campos et al. [239] evaluate the effectiveness of federated learning (FL) for intrusion detection systems (NIDS) in IoT environments. They conduct a comprehensive review focusing on multiclass classification tasks using the ToN_IoT dataset, segmented by IoT device IP addresses and attack types. The study investigates the impact of different aggregation functions within the IBMFL framework, assessing FL's capability to mitigate privacy concerns and enhance detection accuracy. Results indicate that FL offers competitive performance compared to centralized approaches while preserving data privacy, making it a promising solution for deploying NIDS in real-world IoT scenarios. The paper identifies key challenges and proposes future research directions to further optimize FL-enabled NIDS implementations.

R C et al. [240] propose a novel approach to network intrusion detection systems (NIDS) using federated learning (FL) enhanced with deep auto-encoders. They address privacy concerns associated with centralized data collection by adopting a federated approach where models are trained locally on device data. The study introduces modifications to the FedAvg algorithm to optimize model aggregation and reduce the transmission of sensitive model weights. Evaluated on the CICIDDoS 2019 dataset, their approach achieves a notable accuracy rate of 98.82% in identifying intrusions, showcasing FL's potential to improve network security while minimizing data exposure. The research underscores the efficacy of deep learning methodologies within federated settings and contributes to advancing privacy-preserving NIDS solutions in networked environments.

Otoum et al. [241] introduce AS-NIDS, a hybrid intrusion detection system (NIDS) tailored for IoT networks, integrating anomaly and signature-based methods. The AS-NIDS model consists of three critical phases: initial traffic filtering at the IoT gateway to reduce incoming packet volume, preprocessing using the target encoder, Z-score normalization, and discrete hessian eigenmap (DHE) for feature extraction and redundancy elimination, respectively. The hybrid NIDS combines a lightweight neural network (LightNet) employing human mental search (HMS) for traffic clustering and the Boyer-Moore algorithm for efficient signature matching. Anomaly detection utilizes deep Q-learning to categorize attacks such as Denial of Service (DoS), Probe, User-to-Root (U2R), Remote-to-Local (R2L), and normal traffic based on signal-to-noise ratio (SNR) and bandwidth metrics. AS-NIDS is evaluated using the NSL-KDD dataset, demonstrating high detection accuracy and reduced computation time.

Man et al. [242] introduce FedACNN, an intelligent intrusion detection mechanism leveraging federated learning in edge-assisted IoT environments. FedACNN integrates a convolutional neural network (CNN) enhanced with an attention mechanism to improve detection accuracy while reducing communication rounds by 50%. The model leverages edge computing capabilities to mitigate network instability and enhance security against various IoT attack vectors.

Liu et al. [243] develop a communication-efficient on-device federated learning framework for deep anomaly detection in Industrial IoT (IIoT). The framework introduces an Attention Mechanism-based Convolutional Neural Network-Long Short-Term Memory (AMCNN-LSTM) model, which effectively captures fine-grained features from time-series data to detect anomalies accurately. A novel gradient compression mechanism based on top-k selection improves communication efficiency by reducing redundant gradients. Experimental results across multiple real-world datasets demonstrate over 92% detection accuracy and a significant reduction in communication overhead compared to traditional FL frameworks.

Huong et al. [244] present a federated learning-based anomaly detection system tailored for industrial control systems in IIoT environments. The system integrates Variational Autoencoder (VAE), LSTM, and Kernel Quantile Estimator (KQE) to detect anomalies in time-series data from SCADA systems. It achieves a high F1-score of 97.9% and reduces bandwidth

consumption by 35% compared to centralized approaches, making it suitable for edge computing deployment.

Friha et al. [245] introduce FELIDS, a federated learning-based intrusion detection system designed for securing agricultural IoT infrastructures. FELIDS employs deep learning classifiers, including deep neural networks, convolutional neural networks, and recurrent neural networks, to protect IoT device data privacy through localized learning and model aggregation. Evaluation across the CSE-CIC-IDS2018, MQTTset, and InSDN datasets demonstrates superior performance over non-federated learning methods.

Qin et al. [246] propose a federated learning (FL) approach enhanced with feature selection for network intrusion detection. They employ a greedy algorithm to optimize feature sets tailored to different attack categories, improving detection accuracy. Multiple global models are generated based on selected features from edge devices, utilizing the NSL-KDD dataset for evaluation. Experimental results demonstrate significant accuracy improvements, reaching up to 70.4% detection accuracy with feature selection, showing a 25.7% enhancement over non-feature-selected models. The approach mitigates aggregation-induced accuracy loss by grouping devices with similar feature sets, ensuring effective model collaboration in resource-constrained environments.

Tian et al. [247] introduce DC-Adam, an asynchronous federated learning (FL) approach designed to address gradient delay and non-IID data challenges in IoT and CPS environments. They propose a Taylor expansion-based compensation mechanism to mitigate asynchronous gradient inconsistencies, enhancing convergence stability. Additionally, a pre-shared data training strategy for non-IID scenarios is implemented to prevent model divergence. Experimental validation across MNIST, CICIDS-2017, and IoT-23 datasets shows superior performance metrics, achieving up to a 92.92% F1-score. DC-Adam outperforms traditional FL methods by 12.8% in accuracy and demonstrates improved precision and recall, making it suitable for distributed threat detection in heterogeneous networks.

Preuveneers et al. [248] propose a permissioned blockchain-based federated learning system for anomaly detection, integrating blockchain to enhance model auditing and preserve

decentralization. The approach chains incremental updates on a distributed ledger, ensuring transparency and accountability without centralized data storage. Tested on the CICIDS-2017 dataset using a vanilla autoencoder, the system achieves 97% accuracy. Blockchain integration minimally impacts performance, enhancing resilience against poisoning attacks. This method offers a scalable solution for collaborative model learning in federated environments, applicable to various neural network architectures beyond anomaly detection.

Cetin et al. [249] evaluate federated learning (FL) for wireless network intrusion detection, emphasizing privacy-preserving collaboration among edge devices. Using a stacked autoencoder (SAE) and FedAVG, the approach achieves effective anomaly detection on the AWID dataset. Experimental results demonstrate FL's efficacy in maintaining classification accuracy while reducing computation and communication costs. The study addresses the challenges of wireless network security, showcasing FL as a viable solution for scalable intrusion detection systems in dynamic and distributed environments.

Zhao et al. [250] propose MT-DNN-FL, a multi-task federated learning approach for network anomaly detection, VPN traffic recognition, and traffic classification. The model simultaneously learns multiple tasks using shared layers and task-specific subnetworks, aggregating model parameters via FedAVG. Tested on the CICIDS-2017, ISCXVPN-2016, and ISCXTor-2016 datasets, MT-DNN-FL achieves high precision and recall across all tasks, outperforming single-task models. The approach addresses data scarcity and privacy concerns by leveraging federated learning, showcasing its efficiency in complex network environments compared to traditional centralized training architectures.

Author	Dataset(s)	Shared Model	Aggregation Function	No. of Clients	FL Framework
Mothukuri et al. [181]	Modbus network dataset	GRU, Random Forest, Decision Tree	FlAverage	Unspecified number of virtual instances replicating IoT devices	PySyft

Belarbi et al. [135]	TON_IoT	DBN, DNN	FedProx, FedYogi	10	PySyft, Flower
Mauricio et al. [114]	IoT-23 Dataset	MLP	FedAVG	23	-
Nguyen et al. [222]	Custom-generated network traffic (Mirai-infected devices)	GRU	Device-type-specific model aggregation	-	-
Li et al. [223]	Real industrial CPS dataset	CNN-GRU	FedAVG	-	-
Chen et al. [224]	KDD-CUP99, CICIDS-2017, WSN-DS	GRU, SVM	FEDAGRU, FedAvg	Up to 50	PySyft
Al-Marri et al. [225]	NSL-KDD	Feedforward MLP	FedAVG	-	-
Oliveira et al. [226]	Various IoT scenarios, Gaussian noise introduced	Not specified	Custom aggregation method, differential privacy techniques	-	-
Bhavsar et al. [227]	NSL-KDD, Car-Hacking datasets	Logistic Regression, CNN	FedAVG	2 clients, 4 clients	-

Rahman et al. [228]	NSL-KDD	Not specified	Custom aggregation method	-	-
Popoola et al. [229]	Bot-IoT, N-BaIoT	DNN	FedAVG	-	-
Huong et al. [230]	BoT-IoT	DNN, PCA	Custom aggregation method	-	-
Sarhan et al. [231]	NF-UNSW-NB15-v2, NF-BoT-IoT-v2	LSTM DNN	FedAVG	-	-
Zhao et al. (2020) [232]	SEA command dataset	LSTM	FedAvg	4	TensorFlow FL
Zhao et al. (2021) [233]	Proprietary synthetic data (solar PV systems)	LSTM	Custom federated learning algorithm	4	Flower
Zhang et al. [234]	N-BaIoT USC, LANDER IoT	Auto-Encoder	Custom aggregation method	9 (Raspberry Pi devices)	-
Chatterjee & Hanawal [235]	NSL-KDD, DS2OS Traffic, Gas Pipeline Data, Water Tank Data	MLP Stacking Ensemble	FedStacking	4	-
Lazzarini et al. [236]	ToN_IoT, CICIDS2017	Shallow ANN	FedAvg, FedAvgM,	-	Flower

			FedAdam, FedAdagrad		
Saha et al. [237]	MNIST	MLP	FogFL	6	-
Zhang et al. [238]	CICIDS2017 CICIDS2018	AdaBoost and Random Forest	Rank aggregation algorithm with weighted voting	5	-
Campos et al. [239]	ToN_IoT	Logistic Regression	FedAvg, Fed+	10	IBMFL
R C et al. [240]	CICIDDoS 2019	Deep auto-encoders	Modified FedAvg	-	-
Otoum et al. [241]	NSL-KDD	LightNet (Neural Network)	Custom aggregation method	-	-
Man et al. [242]	Not specified	CNN with attention	Custom aggregation method	-	-
Liu et al. [243]	Multiple real-world datasets (IIoT)	AMCNN-LSTM	Gradient compression mechanism	-	-
Huong et al. [244]	Not specified (SCADA systems)	VAE, LSTM, KQE	Custom aggregation method	-	-
Friha et al. [245]	CSE-CIC-IDS2018,	DNN, CNN, RNN	Custom aggregation method	-	-

	MQTTset, InSDN				
Qin et al. [246]	NSL-KDD	Not specified	Greedy algorithm for feature selection	-	-
Tian et al. [247]	MNIST, CICIDS-2017, IoT-23	Not specified	DC-Adam	-	-
Preuveneers et al. [248]	CICIDS-2017	Vanilla Autoencoder	Blockchain (Hyperledger Fabric)	-	-
Cetin et al. [249]	AWID	Stacked Autoencoder (SAE)	FedAVG	-	-
Zhao et al. [250]	CICIDS-2017, ISCXVPN-2016, ISCXTor-2016	Multi-task DNN (MT-DNN-FL)	FedAVG	-	-

Table 5 : Summary of federated learning (FL) being applied for IoT Network Intrusion Detection..

In this subsection, we reviewed several studies on federated learning (FL) approaches for intrusion detection systems (NIDS), anomaly detection in IoT networks, and other related environments, exploring how FL can enhance security and maintain data privacy by leveraging decentralized data across various devices. We examined diverse methodologies and architectures combining various machine learning and deep learning models such as GRUs, CNNs, LSTMs, autoencoders, and ensemble methods, emphasizing privacy-preserving techniques, differential privacy, and the use of blockchain to ensure data integrity and security. Integration of edge and

fog computing was also discussed to optimize communication and computational efficiency in resource-constrained environments. Realistic FL scenarios often involve non-IID data distributions [135], where data generated by end devices varies significantly, posing challenges for model training and convergence. Studies addressing non-IID data highlighted the difficulties in maintaining model performance and stability, proposing techniques like transfer learning, instance-based learning, and specialized aggregation algorithms to mitigate these issues. Notable studies in this area include those by Houng et al. (LocKedge) [230], Zhang et al. (FedIoT with transfer learning) [234], and Tian et al. (DC-Adam) [247]. Performance metrics such as accuracy, precision, recall, F1-score, and communication overhead were commonly used to evaluate the effectiveness of FL-based NIDS, demonstrating that FL approaches could achieve performance comparable to or better than centralized methods while preserving data privacy. Evaluations across various datasets, including IoT-23, NSL-KDD, CICIDS-2017, Bot-IoT, and more, provided a comprehensive understanding of the FL models' robustness and applicability in different scenarios. Techniques to safeguard against model poisoning and adversarial attacks and ensure data confidentiality were discussed, with methods like differential privacy, attention mechanisms, and blockchain integration enhancing the security and reliability of FL systems. By combining advanced machine learning techniques, privacy-preserving methods, and robust evaluation metrics, these studies collectively advance the field of federated learning for intrusion detection in IoT and related environments.

4.3.4 Evaluation Metrics Used in FL for IoT NIDS

Evaluating the performance of federated learning (FL) ML and DL models for Network Intrusion Detection Systems (NIDS) in IoT environments is crucial and requires efficient metrics. These metrics help in understanding and assessing the effectiveness of the models in detecting and classifying network intrusions accurately and efficiently. This section introduces the primary evaluation metrics used in FL for IoT NIDS, including true positive (TP), false negative (FN), false positive (FP), and true negative (TN) [219]. The section will also cover the confusion matrix and derived metrics such as precision, recall, F1 score, accuracy, false alarm rate, true negative rate, et cetera, along with their respective formulas and significance.

The foundation of most evaluation metrics in NIDS is based on four fundamental outcomes of a binary classification problem: true positive (TP), false negative (FN), false positive (FP), and true negative (TN). These outcomes are typically represented in a confusion matrix, which provides a tabular summary of how often a model is capable of correctly classifying data samples with their real labels. [236, 296]. In the context of anomaly or intrusion detection in IoT networks, a CM can be used to verify the rate at which a model manages to:

- Correctly identify intrusion instances, i.e., True Positives (TP).
- Correctly identify benign or normal traffic, i.e., True Negatives (TN).
- Incorrectly confuse or classify non-intrusion instances as intrusion, i.e., False Positives (FP).
- Incorrectly confuse or classify intrusion instances as non-intrusion, i.e., False Negatives (FN).

These four metrics form the basis of the confusion matrix, a fundamental starting point for deriving various performance metrics, where an ideal model would identify all TP and TN correctly and never confuse one class of traffic for the other. Of course, this is not realistically achievable. However, the rates of FP and FN should be kept to a minimum.

Confusion Matrix

The confusion matrix is often represented in a tabular representation that summarizes the performance of a classification model by comparing the actual and predicted classifications. The diagonal of the confusion matrix denotes the correct predictions, while nondiagonal elements are the wrong predictions of a certain classifier [236]. Table 6 depicts these attributes of the confusion matrix.

		Predicted Class	
		Attack	**Normal**
Actual Class	**Attack**	True Positive	False Negative
	Normal	False Positive	True Negative

Table 6 : Confusion Matrix.

The confusion matrix helps in visualizing the types of errors made by the classifier and forms the basis for calculating several important evaluation metrics [296]. Further, the CM allows for different evaluation metrics to be calculated, with some of them used in the recent studies, including:

- **Precision (P)**: Precision is obtained by dividing correctly classified intrusions (TP) by the total number of positive instances ($FP + TP$). This provides the rate of traffic that has been correctly classified as positive, and that is positive [236]. It is calculated as:

$$P = \frac{TP}{TP + FP}$$

[13]

- **Recall (R) or True Positive Rate (TPR)**: Recall, also defined as sensitivity or true positive rate (TPR), measures the ability of the model to correctly identify all positive instances (i.e., attacks) [236]. It is obtained from the correctly identified intrusion instances (TP) divided by the total number of intrusion instances and is calculated as:

$$R = \frac{TP}{TP + FN}$$

[13]

- **F1 Score**: The F1 Score is the harmonic mean of precision and recall, providing a balance between the two metrics. It is ideal for the $F1$ score to have a high score, as the higher the score, the better [236]. It is calculated as:

$$F1 = 2 * \frac{P * R}{P + R}$$

[13]

- **Accuracy (Acc)**: Accuracy is also called detection accuracy and is mostly useful when dealing with a balanced dataset. It measures the overall correctness of the model's

predictions [236]. It is the ratio of correctly classified instances among the total number of instances and is calculated as follows:

$$Acc = \frac{TP + TN}{TP\ +\ TN\ +\ FP\ +\ FN}$$

[13]

- **Receiver operating characteristic curve (ROC Curve**): ROC curve is a commonly used graph that summarizes the performance of a classifier over all possible thresholds. [295] It is generated by plotting the True Positive Rate against the False Positive Rate as the value of the threshold is varied for assigning observations to a given class [296]. The calculation of True Positive Rate and False Positive Rate are given in the following equations:

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

[295]

- **False Alarm Rate (FAR) or False Positive Rate (FPR)**: FAR measures the rate of false positive intrusion instances relative to the total actual negative instances (benign or true negatives).

$$FAR = \frac{FP}{FP + TN}$$

[254]

- **True Negative Rate (TNR) or Specificity**: TNR measures the proportion of actual negatives (correctly identified instances) to all the benign instances.

$$TNR = \frac{TN}{TN + FP}$$

[254]

- **False Negative Rate (FNR)**: FNR defined as the ratio of malicious behavior that have been designated as non-intrusive [297],

$$FNR = \frac{FN}{FN + TP}$$

[297, 298]

- **F-Measure**: It is defined as the harmonic mean of precision and recall. In other words, it is a statistical technique for examining the accuracy of a system by considering both precision and recall of the system [13].

$$F - Measure = 2 * \left(\frac{P * R}{P + R}\right)$$

[13]

In multiclass classification, evaluating model performance becomes more complex than in binary scenarios, requiring a nuanced approach to metric selection and interpretation. As Grandini et al. (2020) [292] describe, the confusion matrix expands to $K\ x\ K$ dimensions for K classes, serving as the foundation for various metrics. While accuracy remains applicable, it may not suffice alone, especially for imbalanced datasets. The authors explain that precision, recall, and F1-score can be extended to multiclass settings using macro-averaging (calculating metrics independently for each class and then averaging) or micro-averaging (aggregating contributions across all classes).

Grandini et al. highlight balanced accuracy as particularly useful for imbalanced datasets by giving equal weight to each class, regardless of size. They also discuss the Matthews Correlation Coefficient (MCC), adapted from binary classification, which provides a balanced measure of the confusion matrix in multiclass scenarios. The paper explores Cohen's Kappa, which measures agreement between predicted and actual classifications while accounting for chance agreement.

These metrics, as presented in the paper, offer different perspectives on model performance, with some like balanced accuracy and MCC being more robust for imbalanced datasets. The authors emphasize the importance of carefully considering the characteristics of specific applications when selecting evaluation metrics, ensuring a comprehensive assessment of model performance across multiple classes.

In conclusion, evaluation metrics play a crucial role in assessing the performance of ML and DL models used in Federated Learning (FL) for IoT Network Intrusion Detection Systems. While traditional metrics like accuracy, precision, recall, and F1-score provide valuable insights, the complexity of multiclass classification in IoT environments necessitates the use of more sophisticated metrics. Balanced accuracy and the Matthews correlation coefficient offer more robust evaluations for imbalanced datasets, which are common in NIDS. Additionally, macro- and micro-averaging techniques allow for a more nuanced understanding of model performance across multiple classes.

As IoT networks continue to evolve and face increasingly complex security threats, the selection and interpretation of these metrics become critical. It is therefore paramount to consider the specific characteristics of the IoT NIDS applications when choosing evaluation metrics, ensuring a comprehensive and accurate assessment of model performance. Naidu et al. (2023) [294] provide a comprehensive examination of evaluation metrics for machine learning, detailing their principles, usage, limitations, and effectiveness under various conditions, helping in making informed decisions. Future work in this area may focus on developing metrics that specifically address the unique challenges of various ML and DL models used for federated learning in distributed IoT environments, such as communication efficiency and privacy preservation.

4.3.5 Benchmark Datasets Used in FL for IoT NIDS

This section reviews various datasets commonly used in Federated Learning for IoT Network Intrusion Detection Systems (NIDS). These datasets play a crucial role in evaluating the performance and effectiveness of proposed methodologies for detecting network intrusions. Each data set has unique characteristics, attack types, and features that make them suitable for different research and implementation objectives.

4.3.5.1 Popular and Widely Used Datasets

- **KDD Cup'99 Dataset [299–301]**: The KDD Cup'99 dataset is one of the most popular and widely used datasets for intrusion detection systems. It contains approximately 5 million records for training and 2 million records for testing. Each record has 41 different features or attributes and is labeled as either normal or attack. The attacks are classified into four main categories: Denial of Service (DoS), Probe, Remote to Local (R2L), and User to Root (U2R). The dataset was derived from the DARPA98 dataset [299], which was created by DARPA (Defense Advanced Research Project Agency) in 1998. It simulates network traffic in a small US Air Force base environment. Despite its popularity, the KDD Cup'99 dataset has been criticized for its age and the presence of redundant records, which can bias machine learning models. Chen et al. (2020) [224] used the KDD Cup '99 as one of the datasets they used to evaluate their Federated Learning-based Attention Gated Recurrent Unit (FedAGRU) for wireless edge networks, which showed improved detection accuracy when compared to other centralized learning algorithms with a lower communication cost of about 70% less than other FL algorithms.
- **NSL-KDD [299–301]**: The NSL-KDD dataset is a refined version of the KDD Cup'99 dataset, addressing some of its limitations. It removes redundant records from the original KDD dataset and creates more sophisticated subsets. NSL-KDD maintains the same 41 features as KDD Cup'99 and the four attack categories (DoS, Probe, R2L, and U2R). The dataset contains about 150,000 data points and is divided into predefined training and test subsets. While NSL-KDD improves upon KDD Cup '99, it's important to note that the underlying network traffic dates to 1998, which may not fully represent modern attack scenarios. Bhavsar et al. [227] used the NSL-KDD dataset to evaluate their FL-NIDS, a federated learning-based intrusion detection system for vehicular networks within IoT environments. Their system achieved an accuracy rate of 94% using this dataset. Padmashani et al. [305] used the NSL-KDD dataset for the evaluation of their federated learning model in their study for network intrusion detection systems. The study presents a novel method that integrates BiLSTM and DNN architectures using TensorFlow Federated Learning to enhance network intrusion detection systems (NIDS). By leveraging decentralized learning and collaborative intelligence, their approach improves detection

accuracy and strengthens defenses against cyber threats. The model achieved high accuracy rates of 99.71% on the IEC dataset and 95.43% on the NSL-KDD dataset. The integration of BiLSTM and DNN architectures through TensorFlow Federated Learning enhances the IDS's ability to detect cyber threats efficiently and adapt to evolving security challenges.

- **UNSW-NB15 [299–301]**: The UNSW-NB15 dataset was created by the Australian Centre for Cyber Security in 2015. It contains approximately 2 million records with a total of 49 features, extracted using Bro-IDS, Argus tools, and some newly developed algorithms. The dataset includes both normal activities and nine types of attacks: worms, shellcode, reconnaissance, port scans, generic, backdoor, DoS, exploits, and fuzzers. UNSW-NB15 was generated using the IXIA PerfectStorm tool in the Cyber Range Lab to create a hybrid of real modern normal activities and synthetic contemporary attack behaviors. It provides a more up-to-date representation of network traffic compared to older datasets. A study by Sarhan et al. (2021) [231] utilized the UNSW-NB15-V2 dataset in a netflow format to evaluate a federated learning-based intrusion detection framework, with results demonstrating the efficiency and effectiveness of the proposed framework with the universal model properly classifying benign and intrusive traffic from multiple organizations while preserving local data privacy.
- **CICIDS 2017 [299, 301]**: The CICIDS 2017 dataset was created by the Canadian Institute for Cybersecurity (CIC) in 2017. It contains both normal flows and updated real-world attacks. The network traffic is analyzed using CICFlowMeter, which extracts information based on timestamps, source and destination IP addresses, protocols, and attacks. CICIDS 2017 includes common attack scenarios such as brute force attacks, Heartbleed attacks, botnets, denial-of-service (DoS) attacks, Distributed DoS (DDoS) Attack, web attacks, and infiltration attacks. The dataset was created over a period of 5 days in an emulated environment and contains network traffic in both packet-based and bidirectional flow-based formats. Over 80 features were extracted for each flow, providing rich information for intrusion detection research. Lazzarini et al. [236] employed the CICIDS2017 dataset to evaluate their federated learning-based NIDS. They used this dataset for both binary and multiclass classification tasks, demonstrating that FL achieves comparable accuracy, precision, recall, and F1-score metrics to centralized approaches while maintaining data privacy.

- **IoT-23 [293, 302]**: The IoT-23 dataset is a labeled dataset with malicious and benign IoT network traffic and a comprehensive collection of network traffic captured from Internet of Things (IoT) devices. It consists of 23 scenarios; these scenarios are divided into twenty network malware captures (pcap files) from infected IoT devices and three benign network captures of real IoT device network traffic, providing a diverse representation of IoT network behavior. The dataset includes various attack types such as Command and Control (C&C), Distributed Denial of Service (DDoS), File Download, and botnet activities (including Mirai, Okiru, and Torii variants). The IoT-23 dataset consists of twenty-three captures (called scenarios) of different IoT network traffic. The IoT-23 network traffic was captured in the Stratosphere Laboratory, AIC Group, FEL, CTU University, Czech Republic, with the goal of offering a large dataset of real and labeled IoT malware infections and IoT benign traffic for researchers to develop machine learning algorithms. This dataset and its research are funded by Avast Software, Prague, and are particularly valuable for researchers focusing on IoT-specific security challenges and the unique characteristics of IoT network traffic. Mauricio et al. [114] utilized the IoT-23 dataset to implement a federated learning approach for anomaly detection in IoT environments. The Federated Learning model allowed local training on IoT devices while only sharing model updates with a central server, thus enhancing data privacy. The evaluation showed device-specific accuracy reaching up to 100%, but the overall system accuracy was limited to 70% due to decentralized training and uneven data distribution. Huang et al. [131] proposed a federated learning strategy using the Aposemat IoT-23 (IoT-23 Dataset for short) for improving the training of non-IID data by allowing mutually trusted nodes to self-aggregate into clusters of trust that will participate in federated learning as peers, and results show that using this strategy is considerably more accurate for federated learning, comparable to proposals that envisage centrally sharing a portion of node data and comparable to centralized machine learning accuracy. Equally, Tian et al. [247] using the IoT-23 dataset propose a delay-compensated Adam (DC-Adam) approach, an asynchronous federated learning-based detection approach, for IoT devices with limited resources. They developed a Taylor expansion-based scheme to compensate for the inconsistency caused by asynchronous communication, which helped overcome the notorious gradient delay problem. They combined both theoretical analysis of convergence

and practical experiments for their evaluation, which showed stable convergence and outperformed barrier-free asynchronous federated learning by 12.8% (accuracy) and 14% (precision). 8.71% (recall) and 11.16% (F1 score) on average compared to the other state-of-the-art approaches.

- **CICIoT 2023 [303, 307]**: The CICIoT2023 is an IoT-based dataset that is the largest (as of 2023) in terms of the number of devices used to set up the network topology and the number of attacks studied. Created by the Canadian Institute for Cybersecurity (CIC) at the University of New Brunswick, the dataset captures a variety of attack scenarios, simulating real-world conditions to provide a robust testing ground for cybersecurity measures. It consists of different types of attack categories, namely DDoS, DoS, Recon, Web-based, brute force, spoofing, Mirai-like, and normal traffic data, ensuring a diverse range of scenarios. All attacks are executed by malicious IoT devices targeting other IoT devices. To accomplish this, 33 attacks are executed in an IoT topology composed of 105 devices. The dataset's primary aim is to facilitate the development and evaluation of intrusion detection systems, enabling researchers to improve IoT security by providing a novel and extensive IoT attack dataset to foster the development of security analytics applications in real IoT operations. Narayan et al. [304], despite not using a federated learning approach, present a framework for an ML-based intrusion detection system tailored for the imbalanced nature of the CICIoT2023 dataset, focusing on IoT security. It employs the Random Forest algorithm and improves detection metrics by applying feature selection and class balancing techniques. The study shows improvements of 3.72% in precision, 3.75% in recall, and 4.69% in F1-score over the base model. Particularly for unsaturated classes, the F1-score improved by 7.9%. The results underscore the effectiveness of combining correlation-based feature selection (CFS) and a balanced random forest classifier (BRFC) in enhancing the detection of IoT security attacks, highlighting the framework's potential for real-world applications.
- **ToN_IoT [306, 307]**: The ToN_IoT dataset is a comprehensive collection of network and operating system data from a realistic IoT and IIoT (industrial IoT) environment that includes telemetry data for IoT and IIoT services, as well as operating system logs and network traffic for the IoT network, collected from a realistic representation of a medium-scale network at the Cyber Range and IoT Labs at UNSW Canberra (Australia). It contains

both normal and various attack scenarios, including DoS, DDoS, ransomware, backdoor, injection, XSS, and password cracking. This dataset explores the interaction of network elements across the edge, fog, and cloud layers and tries to provide data for intrusion detection in large-scale IoT network scenarios. It also provides a rich set of features extracted from network traffic, operating systems, and IoT telemetry data, making it particularly valuable for evaluating machine learning-based intrusion detection systems in IoT contexts. With data heterogeneity being a core lacking in most state-of-the-art datasets, this dataset includes heterogeneous data sources. Belarbi et al. [135] used the TON_IoT dataset to design an experiment representative of the real world and evaluate the performance of an FL-based IDS. Their study explores pre-training and investigates various aggregation methods to mitigate the impact of data heterogeneity. They compared their approach to a centralized solution. The comparison shows that the heterogeneous nature of the data has a considerable negative impact on the model's performance when trained in a distributed manner. However, in the case of a pre-trained initial global FL model, their approach demonstrated a performance improvement of over 20% (F1-score) compared to a randomly initiated global model. Equally, Campos et al. [239] used the ToN_IoT dataset to evaluate the effectiveness of federated learning for intrusion detection systems in IoT environments. Their study focused on multiclass classification tasks, demonstrating that FL offers competitive performance compared to centralized approaches while preserving data privacy.

- **Modbus-based Network Dataset [181]**: This dataset is specifically designed to represent network traffic in industrial IoT applications that use the Modbus protocol. It includes a variety of attack scenarios relevant to industrial control systems, such as man-in-the-middle attacks, Ping DDoS floods, Modbus Query floods, and SYN DDoS attacks. The dataset is particularly useful for researchers focusing on security in industrial IoT environments and SCADA systems. Mothukuri et al. [181] used a Modbus network dataset to evaluate their decentralized federated learning approach for anomaly detection in IoT networks. Their GRU-based model achieved an average detection accuracy of 90.25% on this dataset, demonstrating the effectiveness of federated learning in industrial IoT settings.

4.3.5.2 Other Notable Datasets

- **Bot-IoT [307–309]**: The Bot-IoT dataset was created by the Cyber Range Lab of UNSW Canberra. It incorporates both normal and botnet traffic in an IoT environment. The dataset is a simulated dataset created to study and analyze network forensics using ML and DL techniques. It is based on five IoT scenarios consisting of a weather station, a smart fridge, motion-activated lights, a remotely activated garage door, and a smart thermostat. These simulated environments were exposed to three categories of attacks: information gathering (port scans, operating system (OS) fingerprinting); denial of service (Transmission Control Protocol (TCP), User Datagram Protocol (UDP), Hypertext Transfer Protocol (HTTP) for both denial of service (DoS) and distributed denial of service (DDoS); and information theft (keylogging and data theft), which are commonly exploited by botnets (bots). This dataset consists of more than 72 million packet capture (PCAP) records. The distribution of attack records is not uniform, however, with the information theft attacks having the least number of records.
- **N-BaIoT [307, 310, 311]**: The Network-based Detection of IoT (N-BaIoT) dataset contains network traffic data from nine commercial IoT devices that consist of two doorbells, one thermostat, one baby monitor, four security cameras, and one webcam infected by two notorious IoT botnet malwares: Mirai and BASHLITE. This dataset comprises both benign and attack traffic intended for the study and detection of botnet attacks and provides a realistic representation of IoT device behavior under normal conditions and when compromised by malware, making it valuable for developing and testing IoT-specific intrusion detection systems.
- **Edge-IIoTset [307, 312]**: This is another dataset that was created to study IoT and IIoT devices and networks. Its design architecture consists of seven layers, including the Cloud Computing Layer, Network Functions Virtualization Layer, Blockchain Network Layer, Fog Computing Layer, Software-Defined Networking Layer, Edge Computing Layer, and IoT and IIoT Perception Layer, and 12 IoT (e.g., sound detection sensor, ultrasonic sensor, water level detection sensor, pH sensor meter, soil moisture sensor, heart rate sensor, flame sensor, etc.) and IIoT devices (servo motor, stepper motor, etc.). The testbed was tested with 15 attacks, which were categorized into five broad attack categories, including DoS/DDoS attacks, information gathering, man-in-the-middle attacks, injection attacks, and malware attacks.

- **IoT Network Intrusion Dataset (IoTNID) [307, 313]**: IoTNID was created using two real devices: a camera and a speaker. The dataset consists of reconnaissance, man-in-the-middle (MiTM), DoS, and Mirai attacks. All the attack packets except those of Mirai were captured using the Nmap tool, while the Mirai attack packets were generated using a laptop. The dataset consists of 42 raw network packet files (pcap) at different time points. The packet files are captured by using the monitor mode of the wireless network adapter. The wireless headers are removed by Aircrack-ng.
- **MedBIoT [307, 314]**: MedBIoT is a dataset that tries to emulate a medium-sized network consisting of 80 simulated devices of the following types: a lock, switch, fan, bulb, and 3 real devices, including the Sonoff Tasmota smart switch, TPLink smart switch, and TPLink light bulb. The devices used were a switch, a light bulb, a lock, and a fan. The setup was exposed to three types of botnets: Mirai, BASHLITE, and Torii. This dataset aims to provide data for the intrusion detection of botnets.
- **MQTT-IoT-IDS2020 [307, 315]**: MQTT-IoT-IDS2020 is a dataset based on a publish/subscribe message protocol called Message Queue Telemetry Transport (MQTT) used in the application/middleware layer. The dataset is generated using a simulated MQTT network architecture. The network comprises twelve sensors, a broker, a simulated camera, and an attacker. Five scenarios are recorded: normal operation, aggressive scan, UDP scan, Sparta SSH brute-force, and MQTT brute-force attack. This dataset was intended to be used for intrusion detection using machine learning techniques.
- **MQTTset [307, 316]**: is another dataset based on the MQTT communication protocol, in this case aimed at aiding the application of ML techniques in MQTT networks. The MQTT broker is instantiated by using Eclipse Mosquitto, and the network setup was simulated using eight different sensors of the following types: temperature, light, humidity, carbon monoxide (CO) gas, motion, smoke, door, and fan to exploit five MQTT network attacks. This data set removes features such as source and destination IP (Internet Protocol) addresses, port addresses, and communication times, among others, that can be found in other datasets and focuses mainly on MQTT-based features.
- **DARPA 1998/1999 [300, 301]**: The DARPA 1998 and 1999 datasets were created at the MIT Lincoln Lab and are among the earliest intrusion detection datasets. They contain seven and five weeks of network traffic, respectively, in packet-based format. These

datasets include various attacks like DoS, buffer overflow, port scans, and rootkits. However, they have been criticized for artificial attack injections and many redundant data points.

- **CAIDA DDoS [299, 300]**: The CAIDA DDoS dataset contains network traffic traces from distributed denial-of-service (DDoS) attacks collected in 2007. It focuses specifically on DDoS attacks but lacks diversity in attack types.
- **ISCX 2012 [299, 301]**: The ISCX 2012 dataset was created by capturing traffic in an emulated network environment over one week. It uses a dynamic approach to generate both normal and malicious network behavior, including attacks like SSH brute force, DoS, and DDoS.
- **UNM Dataset [300]**: The UNM dataset, proposed in 2004, contains system call data with a rich set of intrusions such as buffer overflows, symbolic link attacks, and trojan programs. However, its scope is limited compared to more comprehensive datasets.
- **CIDDS-001 and CIDDS-002 [301]**: These datasets were captured within an emulated small business environment. CIDDS-001 contains four weeks of unidirectional flow-based network traffic and includes SSH brute force, DoS, and port scan (ping-scan, SYN-Scan) attacks. CIDDS-002 focuses specifically on port scan (ACK-Scan, FIN-Scan, ping-Scan, UDP-Scan, and SYN-Scan) attacks.
- **CTU-13 [301]**: The CTU-13 dataset was captured on a university network and distinguishes 13 scenarios containing different botnet attacks, including Menti, Murlo, Neris, NSIS, Rbot, Sogou, and Virut. It's available in packet, unidirectional flow, and bidirectional flow formats. In the first stage, all traffic to and from infected hosts is labeled as a botnet. In the second stage, traffic that matches specific filters is labeled as normal. The remaining traffic is labeled as background. Consequently, background traffic could be normal or malicious network traffic.

These datasets provide researchers with a wide range of options for evaluating intrusion detection systems in various IoT network environments, from traditional IoT networks to industrial IoT control systems. When selecting a dataset for research in application of federated learning in IoT NIDS, it's important to consider factors such as the recency of the data, the IoT devices used,

the types of attacks represented, the network environment simulated, and the specific features provided.

Dataset	Year	Attack types	Attacks
KDD Cup'99 [299–301]	1999	DoS, Probe, R2L, U2R	Approximately 5 million records for training and 2 million records for testing.
NSL-KDD [299–301]	2009	DoS, Probe, R2L, U2R	Refined version of KDD Cup'99, about 150,000 data points.
UNSW-NB15 [299–301]	2015	Worms, Shellcode, Reconnaissance, Port scans, Generic, Backdoor, DoS, Exploits, Fuzzers	Approximately 2 million records, 49 features.
CICIDS 2017 [299, 301]	2017	Brute force, Heartbleed, Botnets, DoS, DDoS, Web attacks, Infiltration attacks	Network traffic in packet-based and bidirectional flow-based formats.
IoT-23 [293, 302]	2020	C&C, DDoS, File Download, Botnet (Mirai, Okiru, Torii)	23 scenarios, both malicious and benign IoT network traffic.
CICIoT 2023 [303, 307]	2023	DDoS, DoS, Recon, Web-based, Brute force, Spoofing, Mirai-like, Normal	Simulates real-world IoT conditions, 33 attacks executed in an IoT topology of 105 devices.
ToN_IoT [306, 307]	2020	DoS, DDoS, Ransomware, Backdoor, Injection, XSS, Password cracking	Comprehensive dataset with telemetry data for IoT and IIoT services.
Modbus-based Network Dataset [181]	2020	Man-in-the-middle, Ping DDoS floods, Modbus Query floods, SYN DDoS	Network traffic for industrial IoT applications using the Modbus protocol.

Bot-IoT [307–309]	2018	Information Gathering, DoS, DDoS, Information Theft	IoT environment with normal and botnet traffic.
N-BaIoT [307, 310, 311]	2019	Mirai, BASHLITE	Network traffic from nine commercial IoT devices.
Edge-IIoTset [307, 312]	2020	DoS/DDoS, Information Gathering, Man-in-the-middle, Injection, Malware	12 IoT and IIoT devices, seven-layer architecture.
IoT Network Intrusion Dataset (IoTNID) [307, 313]	2020	Reconnaissance, Man-in-the-middle, DoS, Mirai	Network traffic captured using camera and speaker.
MedBIoT [307, 314]	2019	Mirai, BASHLITE, Torii	Emulates a medium-sized network, 80 simulated devices.
MQTT-IoT-IDS2020 [307, 315]	2020	Aggressive Scan, UDP Scan, Sparta SSH Brute-force, MQTT Brute-force	Simulated MQTT network architecture.
MQTTset [307, 316]	2020	Various MQTT Network Attacks	Simulated MQTT network with eight different sensors.
DARPA 1998/1999 [300, 301]	1998/1999	DoS, Buffer Overflow, Port Scans, Rootkits	Early intrusion detection datasets, packet-based format.
CAIDA DDoS [299, 300]	2007	DDoS	Network traffic traces from DDoS attacks.
ISCX 2012 [299, 301]	2012	SSH Brute Force, DoS, DDoS	Emulated network environment, one week of traffic.
UNM Dataset [300]	2004	Buffer Overflows, Symbolic Link Attacks, Trojan Programs	System call data with rich set of intrusions.

CIDDS-001 and CIDDS-002 [301]	2017/2018	SSH Brute Force, DoS, Port Scans	Emulated small business environment.
CTU-13 [301]	2011	Various Botnet Attacks	Captured on a university network, 13 scenarios.

Table 7 : Table 7: Summary of some benchmark datasets used in FL for IoT NIDS.

4.3.6 Observations, Research Gaps, and Future Trends in FL for IoT NIDS

The integration of Federated Learning (FL) into Network Intrusion Detection Systems (NIDS) for the Internet of Things (IoT) ecosystems represents a significant advancement in addressing the unique security challenges posed by these networks. This section provides a comprehensive analysis of the current state of FL in IoT NIDS, highlighting recent trends and observations, identifying critical research gaps, and discussing future trends that will shape the development and improvement of FL-based NIDS for IoT networks. The insights gathered here summarize the work done in the previous sections, findings from recent studies, and practical implementations. We present a global view of how FL is transforming IoT NIDS and what steps need to be taken to further enhance its effectiveness and efficiency. This will also serve as a foundation for future research and development efforts, ensuring that FL-based IoT NIDS continue to evolve to meet the growing demands of IoT security.

4.3.6.1 Recent Trends and Observations

In recent years, the integration of Federated Learning (FL) into Network Intrusion Detection Systems (NIDS) for IoT ecosystems has gained considerable attention due to its potential to address privacy concerns and enhance security. In the previous sections and subsections, we thoroughly explored the application of FL in IoT NIDS, revealing several significant trends and observations. One prominent trend is the shift towards decentralized learning, where FL enables the training of models across multiple IoT devices without the need to centralize data. This approach not only mitigates the risk of data breaches but also complies with stringent privacy regulations such as GDPR and HIPAA by ensuring that sensitive data remains on the device.

Furthermore, FL has been instrumental in enhancing the privacy of IoT networks. By keeping data localized, FL addresses the privacy concerns that arise from traditional centralized machine learning methods. This decentralized nature of FL has been crucial in building trust among users and stakeholders in IoT ecosystems, where data sensitivity is a major concern. Despite these advantages, the chapter highlights the ongoing challenge of optimizing communication efficiency. The communication overhead between IoT devices and the central server can be substantial, necessitating the exploration of model compression techniques and efficient aggregation methods to minimize bandwidth usage and latency.

In this chapter, we also discussed the issue of handling heterogeneous data in IoT environments. IoT ecosystems are characterized by devices with varying capabilities and data distributions, making it imperative for FL techniques to adapt to non-IID (non-independent and identically distributed) data. The research presented in this chapter indicates that significant progress has been made in developing FL methods that can handle such diversity without compromising performance. Additionally, hybrid approaches that combine FL with other machine learning techniques, such as transfer learning and reinforcement learning, have shown promise in enhancing the accuracy and robustness of NIDS in IoT ecosystems.

Regarding the models, frameworks, and resources discussed in this chapter, several noteworthy trends have emerged. The chapter highlights the increasing adoption of various FL models, such as Federated Averaging (FedAvg) and its variants, which have been widely used due to their simplicity and effectiveness. These models facilitate the aggregation of local updates from IoT devices, ensuring that the global model improves iteratively while maintaining data privacy. Additionally, the use of secure multiparty computation and differential privacy techniques has been observed to enhance the security and robustness of FL models in IoT NIDS. Various frameworks, such as TensorFlow Federated, Flower, and PySyft, have been employed to implement these models, providing the necessary tools and libraries to support FL in resource-constrained IoT environments.

Another important observation is the role of hardware advancements in facilitating the implementation of FL in IoT NIDS. The advent of more powerful GPUs and cloud-based platforms has made it feasible to deploy complex FL models that were previously too resource-intensive for

IoT devices. This has led to a more widespread adoption of FL in the design of NIDS solutions. However, our work in this chapter also notes that the complexity of these models can lead to higher computational costs, emphasizing the need for ongoing research into optimizing FL models for resource-constrained environments.

The analysis of the performance metrics used by the researchers for the evaluation of their various methodologies reflects on the performance metrics used to evaluate FL-based NIDS solutions. Accuracy, recall, and F-measure have been the most used metrics, indicating a focus on the detection capabilities of the models. The reviewed literature suggests that while FL has demonstrated significant potential for improving the detection accuracy of NIDS, there is still room for improvement in terms of reducing false positives and enhancing real-time detection capabilities.

Benchmark datasets are a crucial component for evaluating the effectiveness of new methodologies. The utilization of public datasets is analyzed in Table 7, with some being more prevalent and extensively used than others. Despite the age of some datasets, they continue to be employed in numerous studies. However, network architectures have significantly evolved from those of 20 years ago. In today's IoT era, the security and privacy of big data and sensor nodes are of utmost importance, given the frequent emergence of new attacks. Testing a novel NIDS methodology designed for contemporary networks with an outdated dataset may result in poor real-world performance. It is evident that a model trained and validated with the most recent dataset is likely to outperform one that uses an older dataset in real-world applications.

In summary, the trends and observations detailed in this chapter reflect the significant efforts made in integrating FL into IoT NIDS. From decentralized learning and privacy preservation to handling heterogeneous data and leveraging hardware advancements, the landscape of FL in IoT NIDS is rapidly evolving. While challenges such as communication efficiency and model complexity remain, the progress made thus far sets a promising foundation for future advancements in this field. The next sections will delve into the research gaps and current challenges that need to be addressed and the future trends that are likely to shape the development of FL-based IoT NIDS.

4.3.6.2 Research Gaps and Current Challenges

Federated Learning (FL) is a growing and constantly evolving field, particularly in the context of developing advanced Network Intrusion Detection Systems (NIDS) for the Internet of Things (IoT). In this section, we will discuss some of the research gaps, challenges, and areas of concern associated with this application of FL.

- **Limited On-device Resources**: IoT devices often face severe constraints in terms of computational power, memory, and energy resources. This limitation significantly impacts the deployment of Federated Learning (FL) for cross-device model training of Network Intrusion Detection Systems (NIDS) on these devices. [107] Training deep learning models, especially those required for effective NIDS, can be extremely time-consuming and inefficient on resource-constrained devices [107] [167]. The memory limitations of IoT devices may not be sufficient to handle both model inference and training, particularly in continual learning scenarios where new data is constantly being processed [167]. Furthermore, many low-power IoT devices may not support high-end communication protocols, leading to increased communication latency and reduced efficiency in the FL process [12]. These resource constraints pose a significant challenge to the widespread adoption of FL-based NIDS in IoT environments.
- **Communication Overhead and Bottleneck**: The cost of communication per round of training is a primary limitation in FL applications, particularly in IoT settings. Limited network bandwidth in IoT environments can lead to inefficient communication between clients and the server, creating bottlenecks in the learning process [107] [167]. The iterative exchange of model parameters and trained models, which is fundamental to FL, introduces a heavy communication cost that can be particularly burdensome for IoT devices [107] [317]. In large-scale IoT networks with potentially millions of devices, this can create significant server traffic and system-wide bottlenecks [12]. For example, even with a client participation ratio of just 1%, a network with millions of devices would require the server to handle traffic from at least 10,000 clients, potentially leading to severe congestion and delays in the FL process [12].
- **Data Heterogeneity and Non-IID Data**: IoT devices often generate non-IID (not independently and identically distributed) data, which presents a significant challenge for

FL-based NIDS. This data heterogeneity can lead to model under-fitting and reduced performance, as the global model may struggle to capture the diverse patterns present in different local datasets [12] [107][128, 167]. The variance in data quality, size, and complexity across edge devices can result in asymmetrically trained local models, further complicating the aggregation process [12]. Specific challenges in data heterogeneity include feature distribution skew, label distribution skew, and quantity skew [107] [317]. For instance, different IoT devices may collect data with varying features or focus on different types of network traffic, leading to inconsistencies in the local models. This heterogeneity can significantly affect the speed of convergence in FL algorithms and potentially lead to biased or inaccurate global models for intrusion detection [128].

- **Intermittent Connectivity and Availability**: The nature of IoT environments often involves unstable network connections, leading to frequent device dropouts during training rounds. This intermittent connectivity poses a significant challenge for FL-based NIDS, as it can disrupt the learning process and lead to incomplete or inconsistent model updates [167]. Asynchronous updates in FL, while potentially addressing some connectivity issues, can endanger the convergence of federated training if not carefully managed [167]. The presence of straggler clients—devices that take significantly longer to complete their local training or fail to submit updates within the allocated time—can have a substantial impact on the overall system performance [167]. These issues are particularly pertinent in IoT-based NIDS, where timely detection of network intrusions is crucial and delays or inconsistencies in model updates could lead to security vulnerabilities.
- **System Heterogeneity**: IoT environments are characterized by a wide diversity in system metrics, both in terms of hardware and software. This heterogeneity complicates the design and implementation of FL-based NIDS [107][167, 317]. Different IoT devices may use various deep learning frameworks and model formats, creating challenges in model aggregation and consistency [107] [167]. The variations in storage, processing, and communication capabilities of devices need to be carefully reconciled in FL implementations to ensure fair and effective participation [107] [317]. For example, more powerful IoT devices might be able to train more complex models or process larger datasets, potentially leading to imbalances in their contributions to the global NIDS model.

Addressing this system heterogeneity while maintaining the effectiveness of the NIDS remains a significant challenge in FL for IoT environments.

- **Security and Privacy Concerns**: IoT devices are attractive targets for adversaries, making them particularly vulnerable to various attacks such as phishing, identity theft, Man-in-the-Middle, and DDoS [107]. This vulnerability extends to FL-based NIDS, where compromised devices could potentially undermine the entire system [107] [167]. Federated data and model poisoning attacks, where malicious participants manipulate local data labels, pose a serious threat. Such attacks could cause the global NIDS model to collapse or show false performance, potentially allowing real intrusions to go undetected [12] [107]. Replay and masquerading attacks are possible even if the weights are encrypted, affecting the performance and accuracy of the model [107]. Privacy or data leakage from on-device sensitive data remains a significant concern, as model gradients or weight parameters shared during the FL process can potentially be used to infer private training data [107] [167]. This is particularly problematic for NIDS, where the training data may contain sensitive network traffic information. Balancing privacy preservation techniques (e.g., differential privacy, secure aggregation) with model performance and fairness is an ongoing challenge in FL-based NIDS for IoT [128]. Implementing these privacy-preserving measures without significantly degrading the intrusion detection capabilities of the system remains a complex problem.

- **Model Performance and Precision**: Maintaining high model performance and precision in FL-based NIDS for IoT environments presents several challenges. The performance of trained models can deteriorate over time, necessitating frequent retraining to maintain effective intrusion detection capabilities [317]. Achieving high model precision while maintaining privacy is particularly challenging, especially given the critical nature of security applications [317]. Hyperparameter tuning, which is crucial for optimizing model performance, becomes significantly more difficult in FL settings due to the lack of insights into decentralized data [317]. This challenge is compounded in IoT environments, where the diverse nature of devices and network traffic patterns requires careful tuning to ensure effective intrusion detection across all participating nodes.

- **Fairness and Incentive Mechanisms**: Ensuring fairness in FL processes for IoT-based NIDS, particularly in terms of device contributions and rewards, is an open challenge [128]. Developing appropriate incentive mechanisms to encourage high-quality data contributions from participants is necessary to maintain the effectiveness of the NIDS [317]. This is particularly important in IoT environments, where devices may have varying capabilities and access to different types of network traffic data. Balancing the trade-off between privacy, security, and fairness in FL systems remains an ongoing research problem [128]. For instance, devices that contribute more valuable data or computational resources to the NIDS might expect greater benefits or influence over the model, but this could potentially lead to biases or vulnerabilities in the system.
- **Temporal Dynamics and Continual Learning**: The continuous data collection by IoT devices poses significant challenges for model updates and lifelong learning in FL-based NIDS [167]. As new types of network threats emerge and traffic patterns evolve, the NIDS must adapt continuously. However, limited memory resources on IoT devices can exacerbate the problem of catastrophic forgetting in continuous learning scenarios [167]. This means that as the model learns to detect new types of intrusions, it may lose its ability to identify previously learned attack patterns. Developing effective strategies for continual learning in resource-constrained IoT environments while maintaining the overall effectiveness of the NIDS remains a significant challenge.
- **Standardization and System Development**: The lack of standardization in FL for IoT hinders widespread deployment and interoperability of FL-based NIDS solutions [167]. There is a pressing need for user-friendly integrated simulation environments and system-level tools specifically designed for FL in IoT settings [167]. These tools would greatly facilitate the development, testing, and deployment of FL-based NIDS across diverse IoT ecosystems. Standardization efforts could help address issues of compatibility between different IoT devices and FL frameworks, enabling more robust and widely applicable NIDS solutions.
- **Client Management**: The admission and management of participating devices in decentralized FL models present significant challenges for IoT-based NIDS [317]. Developing effective strategies for client selection and aggregation in heterogeneous IoT environments is crucial for maintaining the effectiveness and efficiency of the NIDS [317].

This includes determining which devices should participate in each round of training, how to handle devices that join or leave the network dynamically, and how to ensure that the contributions of each device are appropriately valued and integrated into the global model. Effective client management is essential for maintaining the integrity and performance of the FL-based NIDS in dynamic IoT environments.

4.3.6.3 Future Trends and Research Opportunities

As Federated Learning (FL) continues to evolve in the context of Network Intrusion Detection Systems (NIDS) for Internet of Things (IoT) environments, several promising research directions and trends are emerging to address the challenges identified in the previous section. This subsection explores these future trends and potential solutions in detail.

Resource-Efficient Algorithms

The development of resource-efficient FL algorithms is crucial for overcoming the limitations of IoT devices. This trend focuses on creating lightweight neural network architectures and optimizing existing algorithms to reduce computational and memory requirements.

Future research in this area is likely to explore advanced model compression techniques specifically tailored for FL in IoT environments. This may include novel pruning methods that consider the distributed nature of FL, adaptive quantization schemes that adjust precision based on device capabilities, and knowledge distillation approaches that enable smaller models to learn from larger ones without direct access to training data.

Another promising direction is the development of hybrid architectures that combine edge computing with on-device processing. These approaches could dynamically partition neural networks, executing resource-intensive layers on nearby edge servers while keeping privacy-sensitive computations on the IoT devices themselves. This could significantly reduce the computational burden on resource-constrained devices while maintaining data privacy and low-latency responses for intrusion detection.

Recent research recognizes both the energy efficiency requirements for future federated learning and the need to compress deep learning, especially during the learning process [318, 319]. Developing ultra-light DNN architectures is essential for device-/edge-learning [318], which aligns with the goals of resource-efficient FL algorithms for IoT-based NIDS.

To facilitate resource management in on-device FL training, some innovative approaches have been proposed. For instance, a resource-aware FL architecture on mobile devices has been developed that considers computation resources during training [320]. This architecture introduces a soft training technique to accelerate the training of stragglers with weak computational capabilities, allowing them to partially train the model by masking resource-intensive neurons during local training but recovering them during parameter aggregation [320].

Communication-Efficient FL Protocols

Reducing communication overhead is essential for the scalability of FL-based NIDS in IoT environments. Future research is likely to focus on developing more efficient communication protocols and compression techniques.

One area of exploration is the development of adaptive compression algorithms that dynamically adjust the level of compression based on network conditions, device capabilities, and the importance of specific model updates. This could involve techniques such as significance-based update filtering, where only the most impactful parameter changes are transmitted, or dimensionality reduction methods that project updates into lower-dimensional spaces while preserving key information.

An important aspect of active research is the reduction of the communication burden during the federated learning process. Studies have emphasized the development of resource allocation strategies, especially to reduce communication requirements between nodes with gossip algorithms [116, 321, 322]. Other research activities focus on the reduction of bandwidth during training through sparsification and quantization methods [321], where machine learning models are sparsified and/or compressed before they are shared with other nodes.

One promising approach is the use of lightweight deep learning models, such as binarized neural networks (BNN). These models convert floating-point weights to binary formats, enabling efficient bit-wise operations and significantly reducing the size of parameters that need to be transferred [13, 323]. This not only facilitates efficient communication but also ensures line-speed packet processing and traffic classification, which is crucial for real-time intrusion detection in IoT networks [323].

Recent research has proposed efficient communication protocols that can compress uplink and downlink communications while remaining robust to an increased number of clients and data distribution [324]. These protocols use a combination of sparsification, ternarization, error accumulation, and optimal Golomb encoding techniques for uplink compression and speeding up parallel training on the global server without compromising learning convergence [324].

The implementation of modern communication technologies, particularly 5G networks, is being explored to improve bandwidth and reduce latency in FL-based IDS systems. These approaches leverage the reliability and connectivity of 5G networks to enhance the performance of FL-based NIDS, allowing for more efficient data transfer and model updates [13, 325].

Handling Non-IID Data and Heterogeneity

Addressing data heterogeneity and non-IID distributions is crucial for effective FL-based NIDS in diverse IoT environments [12]. Future research in this area is likely to focus on developing more sophisticated algorithms that can adapt to and leverage the inherent diversity of IoT data.

One potential approach is the development of personalized FL frameworks that maintain a balance between global knowledge sharing and local specialization. These systems might employ meta-learning techniques to quickly adapt global models to device-specific data distributions or use multi-task learning approaches to jointly optimize for both common and device-specific intrusion detection tasks.

An active direction of research is to develop federated learning for training heterogeneous local models with varying computation complexities and producing a single powerful global

inference model [326]. This approach could be particularly beneficial for IoT-based NIDS, where devices with different capabilities need to collaborate effectively.

Another area of exploration could be the design of advanced data augmentation and synthesis techniques specifically for FL scenarios. These methods could help address data imbalances and scarcity issues in a privacy-preserving manner, potentially using techniques like federated generative adversarial networks (GANs) to create realistic synthetic intrusion data that can be shared across devices without compromising privacy.

To handle non-IID data more effectively, some researchers have proposed grouping client models into smaller clusters based on the similarity of weight updates [134]. This hierarchical clustering approach attempts to preserve uniformity within clusters, potentially leading to higher convergence rates and reduced communication costs compared to traditional FL architectures [134].

Reinforcement learning-based approaches for intelligent device selection during aggregation are also being developed to counteract irregularities caused by non-IID data. These frameworks use reinforcement learning to create smart agents that select subsets of devices from received weight updates during each round, maximizing the accuracy of the server model [134].

Asynchronous and Fault-Tolerant FL

To address challenges related to intermittent connectivity and device availability, future research is likely to focus on developing more robust asynchronous FL algorithms and fault-tolerant mechanisms.

Asynchronous FL approaches that can handle varying degrees of staleness in device updates are an important area of investigation. These algorithms might incorporate adaptive learning rate schemes that adjust based on update freshness or employ novel aggregation methods that can effectively combine updates from different time steps while maintaining model consistency.

Asynchronous Federated Learning (AFL) [12] is gaining traction as a solution to reduce server-side traffic accumulation and ensure continuous data flow. AFL approaches have shown promise in achieving faster model convergence and higher system throughput. By allowing clients to update models asynchronously, AFL can reduce system lag and improve overall efficiency in FL-IoT systems [12]. ASO-Fed [327], an asynchronous online federated learning method, is introduced. This algorithm updates the central server asynchronously to eliminate system lag. Its performance is evaluated using an image dataset and three real-world datasets with non-IID data, resulting in quicker model convergence and significantly improved system throughput.

Fault-tolerance is another critical aspect, especially in the context of IoT-based NIDS, where devices may be compromised or behave erratically. Future research might explore Byzantine-robust aggregation rules that can maintain model integrity even in the presence of malicious or faulty clients. This could involve the development of reputation systems that track the reliability of device contributions over time or the use of blockchain-inspired consensus mechanisms to ensure the validity of model updates.

Recent research advancements are starting to consider real-world propagating channels [328], as previous implementations often assumed ideal channels. This consideration is crucial for developing practical and robust FL-based NIDS for IoT environments.

Enhanced Security and Privacy Measures

As security and privacy concerns remain paramount, future research is likely to focus on developing more robust protection mechanisms for FL-based NIDS.

One important direction is the integration of advanced differential privacy [209] techniques into FL frameworks for IoT security. Future research might explore adaptive privacy mechanisms that dynamically adjust the level of noise added to model updates based on the current threat landscape, data sensitivity, and desired privacy guarantees. This could involve developing novel privacy accounting methods that provide tighter bounds on privacy loss in federated settings or exploring the use of local differential privacy techniques that offer stronger individual-level protections.

Recent studies have emphasized the characterization of FL systems' robustness to differential privacy attacks [329] and proposed perturbation techniques, such as differential privacy, to protect training datasets against data breaches [330, 331]. For example, artificial noise (e.g., Gaussian noise) can be inserted into the gradients of neural network layers to preserve training data and hidden personal information against external threats while maintaining convergence properties [331].

Another promising area is the development of more efficient secure multi-party computation (SMPC) [332, 333] and homomorphic encryption (HE) [334] techniques specifically designed for FL in resource-constrained IoT environments [335, 336]. This might include research into lightweight cryptographic protocols that can provide strong security guarantees with minimal computational overhead, or the design of hybrid systems that combine different privacy-preserving techniques to achieve an optimal balance between security, efficiency, and model performance.

In [337], the paper investigates the practical combination and application of Differential Privacy (DP) and Homomorphic Encryption (HE) techniques in Federated Learning Protocols. These techniques address distinct confidentiality threats: DP focuses on protecting data from other clients or end-users, while HE aims to secure data during aggregation at the server. To bridge the gap toward more realistic deployment scenarios, the paper introduces a novel stochastic quantization operator. This operator establishes DP guarantees even when the noise is both quantized and bounded due to the use of HE. Experimental results on the FEMNIST dataset demonstrate that achieving state-of-the-art privacy and utility trade-offs impacts HE parameters, resulting in a computation time overhead of 0.2% to 1.1% during the training of a 500k-parameter model.

Homomorphic encryption (HE) allows encrypted values to be involved in computations. While HE improves the security of the FL process, it requires large communication and computational overhead. To address this, researchers have proposed batch encryption-based FL frameworks that use batch encoding and quantization schemes to lower communication and computational costs [134].

Blockchain technology is also being integrated into FL frameworks to create secure communication and transaction channels. This approach not only secures data from poisoning and inference attacks but also achieves good accuracy in benchmark datasets. By utilizing secure transaction records, selective model adjustments can be made based on improvement, protecting the system from various types of attacks [134].

Edge Computing and Resource Management

The integration of edge computing in FL-IoT systems is a growing trend to address the resource constraints of IoT devices. Researchers are developing efficient resource management algorithms and architectures that consider data importance and communication awareness to optimize training accuracy, fairness, and convergence time. These algorithms or architectures prioritize clients based on their resources in the presence of non-I.I.D. data across clients, significantly reducing overall training time without compromising model performance [134].

To facilitate resource management in on-device FL training, some approaches have been provided. The work done in [320] addresses challenges in training AI/ML models on heterogeneous edge devices within a federated network. RaFL allocates resource-aware models to edge devices using Neural Architecture Search (NAS). It allows customized model deployment based on the diverse computational resources available on these devices. Also, in [338], they proposed DeepReBirthis, an improved deep neural network (DNN) architecture designed for mobile AI training. It integrates a streamlined slimming model with consecutive non-tensor and tensor layers. This architecture speeds up training and model execution, resulting in 3x faster training and 2.5x memory savings. Despite a minor 0.4% drop in top 5 accuracy, the performance remains high on the ImageNet validation dataset. Similar, [339] proposes algorithms to optimize training accuracy, fairness, and convergence time in federated learning. By prioritizing clients based on their resources (using importance sampling and rank ordering), this approach significantly reduces overall training time without sacrificing model performance.

Lightweight deep learning models and software-based deep learning accelerators are being designed to support AI/DL training on mobile hardware with limited resources. These approaches aim to optimize hardware usage for data training without compromising performance accuracy,

enabled by techniques such as runtime layer compression and deep architecture decomposition [134].

Advanced Client Management Strategies

Sophisticated client management strategies are crucial for optimizing the performance and efficiency of FL-based NIDS in heterogeneous IoT networks. Future research in this area is likely to focus on developing more intelligent and adaptive selection and aggregation mechanisms.

One promising direction is the exploration of multi-factor client selection algorithms that consider a wide range of criteria, including data quality, computational capabilities, network conditions, and historical performance. These approaches might incorporate elements of reinforcement learning to dynamically optimize client selection strategies based on evolving system conditions and performance metrics.

Another area of investigation could be the development of fairness-aware aggregation mechanisms that ensure equitable participation and influence across diverse IoT devices. This might involve designing novel weighting schemes that balance the contributions of different clients based on factors like data quantity, quality, and relevance to current threat detection tasks.

Recent research has proposed data-importance and communication-aware resource management algorithms to optimize training accuracy, fairness, and convergence time [339]. These algorithms prioritize clients based on their resources in the presence of non-IID data across clients, potentially reducing overall training time without compromising model performance [339].

Continual Learning and Model Adaptation

Addressing temporal dynamics and evolving threats requires FL models capable of continual learning and adaptation. Future research in this area is likely to focus on developing more sophisticated techniques for ongoing model updates without compromising performance on previously learned tasks.

One potential direction is the exploration of elastic architecture techniques that allow FL models to dynamically expand their capacity to accommodate new attack patterns or network

behaviors. This could involve the development of federated architecture search algorithms that can discover optimal model structures in a distributed manner or the design of modular FL frameworks that can seamlessly integrate new components to handle emerging threats.

Another important area of investigation is the development of advanced regularization techniques specifically tailored for continual learning in federated settings. These methods might build upon concepts like elastic weight consolidation or synaptic intelligence, adapting them to the unique challenges of FL-based NIDS. Future research could also explore the use of federated replay mechanisms that allow the system to retain knowledge of past experiences without explicitly sharing sensitive data between devices.

Standardization and Development Frameworks

The trend towards standardization in FL for IoT is likely to gain momentum, with future research focusing on developing comprehensive, open-source platforms that can accelerate the deployment of FL-based NIDS solutions.

Future standardization efforts might focus on creating modular, extensible frameworks that can support a wide range of FL algorithms, aggregation methods, and privacy-preserving techniques. These platforms could provide standardized interfaces for integrating custom NIDS models and datasets, as well as tools for benchmarking and comparing different FL approaches in IoT security contexts.

Another important direction is the development of simulation environments and testing frameworks specifically designed for FL-based NIDS in the IoT. These tools could help researchers and practitioners evaluate the performance, security, and efficiency of their systems under realistic conditions, considering factors like device heterogeneity, network dynamics, and evolving attack patterns.

The development of standard specifications for FL-IoT ecosystems is crucial for large-scale deployment. Initiatives like the ETSI Multi-Access Edge Computing (MEC) [134, 340] and communication protocols such as OPC-UA [134, 341] and IEEE 802.11be Extremely High

Throughput [134, 342] are expected to support service providers in deploying intelligent IoT services at the network edge with FL-IoT components and devices.

Explainable AI in FL-based NIDS

Incorporating explainable AI techniques in FL-based NIDS is essential for enhancing trust and interpretability in complex IoT security systems. Future research in this area is likely to focus on developing privacy-preserving explanation methods that can provide insights into model decisions without compromising the confidentiality of local data.

One potential direction is the exploration of federated explanation generation techniques that can produce human-interpretable justifications for intrusion detection decisions across distributed IoT devices. This might involve developing novel aggregation methods for local explanations that can provide global insights while preserving privacy or designing explanation mechanisms that can adapt to different levels of user expertise and device capabilities.

Another promising area is the integration of causal inference techniques into FL-based NIDS. Future research might explore ways to infer causal relationships between network behaviors and security incidents across distributed IoT environments, potentially leading to more robust and interpretable intrusion detection models.

Advanced Optimization Techniques

To improve the convergence and performance of FL algorithms, researchers are exploring advanced optimization techniques. For instance, momentum-federated learning incorporates momentum gradient descent (MGD) at the central server to minimize the loss function with less computation. This approach optimizes learning parameters, leading to faster convergence compared to traditional FL algorithms with first-order gradient descent [12].

Energy-Efficient Learning

Addressing the energy consumption challenges in FL-IoT systems, especially for lightweight mobile devices, is a critical area of research. Techniques such as gradient sparsification, gradient quantization, weight quantization, and dynamic batch sizes are being

integrated into FL training procedures to mitigate energy costs. These methods aim to balance the energy consumption of local computing for model training and wireless communications for model exchange, boosting overall energy efficiency [134].

4.4 Conclusion

In conclusion, the future of FL for IoT NIDS is rich with research opportunities and promising trends. Federated learning started to emerge as an important research topic in 2015 and 2016 [343, 344], with the first publications on federated averaging in telecommunication settings. Since then, the field has expanded rapidly, addressing challenges in resource efficiency, communication overhead, data heterogeneity, and privacy.

In this chapter, we explored the application of Federated Learning (FL) in securing IoT ecosystems, particularly within Network Intrusion Detection Systems (NIDS). FL emerged as a solution to the privacy concerns inherent in traditional centralized learning models by allowing local data processing on IoT devices, thus avoiding the transfer of sensitive data to central servers. This chapter extensively reviewed how FL can overcome the significant security challenges in IoT environments, such as data privacy, scalability, and the heterogeneity of IoT devices.

We presented FL's potential to significantly enhance NIDS in IoT by providing a decentralized yet collaborative approach to intrusion detection. FL's ability to process data locally on distributed devices, while aggregating model updates centrally, ensures that privacy is maintained without compromising the detection performance. The unique characteristics of IoT systems, including resource constraints and non-IID (Independent and Identically Distributed) data, pose substantial challenges for FL, but we identified that techniques like differential privacy, secure aggregation, and adaptive learning can mitigate these issues.

The chapter also delved into the practical implementation of FL in real-world IoT systems, discussing its capacity to handle communication overhead and model aggregation. Federated Averaging and other advanced optimization techniques were identified as critical to improving FL's efficiency in IoT networks, particularly when dealing with large-scale deployments. Furthermore, we examined the importance of customizing FL architectures to address varying

device capabilities and network conditions in IoT environments, suggesting strategies like model pruning and lightweight architectures.

This chapter concludes by emphasizing that while FL presents a promising paradigm for IoT security, there are still research gaps, particularly in ensuring that FL systems can operate effectively in highly heterogeneous, large-scale IoT networks. Future work should focus on optimizing communication protocols, enhancing privacy-preserving mechanisms, and developing more efficient models for real-time intrusion detection in IoT.

PART 2

CONTRIBUTION

Chapter 5: Proposed FED-NIDS Architecture and Design

In the previous chapter, we explored comprehensively how traditional Network Intrusion Detection Systems (NIDS) struggle to cope with the scale, heterogeneity, and privacy concerns inherent in IoT ecosystems. Our comprehensive review of the state-of-the-art revealed several key insights that inform the design of FED-NIDS.

Federated Learning (FL) emerged as a promising paradigm for collaborative model training without centralizing sensitive data, addressing critical privacy concerns in IoT environments. Traditional NIDS architectures face significant scalability issues when applied to the vast and diverse landscape of IoT devices. Many IoT devices operate under severe computational and energy limitations, necessitating lightweight and efficient security solutions. The non-IID (independent and identically distributed) nature of data across IoT devices poses significant challenges for model training and performance. The dynamic and evolving nature of cyber threats in IoT ecosystems requires adaptive and self-improving intrusion detection mechanisms. Nature-inspired algorithms showed promise in enhancing the performance and adaptability of machine learning models in resource-constrained environments.

These results show that we need a new NIDS architecture that can handle the special problems of IoT security and make the most of the benefits of federated learning and bio-inspired optimization methods. This chapter presents the design of FED-NIDS, a novel Bio-Inspired Federated Deep Learning-Based Network Intrusion Detection System specifically tailored for IoT ecosystem security.

5.1 Design Objectives and Specifications of Proposed Architecture

Building upon the insights gained from our literature review, we have identified the following key objectives and specifications for FED-NIDS:

- **Privacy-Preserving Collaboration**: Design a federated learning framework that enables collaborative model training across diverse IoT environments without compromising data privacy or security.

- **Scalability and Adaptability**: Develop an architecture capable of seamlessly integrating new IoT devices and adapting to changing network topologies and data distributions.
- **Resource Efficiency**: Optimize the system to operate efficiently on resource-constrained IoT devices, minimizing computational overhead and energy consumption.
- **Heterogeneous Data Handling**: Implement mechanisms to effectively manage and learn from non-IID data across different IoT devices and environments.
- **Bio-Inspired Optimization**: Incorporate nature-inspired algorithms to enhance model performance, feature selection, and overall system adaptability.
- **Real-Time Detection**: Ensure the system can perform intrusion detection in real time, promptly identifying and responding to potential security threats.
- **Comprehensive Threat Coverage**: Design the system to detect a wide range of IoT-specific attacks, including both known and zero-day threats.
- **Minimal Communication Overhead**: Optimize the federated learning process to minimize data transfer between devices and the central server, reducing bandwidth usage and latency.
- **Interpretability**: Develop mechanisms to provide explainable insights into detected intrusions, facilitating effective response and mitigation strategies.
- **Robustness and Fault Tolerance**: Ensure the system remains operational and effective even in the face of device failures or adversarial attacks on the federated learning process.

These goals help shape the creation of FED-NIDS, making sure that the final system meets the important safety needs of IoT ecosystems and gets around the problems with current NIDS options.

5.2 Overall Architecture of FED-NIDS

We design the FED-NIDS architecture to meet the objectives and specifications outlined above, leveraging the power of federated learning, deep neural networks, and bio-inspired optimization techniques. Figure 20 provides a high-level overview of the FED-NIDS architecture.

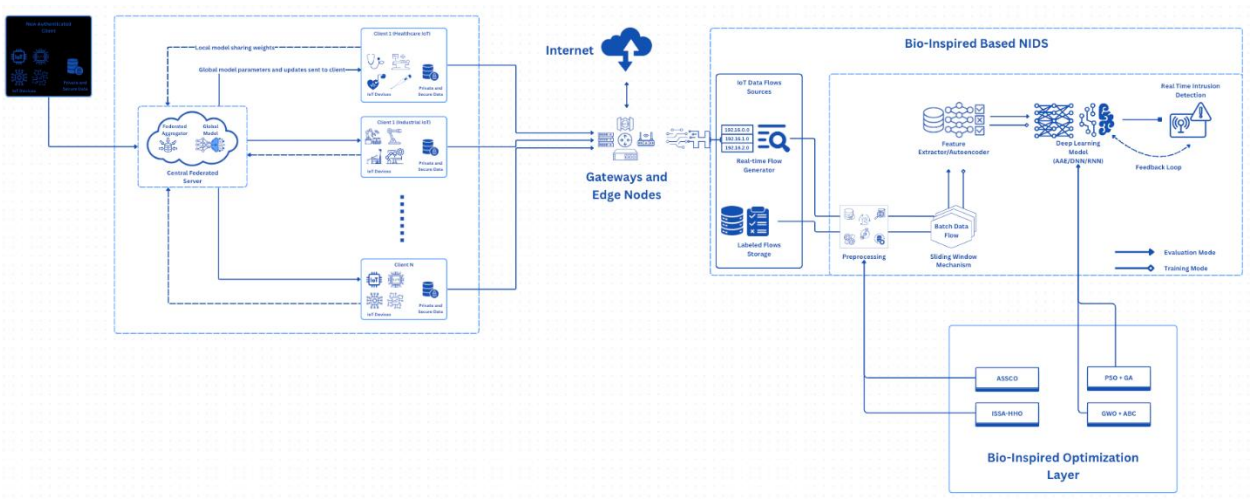


Figure 20 : Global Architecture of the proposed FED-NIDS System.

5.2.1 Operation Modes

FED-NIDS operates in two distinct modes: training mode and evaluation mode. These modes allow the system to continuously learn from distributed data sources while providing real-time intrusion detection capabilities.

5.2.1.1 Training Mode

FED-NIDS uses federated learning to train a global intrusion detection model across multiple IoT environments in Training Mode. It does this without storing sensitive client data centrally, so it can't see it. The training process operates as follows:

- **Local Data Processing**: Each participating IoT environment (client) preprocesses its local network traffic data, extracting relevant features and labeling flows as benign or malicious based on known patterns or expert knowledge.
- **Local Model Training**: Clients train local deep learning models on their preprocessed data, using the current global model as an initialization point.
- **Model Update Sharing**: Instead of sharing raw data, clients send only the model updates (e.g., gradients or model parameters) to the central federated server.

- **Secure Aggregation**: The central server aggregates the model updates from all participating clients using secure aggregation techniques to protect individual contributions.
- **Global Model Update**: We use the aggregated updates to enhance the global model, then distribute it back to all clients.
- **Bio-Inspired Optimization**: We apply nature-inspired algorithms at both local and global levels to optimize model architectures, hyperparameters, and feature selection.

This iterative process allows FED-NIDS to continuously improve its detection capabilities while preserving the privacy of individual IoT environments.

5.2.1.2 Evaluation Mode

In Evaluation Mode, FED-NIDS deploys the trained global model to perform real-time intrusion detection on live IoT network traffic. The evaluation process includes:

- **Real-Time Data Capture**: The edge or gateway level captures and preprocesses network traffic from IoT devices in real time.
- **Feature Extraction**: The process replicates the training mode by extracting relevant features from the captured traffic flows.
- **Model Inference**: The locally deployed global model receives the preprocessed data for inference.
- **Intrusion Detection**: The model classifies each flow as benign or potentially malicious, with additional granularity for different types of attacks.
- **Alert Generation**: The system generates alerts with relevant details to facilitate rapid response when it detects potential intrusions.
- **Continuous Learning**: The system adapts to new threat patterns by feeding back detected anomalies and confirmed intrusions into the training process.

The Evaluation Mode ensures that FED-NIDS provides continuous protection against both known and emerging threats in IoT ecosystems.

5.2.2 General Description of the Architecture

The innovative FED-NIDS architecture combines federated learning, deep learning, and bio-inspired optimization techniques to secure IoT ecosystems. Several components work in harmony to compose this privacy-preserving and adaptive intrusion detection system. The architecture encompasses the IoT ecosystem and edge layer, a collection of diverse IoT devices and edge nodes responsible for generating and processing data. The Federated Learning Framework manages distributed training, secure aggregation, and model distribution. The system embeds privacy-preserving mechanisms to secure data and models, and specialized components manage the unique data flows and communication protocols in IoT environments. The bio-inspired NIDS forms the core of the detection system, with deep learning models enhanced by bio-inspired architectural elements. The bio-inspired optimization layer continuously improves system performance by using nature-inspired algorithms to optimize features and model architectures. Together, these components create a comprehensive and adaptive security solution for IoT environments.

We designed these components to work seamlessly together, addressing the unique challenges of IoT security while leveraging the strengths of federated learning and bio-inspired computing. The following sections will delve into the details of each component, explaining their design, functionality, and integration within the overall FED-NIDS architecture.

5.3 Architecture Components

The different parts that make up our FED-NIDS are the IoT ecosystem and edge layer, the federated learning framework, privacy-preserving mechanisms built into the system to protect data and models, specialized components that handle the specific data flows and communication protocols in IoT environments, and the bio-inspired optimization layer. Throughout the following subsections, we will explain the reasons for the choices made, thus justifying the decisions made.

5.3.1 IoT Ecosystem and Edge Layer

The foundation of the FED-NIDS architecture is the diverse array of IoT devices that generate data. These devices span various sectors, including healthcare IoT (e.g., medical sensors, patient monitoring devices), industrial IoT (e.g., manufacturing equipment, process control

sensors), smart home devices (e.g., thermostats, security cameras), and other IoT applications. Each device produces potentially sensitive data that requires protection while maintaining its usefulness for intrusion detection.

The architecture works with and incorporates network gateways and edge nodes that serve as intermediaries between IoT devices and the broader network. These components play crucial roles in data aggregation, initial processing, security enforcement, and communication management. They collect data from multiple IoT devices, perform preliminary analysis and filtering, implement first-line defense mechanisms, and handle network protocols and data transmission.

To ensure privacy and reduce network load, each client (representing a distinct IoT environment or organization) maintains its own private and secure data storage. This local storage contains raw IoT device data, preprocessed data ready for analysis, and local model parameters and updates.

5.3.2 Federated Learning Layer

At the core of the federated learning layer is the Central Federated Server, which consists of two main components: the Federated Aggregator and the Global Model. The Federated Aggregator is responsible for coordinating the federated learning process, including initiating training rounds, collecting model updates from clients, aggregating updates to improve the global model, and distributing updated global model parameters to clients. The Global Model is the centralized deep learning model that benefits from the collective knowledge of all participating clients without directly accessing their private data. Figure 21 provides an overview of the federated learning layer.

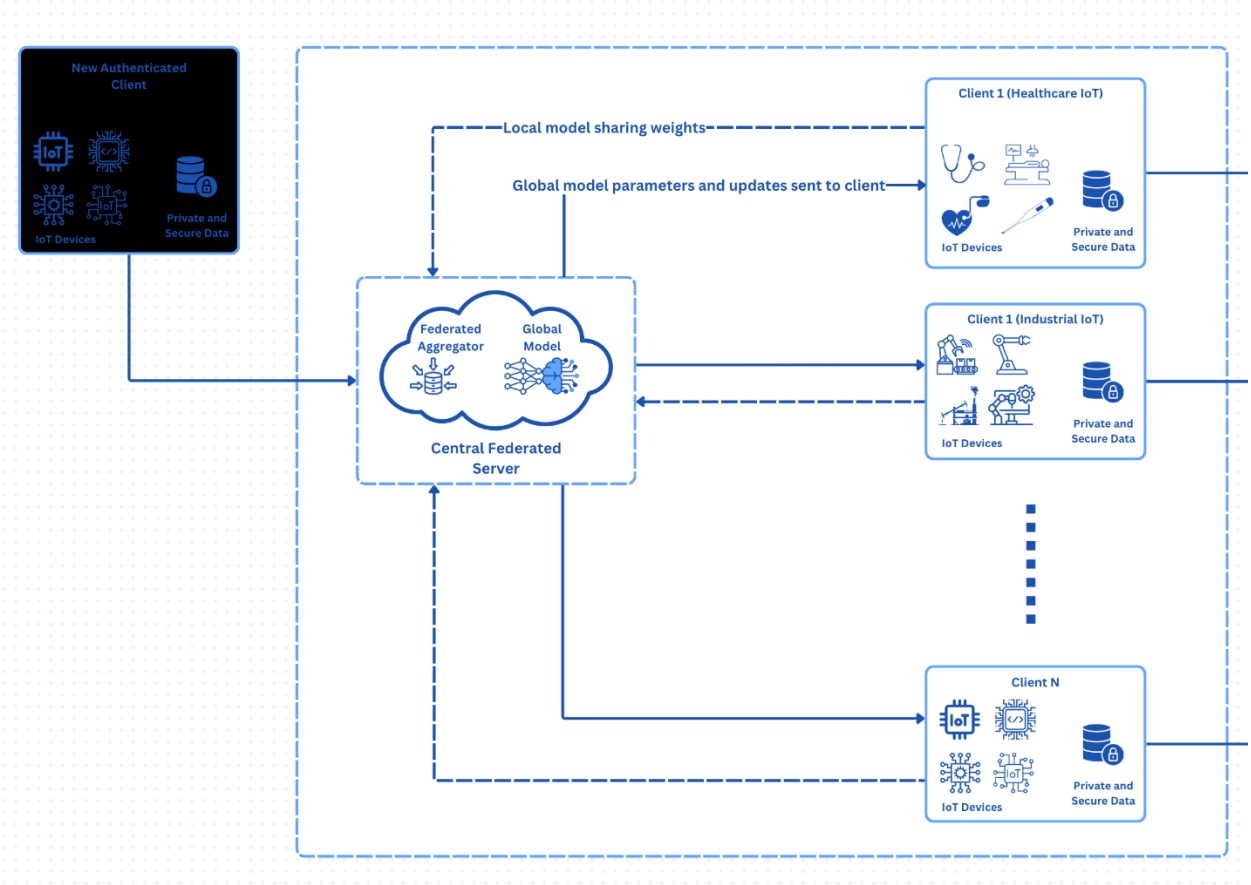


Figure 21 : Federated Learning Layer.

The federated learning process in FED-NIDS follows a circular pattern. Initially, the Central Federated Server initializes the global model and distributes it to all clients. Each client then trains the model on their local data, generating model updates. These updates, rather than raw data, are sent to the Federated Aggregator. The Aggregator integrates updates from all clients to enhance the global model, subsequently returning the updated global model parameters to each client. This process repeats in multiple rounds to continuously improve the model's performance. Figure 22 provides an overview of the federated learning client-server exchange and iteration.

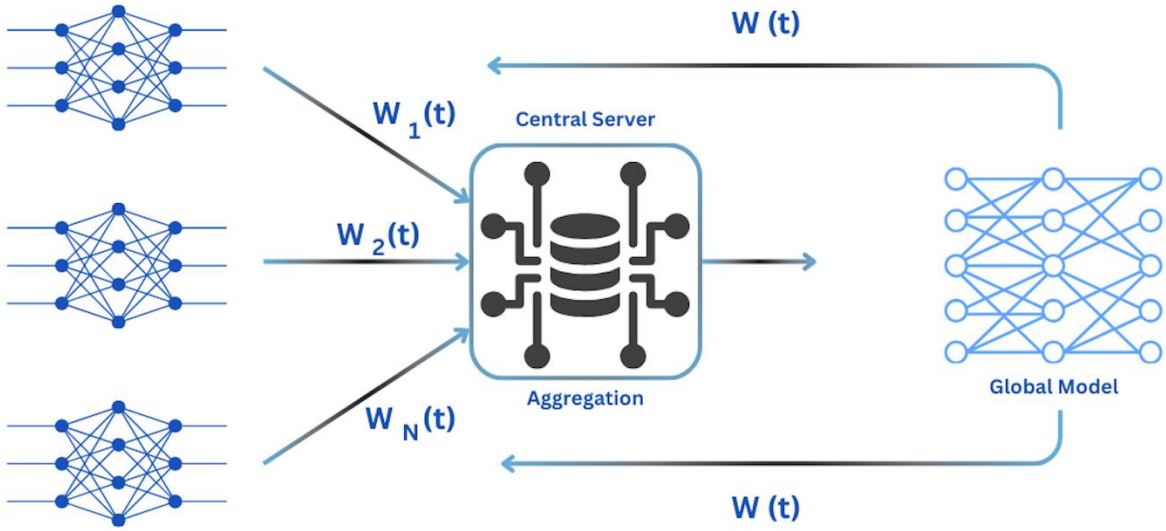


Figure 22 : Federated Learning Client-Server Exchange Overview.

5.3.3 Privacy-Preserving Mechanisms

Our proposed FED-NIDS system implements a multi-layered approach to ensure data privacy and security throughout the federated learning process. This includes encrypted communication, differential privacy, and secure aggregation.

Encrypted Communication

State-of-the-art encryption protocols, primarily Transport Layer Security (TLS) 1.3 [354], protect all data transfers between clients and the central server. This protocol provides end-to-end encryption for all network communications, using a combination of symmetric and asymmetric cryptography [357]. Asymmetric encryption, like RSA or Elliptic Curve Cryptography (ECC), establishes a shared secret key during the handshake phase [355, 356]. Symmetric encryption, typically AES-256 in GCM mode, ensures efficient and secure data transmission during data transfer [355, 356].

Key management is handled through a public key infrastructure (PKI) with regularly rotated certificates to manage encryption keys securely. The system also implements Perfect Forward Secrecy (PFS) [358, 359] using ephemeral Diffie-Hellman (DH) [355] key exchange to ensure that compromised long-term keys cannot decrypt past communications.

Differential Privacy

Differential privacy techniques are critical in preventing the extraction of individual data points from model updates, particularly in federated learning environments. The use of differential privacy in federated learning, as explored in [360], enhances privacy by adding noise not only to the outputs but also to the intermediate computations, which is critical for protecting sensitive data in distributed systems. Similarly, privacy accounting methods are introduced, such as moments accountants, which allow for tracking the cumulative privacy loss over multiple training rounds, ensuring the architecture remains within an acceptable privacy budget [361].

These methods, applied in FED-NIDS, help manage privacy expenditure efficiently, making the system scalable and secure for IoT ecosystems. In FED-NIDS, we implement local differential privacy (LDP) at the client level before transmitting model updates to preserve user privacy. This approach leverages the concepts of ε-differential privacy and the Laplace mechanism, which ensure that the presence or absence of a single record in the dataset does not significantly influence the outcome of the model.

A randomized algorithm $A(D)$ is said to provide ε-differential privacy if for all datasets D and D', differing by a single record, and for any output S in the range of the algorithm [362]:

$$\Pr[A(D) \in S] \leq e^{\varepsilon} \cdot \Pr[A(D') \in S]$$

In this framework, ε serves as the privacy budget, controlling the trade-off between privacy and utility, where a smaller ε offers stronger privacy guarantees. To achieve this, our architecture applies the gradient clipping technique to bound the L2 norm of model updates, followed by adding noise drawn from a Laplace distribution [363] to the gradients before sharing them with the central server. We apply the Laplace mechanism by adding noise from the Laplace distribution to the model updates. The noise is drawn with a scale parameter λ , which is calculated as $\lambda = \frac{\Delta f}{\varepsilon}$ where Δf represents the sensitivity of the function and ε is the privacy parameter (privacy budget). This ensures that the probability of a given output remains almost the same, even if a single record in the dataset is modified, thus maintaining ε-differential privacy. The added noise ensures that the model updates do not reveal sensitive information, striking a balance between privacy and utility. This noise ensures that the presence or absence of a single individual in the dataset does not significantly affect the result, preserving privacy while maintaining utility.

Mathematically, for a real-valued query or function f, the Laplace mechanism ensures that the function output with added noise follows $T_{A(x)} = f(x) + Y$, where $Y \sim Lap(\lambda)$, and λ is the

scale parameter that controls the amount of noise added. The Laplace distribution is centered at zero, with its probability density function (PDF) defined as:

$$noise(y) \propto \exp\left(-\frac{|y|}{\lambda}\right)$$

Here, $\lambda = \frac{\Delta f}{\varepsilon}$, where:

- Δf is the sensitivity of the function f, representing the maximum change in f when a single data point is added or removed from the dataset.
- ε is the privacy parameter, also known as the privacy budget, where a smaller ε implies stronger privacy but less accurate results.

The mechanism ensures that the probability of obtaining a certain output does not change significantly when a single record is modified. This is formalized as follows:

$$\frac{\Pr\left[T_{A(D_1)} = t\right]}{\Pr\left[T_{A(D_2)} = t\right]} \leq e^{\varepsilon}$$

In this equation:

- D_1 and D_2 are datasets that differ by a single record.
- t represents any potential outcome of the function.

By utilizing advanced composition theorems and privacy accounting techniques, our FED-NIDS system can track and control privacy loss over time. This makes the implementation robust for continuous training, as it balances the trade-offs between model accuracy and privacy guarantees, particularly in environments with sensitive IoT data flows.

Secure Aggregation

Secure Aggregation is a type of Secure Multi-Party Computation algorithm where a group of parties $u \in U$ who don't trust each other hold private value x_u and work together to find an aggregate value, such as the sum $\sum_{u \in U} x_u$, without telling each other anything about their private values other than what can be learned from the aggregate value [364].

Our FED-NIDS architecture relies heavily on secure aggregation to aggregate model updates from IoT devices without revealing individual contributions. The architecture leverages **pairwise masking [365, 366]**, where each IoT device generates random masks for every other device participating in the federated learning process. This technique protects sensitive data within

the network intrusion detection system (NIDS) while improving the global intrusion detection model.

For every client i in the FED-NIDS proposed architecture, a **Diffie-Hellman secret key exchange** is used to establish secure channels between clients [364]. Each client pairs with every other client to compute shared pairwise masks p_{ij} using the following equation:

$$p_{ij} = g^{k_i k_j \, mod \, p}$$

Where g is the generator and k_i, k_j are the private keys of clients i and j [364, 366]. Client i masks its local update x_i by adding the sum of the masks it shares with other clients:

$$m_i = x_i + \sum_i p_{ij}$$

The server receives these masked updates and sums them from all clients. As the pairwise masks cancel each other out (i.e., $p_{ij} = -p_{ji}$), the server is left with the sum of true updates:

$$\sum_i m_i = \sum_i x_i$$

This mechanism ensures that no individual update can be inferred, even if the server colludes with up to $n - 2$ clients, making it resilient against privacy attacks [365].

To handle **client dropouts**, which are common in IoT environments, FED-NIDS employs **Shamir's secret sharing**. Each client distributes shares of its masks to other clients, ensuring that even if a client drops out, the remaining clients can recover the necessary masks and complete the aggregation securely [364-366].

Some applications may find secure aggregation alone sufficient, but others may require stronger guarantees due to the failures of ad-hoc anonymization techniques [368, 369, 370] and the demonstrated ability to extract information about individual training data from fully trained models (essential aggregates) [371, 372, 373]. In such cases, secure aggregation combines well with differential privacy [374]. This is particularly beneficial for local privacy settings [375], which offer provable guarantees for the protection of individual training examples [361, 376] even when the data aggregator is not assumed to be trusted [378, 378]. To further enhance privacy in our proposed architecture, we integrate differential privacy (DP) as explained in the previous

section, which works best in combination with secure aggregation by adding noise to the updates. This ensures that individual contributions remain obfuscated even if some information about the updates leaks. This is seen with the integration of the **Laplace mechanism, which** is used to add noise N to the aggregated sum:

$$Aggregated\ Output\ = \sum_i x_i + N, N \sim Lap(\lambda)$$

Where $\lambda = \frac{\Delta f}{\varepsilon}$ is the scale of the Laplace distribution, Δf is the sensitivity of the function, and ε is the privacy budget.

The security of Diffie-Hellman-based key exchanges rests on the **Decisional Diffie-Hellman (DDH)** assumption [366]. The key exchange between two clients ensures that the shared mask p_{ij} remains private, as the adversary cannot distinguish it from a random value:

$$Adv_{DDH}(G) \leq \varepsilon$$

Where ε is a negligible probability under the DDH assumption.

Finally, the communication efficiency of secure aggregation protocols varies with the number of clients and the dimensionality of model updates. Recent improvements, such as e-SeaFL, reduce communication overhead by limiting the number of communication rounds to just one, significantly improving scalability for high-dimensional models [365].

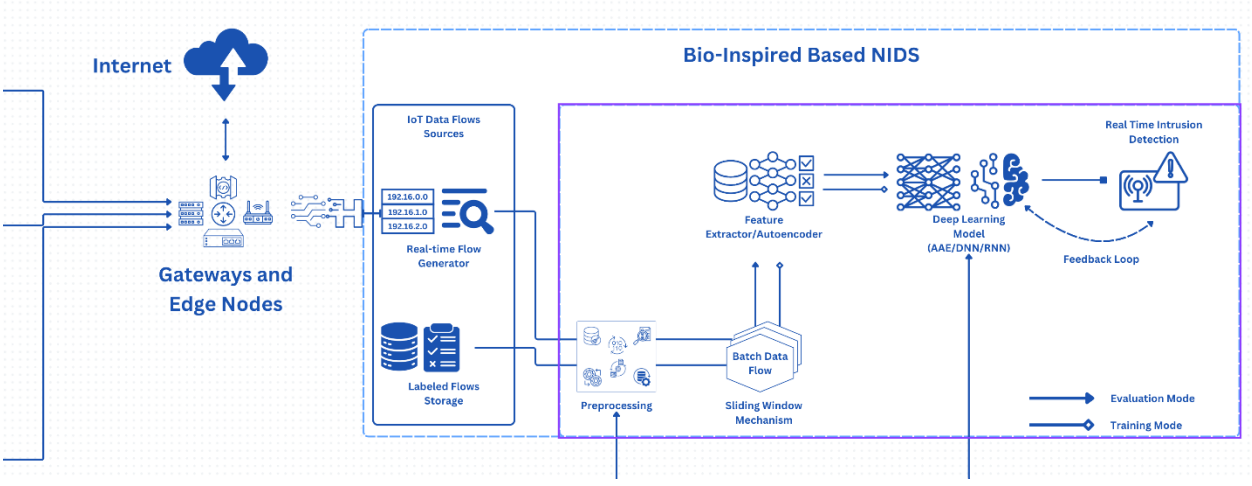


Figure 23 : Bio-Inspired NIDS and IoT Data Flows Sources.

5.3.4 IoT Data Flows and Communication Protocols

The Internet of Things (IoT) ecosystem is inherently vulnerable to cyberattacks due to the heterogeneity and complexity of connected devices, making Network Intrusion Detection Systems (NIDS) crucial for maintaining network security. While machine learning (ML) and deep learning (DL) approaches have advanced intrusion detection, they face limitations in feature engineering and fail to fully capture the structural dependencies of IoT networks. In response, we proposed FED-NIDS, which introduces a novel architecture that leverages federated learning and bio-inspired optimization methods to enhance intrusion detection. To design a more efficient architecture, we had to answer the following questions:

- How are IoT network packets captured, stored, and processed into data flows?
- What are the specific characteristics of IoT data streams?
- How do we analyze and model these data streams for intrusion detection?

This section delves into the process of IoT data flow generation, characterizes these flows, and presents strategies for analyzing IoT data streams in the context of NIDS. We also explore the protocols used for efficient communication in IoT environments.

5.3.4.1 Study of IoT Data Flows

In any computer network, including IoT networks, information is exchanged in the form of packets between different nodes. These packets contain both the message being transmitted and metadata such as IP addresses, ports, and timestamps. Due to the high volume of IoT devices, analyzing individual packets is impractical. Therefore, packets are grouped into data flows, which are easier to process and analyze.

A data flow represents a collection of packets that share common characteristics over a specific time interval, typically captured and aggregated at gateways or edge nodes before being sent to the cloud. Tools such as libpcap, Wireshark, or TCPdump are used to capture these packets and store them in PCAP files. These PCAP files are processed to extract flows, which can be further analyzed for security threats.

5.3.4.2 IoT Data Flow Generation Process

In our proposed FED-NIDS architecture, IoT data flows are generated or captured from and used from two main sources of data streams: a labeled data stream repository and a real-time data stream generator. Figure 23 captures an overview of the IoT data flows.

Labeled IoT Data Streams

The Labeled Flow Storage component serves as a repository for organizing both real-time and historical IoT data flows, particularly for training and validating machine learning and deep learning models. These labeled flows are crucial because they provide ground-truth information for distinguishing between benign and anomalous network behaviors. These labeled flows come from various sources, including historical network traffic logs, publicly available IoT datasets (such as CICIDS2017, IoT23, and TON-IoT, et cetera), and simulated IoT networks. Additionally, edge gateways may contribute real-time annotations of suspicious behavior. These labeled flows are stored in databases or suitable file formats for efficient querying, model training, and evaluation, providing a strong foundation for detecting known and emerging threats.

Real-Time Flow Generator

This component utilizes libraries like libpcap to monitor and store network traffic in real time. The generator supports multiple IoT communication protocols, including MQTT and CoAP, as well as standard TCP/IP traffic.

Captured traffic is then processed using flow generation tools like CICFlowMeter (formerly ISCXFlowMeter) [379], which aggregates packets into meaningful flows. CICFlowMeter is a network traffic flow generator and analyzer. It can be used to generate bidirectional flows, where the first packet determines the forward (source to destination) and backward (destination to source) directions; hence, more than 80 statistical network traffic features such as duration, number of packets, number of bytes, length of packets, etc. can be calculated separately in the forward and backward directions [380, 381]. CICFlowMeter extracts a comprehensive set of flow-level features, including:

- **Basic flow metrics**: duration, packet count, byte count.
- **Statistical measures**: mean, standard deviation, and maximum/minimum values of packet sizes and interarrival times.
- **Flags and advanced features**: SYN/ACK flag counts, initial window sizes, and idle time.

The flow data is stored in CSV format for subsequent analysis and learning tasks. Labeled Flow Storage is implemented as a scalable database (e.g., InfluxDB) [382] to organize flows by timestamp, source/destination, protocol, and known attack types.

5.3.4.3 Characteristics of IoT Data Streams

Once captured and converted into a usable format (CSV), each IoT data flow represents a distinct sample characterized by metadata and statistical properties:

- **Metadata**: Information such as IP addresses, ports, protocols, and capture timestamps.
- **Statistical Properties: These are generated by analysis tools and include the number of packets, bytes, and flow duration.**

These characteristics are essential for understanding the behavior of network traffic and detecting anomalous patterns. We employ feature selection techniques (mutual information) to identify the most relevant attributes for intrusion detection. Figure 24 depicts a flow-wise illustration of the IoT streams characteristics.

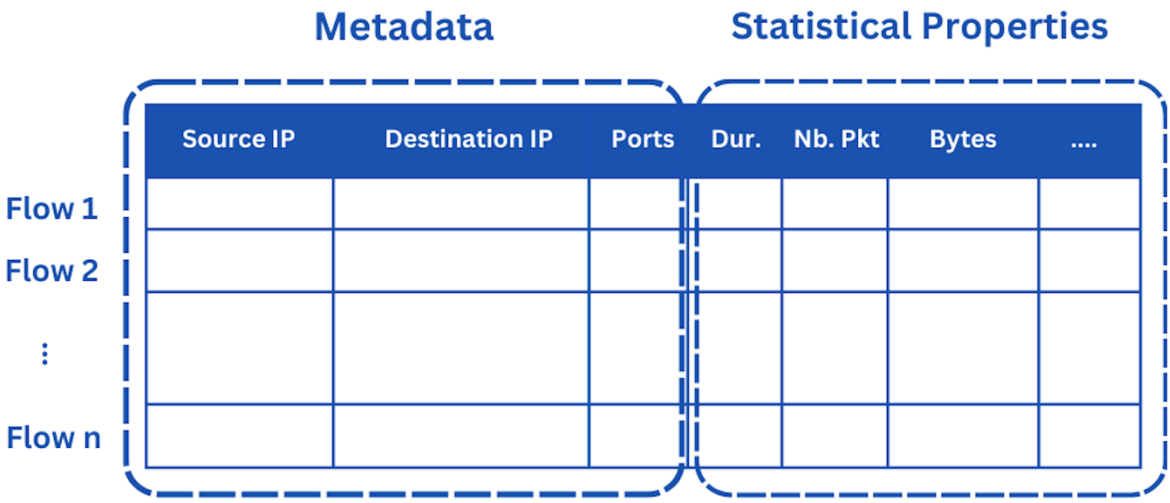


Figure 24 : Characteristics IoT Data Streams.

5.3.4.4 Strategies for Analyzing IoT Data Streams

You can perform network traffic analysis offline or online.

- **Offline Analysis**: This method involves analyzing log files containing historical data flows. It is typically used for post-event analysis to evaluate the security status over a specific period.
- **Online Analysis**: In contrast, online analysis occurs in real-time, where NIDS analyzes incoming IoT data flows in batches. Online NIDS systems leverage batch processing to reduce resource consumption, making them more suitable for environments with high data volumes.

5.3.4.5 Sliding Window

The proposed FED-NIDS architecture implements a sliding window [383] mechanism to analyze IoT data streams. This mechanism divides data sequences into time-based segments, allowing for short-term trend analysis. A sliding window has two characteristics:

- Window size: It determines the number of samples (streams) or the duration of the portions (segments) into which the data sequence is divided.
- Overlap rate: It allows sharing a portion of the samples between consecutive segments, which makes it easier to capture transitions and variations and facilitates the analysis of temporal trends.

We dynamically adjust the window size and overlap rate based on network conditions to ensure we don't miss any data during analysis. This sliding window mechanism is particularly effective for detecting slow and low attacks in real time.

5.3.4.6 Communication Protocols in IoT

The proposed FED-NIDS system supports various IoT communication protocols to accommodate the diversity of IoT devices:

- **MQTT**: A lightweight, publish-subscribe protocol ideal for resource-constrained devices. It supports multiple Quality of Service (QoS) levels to ensure reliable message delivery and uses TLS for secure communication. MQTT's design allows it to function on low-end, battery-operated sensor/actuator devices and span bandwidth-constraint WSNs like ZigBee-based networks [384].
- **Modbus**: Modbus is a decades-old protocol that provides an efficient way to communicate with physical devices that lack an inbuilt communication protocol [112]. Industrial applications support both Modbus RTU and Modbus TCP/IP. The system parses Modbus frames and analyzes them for potential intrusions.

The FED-NIDS also employs gRPC for federated learning communication. This custom protocol uses TLS with mutual authentication for secure message exchanges between clients and the central server. We incorporate compression techniques like Blosc to minimize bandwidth usage during model updates.

5.3.4.7 Data Preprocessing

Any machine learning project must include data preprocessing, which prepares raw data so that models can use it effectively. We apply preprocessing steps, specifically data cleaning and data transformation, to guarantee the integrity of the training data.

Data Cleaning: It is an integral part of data preprocessing with the primary goal of ensuring data integrity and reliability. This step involves:

- Ensuring data consistency, addressing inconsistencies, and examining duplicates.
- Handling missing values includes strategies like deletion of rows or columns, mean/median imputation, or advanced machine learning methods.
- Removing irrelevant attributes is used to enhance data quality, boosting confidence and validity in learning processes.

Data Transformation: Once the data has been cleaned, it undergoes transformations to suit the model. Among the well-known transformations are:

- Categorical feature encoding, such as ordinal encoding, assigns integers from 0 to $(N-1)$, where N is the number of categories, to categorical data (often strings).
- Normalization (standardization) is a critical transformation, especially for algorithms assuming normally distributed data. We perform normalization for a given feature X as follows: $\tilde{X} = \frac{X-\overline{X}}{\sigma}$, where σ is the standard deviation, $\tilde{X}$ is the normalized feature, and $\overline{X}$ is the mean.
- Scaling confines numerical values between 0 and 1. This is particularly useful in neural network training, especially in backpropagation, which is sensitive to large values. Scaling is done as follows: $\tilde{X} = \frac{X-\min}{\max-\min}$ Where $\tilde{X}$ is the scaled feature, min is the minimum value, and max as its maximum value of X.

By employing these preprocessing techniques, the FED-NIDS system ensures the data is well-prepared for training and model evaluation.

5.3.5 Bio-Inspired Based NIDS

At the core of our FED-NIDS architecture lies a sophisticated bio-inspired-based NIDS, which leverages the power of autoencoders for efficient feature extraction before passing the data to the deep learning model for intrusion detection (see Figure 23). This approach draws inspiration

from biological systems' ability to process and compress complex sensory inputs into essential features.

5.3.5.1 Autoencoder Integration for Feature Extraction

As an unsupervised learning tool for finding efficient data representations of IoT network traffic, the autoencoder is an important part of our FED-NIDS system. The autoencoder gets the most important parts of the network flows by squeezing the high-dimensional input data into a lower-dimensional latent space. This gets rid of noise and unnecessary data.

The autoencoder in our system consists of two main parts:

- **Encoder**: This part of the network takes the raw input data, such as IoT network traffic flows, and compresses it into a lower-dimensional latent representation. The encoder progressively reduces the dimensionality of the data, ensuring that only the most critical features are retained. We can describe the encoder function f(x) mathematically as follows:

$$h = f(x) = \sigma(Wx + b)$$

  Where x is the input, W are the weight matrices, b are the biases, and σ is the activation function (typically non-linear, such as ReLU or sigmoid). The resulting output (h) is the latent space or bottleneck representation.

- **Decoder**: The decoder takes the compressed latent representation and reconstructs it back into the original data format. The goal is to minimize the reconstruction error between the input and the output by training the autoencoder. The decoder function g(h) reconstructs the data as follows:

$$\hat{x} = g(h) = \sigma(W'h + b')$$

  Here, W' and b' are the decoder weights and biases, and $\hat{x}$ is the reconstructed output of the activation function.

Here is a representation of our autoencoder's architecture:

Input Layer → Encoder (Hidden Layers) → Latent Space → Decoder (Hidden Layers) → Output Layer

Training Process

We train the autoencoder to minimize the reconstruction error between the input x and the reconstructed output $\hat{x}$. This is typically done using the mean squared error (MSE) loss function:

$$L(x, \hat{x}) = \| x - \hat{x} \|^2$$

The training process involves the following steps:

- **Forward Pass**: The encoder passes the input data to the latent space, and the decoder then produces the reconstructed output.
- **Loss Calculation**: The reconstruction error is calculated using the MSE loss function.
- **Backward Pass**: The gradients of the loss with respect to the network parameters are computed using backpropagation and used to update the weights using an optimizer like stochastic gradient descent (SGD).

Benefits for FED-NIDS

The integration of autoencoders in our FED-NIDS offers several advantages:

- **Dimensionality Reduction**: By compressing the high-dimensional network traffic data into a lower-dimensional latent space, autoencoders help reduce the computational complexity of the subsequent deep learning model.
- **Noise Reduction**: Autoencoders can effectively filter out noise and irrelevant information from the input data, allowing the system to focus on the most important features for intrusion detection.
- **Feature Learning**: Through the process of reconstructing the input data, autoencoders learn to capture the most salient features of the network traffic, which can be more informative for intrusion detection than manually engineered features.

- **Anomaly Detection**: The reconstruction error of the autoencoder can be used as an additional signal for detecting anomalies. Unusual network patterns that the autoencoder struggles to reconstruct accurately may indicate potential intrusions.
- **Privacy Preservation**: By working with the compressed representations in the latent space, autoencoders contribute to the privacy-preserving nature of FED-NIDS, as the original raw data is not directly used in the federated learning process.

In the context of our FED-NIDS, the autoencoder is applied to the preprocessed network flow data at each client site. We then feed the encoded representations from the latent space into the main deep learning model for intrusion detection. This two-stage approach allows for more efficient and effective learning, especially in the distributed setting of federated learning.

Using autoencoders in this way makes our FED-NIDS system better at processing and analyzing the huge amounts of different data that IoT devices produce. This makes the system better at finding problems because the deep learning models can now work with a smaller, easier-to-handle dataset that focuses on the most important features. This lowers the computational load and improves the accuracy of detection.

The weights of the autoencoder are also updated through the federated learning process, allowing the feature extraction mechanism to improve over time based on data from all participating clients without directly sharing raw network data.

5.3.5.2 Deep Learning Models

The heart of the detection system is a sophisticated deep learning model that incorporates various architectures. This includes autoencoders (AAE) for unsupervised anomaly detection, capable of learning normal behavior patterns and flagging deviations as potential intrusions. Deep Neural Networks (DNN) are employed for complex pattern recognition and classification, allowing the system to identify intricate attack signatures that might be missed by traditional rule-based systems. Recurrent neural networks (RNN), particularly Long Short-Term Memory (LSTM) variants and Gated Recurrent Units (GRUs), are used for analyzing sequential data and identifying time-dependent attack patterns, which is crucial for detecting sophisticated, multi-stage attacks.

The system provides real-time intrusion detection capabilities, continuously monitoring network traffic and rapidly classifying network flows as benign or malicious. When an intrusion is detected, the system generates immediate alerts, allowing for quick responses to potential threats. We implement a feedback loop to continuously improve the detection model, incorporating new attack patterns and reducing false positives over time.

5.3.6 Bio-Inspired Optimization Layer

The Bio-Inspired Optimization Layer is an important part of our proposed FED-NIDS architecture. We aim to enhance the system's ability to detect network intrusions in IoT environments by increasing its flexibility and effectiveness. This layer incorporates four carefully selected bio-inspired optimization algorithms: ASSCO, ISSA-HHO, PSO-GA, and GWO-ABC. Figure 25 shows an overview of this layer. Each of these algorithms brings unique strengths to the optimization process of the proposed FED-NIDS model, collectively contributing to a robust and adaptive NIDS.

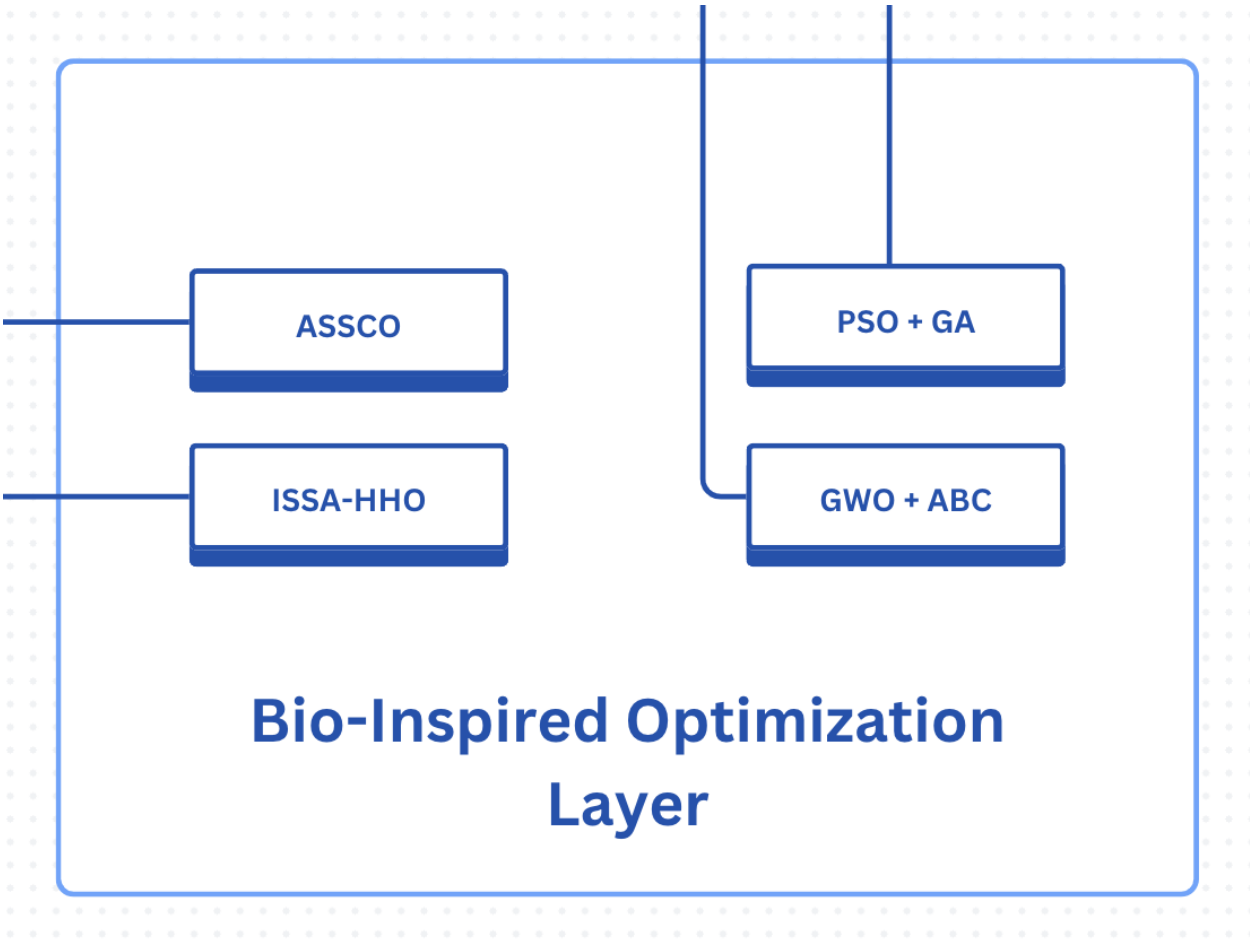


Figure 25 : Bio-Inspired Optimization Layer.

Bio-inspired optimization algorithms are metaheuristics that mimic the nature of solving optimization problems [385]. In the context of our FED-NIDS, these algorithms offer powerful approaches to optimize various aspects of the system, including feature selection, parameter tuning, and adaptive threat detection.

5.3.6.1 Adaptive Social Spider Colony Optimization (ASSCO)

ASSCO (Adaptive Social Spider Colony Optimization): ASSCO is inspired by the foraging behavior of social spiders [387]. In our FED-NIDS, ASSCO is primarily used for feature selection and parameter optimization. The algorithm models the prey capture behavior of social spiders, where spiders sense the presence of prey through vibrations transmitted along the web thread. The

spiders move through the solution space, leaving information trails that influence the movement of other spiders, like pheromone trails in ant colony optimization [388].

In practice, the frequency of vibration (that is, the number of oscillations per unit time) is the factor that differentiates the spider's vibration from that of the prey; it enables a spider to react to the prey in a different way from a neighboring spider on a web. Therefore, the frequency represented as $f_{1,t}$ is proportional to the fitness value (solution quality) of the objective function to be optimized $f(X_{i,t})$.

Hence, the frequency of vibration is represented as follows:

$$f_{1,t} = \left(\frac{f(x_{i,t}) - f_{\min}}{f_{\max} - f_{\min}}\right) + C$$

Where:

$f(X_{i,t})$ is the fitness value of the evaluated objective function with regards to the spider's position $X_{i,t}$. Considering optimization theory, the objective function is any of the standard benchmark functions used to validate global optimization problems. The variables $f_{\max}$ and $f_{\min}$ represent the maximum fitness and minimum fitness values of the entire population obtained by evaluating the objective function with regards to the spider's positions. These frequencies are also defined, considering optimization, as:

max

$$f_{max} = {}_{j=1,2,\ldots,n}\left(f(X_j)\right)$$

and

min

$f_{min} = {}_{j=1,2,\ldots,n}\left(f(X_j)\right)$ min

C is the proportionality constant defined as a small exponential decay function for the prey because the frequency fades or decreases with time (that is, the frequency at which the prey vibrates the

web as soon as it is caught is at maximum and fades with time as it gets tired or gets incapacitated by the spider), and for the spider, the value increases as it approaches the prey and becomes equal to that of the prey (that is, the frequency of the spider increases as it approaches the prey, and at the point of the global position, which is the position of the prey, it equals the frequency of the prey. This is because at that point the spider, incapacitating the prey, vibrates the web more than when it moves cautiously to get the prey).

5.3.6.2 Improved Salp Swarm Algorithm with Harris Hawk Optimization (ISSA-HHO)

ISSA-HHO is a hybrid algorithm that combines the strengths of the improved Salp Swarm Algorithm and Harris Hawks Optimization [389]. In our FED-NIDS, ISSA-HHO is utilized for balancing exploration and exploitation in the search space, crucial for optimizing NIDS parameters and adapting to new threats. The bio-inspired algorithms are executed in the exploration and exploitation phase. The exploration part of ISSA is modified using HHO algorithm. The trigonometric functions with HHO exploration equations are used to enhance the exploration phase of the ISSA. Afterward, levy flight is used to reform the leader position of the salp. This function controls the step taken by salps. During exploration, the salps perform the global search without falling into local optima. Here, HHO exploration equations are used to perform the global search. The opposition-based learning [389] is used for the Hawks population initialization.

$$H_{mean}(it) = \sum_{i=1}^{N} \frac{H_i(it)}{N} \qquad (i)$$

$$LeF = \frac{ur \times \sigma}{|vr|^{\frac{1}{\beta}}},\ \sigma = \left(\frac{\Gamma(1+\beta) \times \sin\left(\frac{\pi\beta}{2}\right)}{\Gamma\left(\frac{1+\beta}{2}\right) \times \beta \times \sin\left(\frac{\beta-1}{2}\right)} \right)^{\frac{1}{\beta}} \qquad (ii)$$

Afterward, Hawks position is updated as seen in (a) and (b):

$$SH(it+1) = H_{rnd}(it) - rn_3 \cdot \sin(rn_0) \cdot |H_{rnd}(it) - 2rn_4\ H(it)| \text{ if } q \geq 0.5 \qquad (a)$$

$$SH(it+1) = H_{pr}(it) - H_{mean}(it) - rn_5 \cdot \cos(rn_0) \cdot \left(LwB + rn_6(UpB - LwB)\right) \text{ if } q < 0.5 \qquad (b)$$

Where $H_{rnd}(it)$ is the random Hawks position obtained using opposition-based learning. q, rn_3, rn_4, rn_5, $and\ rn_6$ are random numbers. $H_{pr}(it)$ is the prey location. $H(it)$ is the current Hawks position. $H_{mean}(it)$ is the mean position obtained from equation (i).

LwB is the lower bound and UpB is the search space upper bound. During exploitation, local searches to obtain an optimal solution are performed. The Lef is used to control the step size of follower salps. Therefore, the salp follower position is updated as:

$$SH_n^m = \frac{1}{2}(SH_n^m - SH_n^{m-1}) + LeF(Dim) \qquad (c)$$

Where SH_n^m is the current salps position and SH_n^{m-1} is the previous salp position. $Lef(Dim)$ is a levy flight function obtained from equation (ii). After that, local search technique [389] is applied to avoid local optima and improve solutions.

5.3.6.3 Particle Swarm Optimization with Genetic Algorithm (PSO + GA)

PSO-GA (Particle Swarm Optimization - Genetic Algorithm): The PSO-GA hybrid combines Particle Swarm Optimization with Genetic Algorithms [385, 386]. In our FED-NIDS, PSO-GA is employed for optimizing complex NIDS parameters, leveraging both global exploration (GA) and local exploitation (PSO) of the solution space. The population-based technique of Particle Swarm Based Optimization was inspired by bird flocks and fish schools. PSO is a computational technique that is quite like evolutionary computation. The system gives random solutions to a population to choose from, and the best one is found after generations of updates. The birds are represented as particles that are persistently searching for the optimized global known position (G_{best}). The best position of the local particle (P_{best}) is used as a guide to finding the G_{best}. If a particle finds another best position, its P_{best} position is updated. As a result of each particle's collaboration mechanism, the G_{best} value is obtained by optimizing the value. In a D-dimensional target space with N uniformly distributed particles. The position and velocity of the ith particle are denoted by Xi and Vi, respectively, as in equation 1.

$$Velocity\ V_i^{k+1} = \omega V_i^k + c_1.rand(\ ).\left(P_{best} - X_i^k\right) + c_2.rand(\ ).\left(G_{best} - X_i^k\right) \qquad (1)$$

Position, as in equation 2. $X_i^{k+1} = X_i^k + V_i^k$ (2)

Where C_1 and C_2 represent the coefficients of social and cognitive acceleration and w is the weight of inertia. It's also appealing because there are only a few frameworks to configure. One version works well in a wide range of implementations with small adjustments.

GA is a population-based optimization algorithm. The potential solutions to an optimization problem are represented as individuals within a population. These individuals are denoted as a genotype of chromosomes and represent potential solutions to the optimization problem, regardless of the field to which they belong (in our case, IoT and NIDS). Their data are encoded as a string of binary digits that can be further manipulated and processed [390].

5.3.6.4 Grey Wolf Optimization with Artificial Bee Colony (GWO + ABC)

GWO-ABC (Grey Wolf Optimization - Artificial Bee Colony): GWO-ABC combines Grey Wolf Optimization with Artificial Bee Colony. In our FED-NIDS, this hybrid approach allows for a hierarchical search strategy (GWO) combined with effective exploitation of promising solutions (ABC), particularly useful for adaptive threat detection. Grey wolf optimization, a meta-heuristic swarm technique, was introduced for the first time by Simon Fong [386]. The GWO algorithm is a swarm-intelligence-based metaheuristic procedure inspired by the social behavior and hunting strategies of wolves in packs [386]. It is assumed that in dependence on their roles in their packs, grey wolves are socially organized as a pyramidal hierarchy, although that may not be so in reality in the wild, and some prestigious sources have even described it as a myth [390].

The standard GWO algorithm is initialized by setting the number of pack members to n, the parameter a that gradually decreases its value from 2 to 0, the maximum number of iterations t_{max}, and the search agents X_i, where $i = 1, 2, \ldots, n$ for a fitness function $f(X_i)$. The three best solutions are, according to the pack hierarchy, denoted as X_α, X_β, X_δ. The distances to the target are described as [390]:

$$D_\alpha = |C_1 X_\alpha - X|$$

$$D_\beta = |C_2 X_\beta - X|$$

$$D_\delta = |C_3 X_\delta - X|,$$

And the search agents (solutions) are

$$X_1 = |X_\alpha - A_1 D_\alpha|$$

$$X_2 = |X_\beta - A_2 D_\beta|$$

$$X_3 = |X_\delta - A_3 D_\delta| .$$

The next iteration step is calculated as [390]:

$$X(t+1) = \frac{X_1 + X_2 + X_3}{3},$$

Where $A = 2a \cdot r_1 \cdot a$ and $C = 2r_2$ are coefficients, while r_1, r_2 are random vectors with intensities in the range from 0 to 1.

ABC's operational model is made up of three essential factors. First is the food source quality, in this case nectar, which is determined by a variety of profitable factors like nest distance, quantity in richness, and removal effort [385]. The foragers are next on the list, also called worker bees, who are tasked with bringing details of the place and food source availability to the nest. Behavior of a swarm of bees in solving unidimensional, multi-dimensional, and other dimensional problems. The problems are classifying the attacks, choosing the features, and finding out the attacks. ABC optimization includes the following phases: phase of initial population, phase of working bees, phase of observer bees, and phase of lookout bees [385].

By integrating these bio-inspired optimization algorithms into our FED-NIDS, we create a system that can efficiently adapt to new threats, optimize its parameters for better performance, and balance between exploration of new solutions and exploitation of known good solutions. This approach significantly enhances the robustness and effectiveness of our NIDS in detecting and responding to network intrusions in complex and evolving IoT environments.

5.3.7 System Integration and Data Flow

The integration of all components in FED-NIDS creates a seamless flow of data and processes. IoT devices generate data, which is collected and processed at the edge layer. This preprocessed data is then used to train local models within each client's environment. The model

updates, rather than raw data, are sent to the Central Federated Server, where they are aggregated using our secure aggregation techniques.

The improved global model is then distributed back to all clients, ensuring that each participant benefits from the collective learning without compromising data privacy. The Bio-Inspired Based NIDS component uses the latest model for real-time intrusion detection, continuously monitoring network traffic for potential threats.

Meanwhile, the Bio-Inspired Optimization Layer works continuously to improve all aspects of the system, from feature selection to model architecture. This creates a dynamic, self-improving system that can adapt to new threats and changing network conditions.

5.3.8 Scalability and Adaptability Features

We designed FED-NIDS to be highly scalable and adaptable to dynamic IoT environments. The system implements a hierarchical architecture, using a tree-like structure of aggregators to handle large numbers of clients efficiently. This allows the system to scale horizontally as the number of IoT devices and clients increases.

Cloud-native technologies like AWS IoT Central and Kubernetes enable dynamic resource allocation by scaling computational resources according to demand. This ensures that the system can handle peak loads without overprovisioning resources during quieter periods.

To adapt to the heterogeneous nature of IoT devices, FED-NIDS employs model compression techniques like pruning and quantization. This enables the deployment of models on resource-constrained devices without significantly compromising performance. The system also tailors model complexity and update frequency based on individual device capabilities, ensuring efficient operation across a wide range of IoT hardware.

The system's adaptability extends to its ability to handle dynamic threat landscapes. It maintains an ensemble of models to capture diverse attack patterns and incorporates online learning techniques to rapidly adapt to newly observed attack vectors. We implement concept drift detection to continuously monitor model performance, detect shifts in data distribution, and trigger necessary model updates.

5.3.9 Handling Non-IID Data in FED -NIDS

A key challenge in federated learning for IoT environments is dealing with non-IID (Independent and Identically Distributed) data. The proposed FED-NIDS will incorporate several strategies to address this issue.

To find out how different non-IID clients are, the system is designed to use methods like Earth's Mover's Distance (EMD) to find out how different the distributions are and Jensen-Shannon divergence to see how different the probability distributions are. This analysis helps understand the extent of data heterogeneity and informs subsequent optimization strategies.

Each client will employ federated data augmentation to generate synthetic samples, which aids in balancing class distributions. This process uses techniques like SMOTE (Synthetic Minority Oversampling Technique) adapted for federated settings, with differential privacy mechanisms ensuring that synthetic samples don't leak information about real data points.

FED-NIDS also implements personalized federated learning, allowing each client to maintain a personalized model alongside the global model. This approach uses techniques like model-agnostic meta-learning (MAML) for quick adaptation to local distributions. Each client's local model can fine-tune with a small amount of local data, thanks to MAML, which enables the global model to serve as a useful initialization point. This balances the benefits of collaborative learning with the need for models tailored to each client's unique data distribution.

The system employs robust aggregation mechanisms that are resilient to non-IID data. One such technique is the FedProx algorithm, which adds a proximal term to the local optimization objective. This proximal term penalizes large deviations from the global model, helping to maintain consistency across clients with divergent data distributions. Additionally, FED-NIDS uses attentive aggregation to weigh client updates based on their utility to the global model. This approach gives more importance to updates that are likely to generalize well across the entire network, mitigating the impact of outlier distributions.

To further address data heterogeneity, FED-NIDS implements a clustered federated learning approach. This involves grouping clients with similar data distributions using federated clustering algorithms. These algorithms can identify groups of clients with similar characteristics without sharing raw data, preserving privacy. The system maintains separate global models for each cluster, which enhances its ability to capture diverse distributions across the IoT network. This clustered

approach strikes a balance between the one-size-fits-all global model and fully personalized local models.

The system also incorporates adaptive learning rate techniques to handle the varying data distributions across clients. The system implements methods such as cyclic learning rates or the Adam optimizer with warm restarts. These techniques allow the learning process to adapt dynamically to the characteristics of each client's data, improving convergence and model performance in the face of non-IID data.

To ensure that the global model remains effective across all clients despite data heterogeneity, FED-NIDS employs a multi-task learning framework. This approach treats each client's task as a related but distinct problem, allowing the model to learn shared representations while still capturing client-specific nuances. We carefully balance the multi-task learning objective with the global objective to ensure the resulting model performs well across all clients.

Lastly, FED-NIDS implements a continual learning mechanism to address the evolving nature of IoT data distributions. This involves techniques like elastic weight consolidation (EWC) to prevent catastrophic forgetting as the model adapts to new data patterns. The continual learning approach allows the system to maintain knowledge of past data distributions while still adapting to new ones, ensuring robust performance over time even as the IoT environment changes.

By incorporating these diverse strategies for handling non-IID data, FED-NIDS can effectively manage the heterogeneous nature of data across various IoT environments. This ensures robust and accurate intrusion detection capabilities across a wide range of scenarios and client types.

5.4 Conclusion

This chapter introduced and went into detail about the proposed FED-NIDS system's architecture. It uses bio-inspired algorithms and Federated Learning (FL) to solve the security problems that are unique to IoT ecosystems. We designed FED-NIDS to provide a scalable, adaptive, and privacy-preserving solution that can handle the vast, heterogeneous, and resource-constrained nature of IoT environments. The design of FED-NIDS takes advantage of FL's collaborative and decentralized features while adding bio-inspired optimization techniques to make the system better at finding intrusions and responding to them in real time.

We outlined the key design objectives of FED-NIDS, which include ensuring privacy-preserving collaboration, scalability, and resource efficiency. Several layers compose the architecture, including the IoT ecosystem and edge layer, the federated learning layer, and the bio-inspired NIDS. Each of these components is crucial to achieving robust, real-time intrusion detection. The architecture's strength lies in its ability to manage non-IID data and adapt to dynamic network conditions through continuous learning and optimization.

We employ bio-inspired algorithms, such as ASSCO, a hybridization of ISSA-HHO, PSO with GA, and GWO with ABC, to enhance model performance and adaptability. These algorithms are particularly suited for optimizing feature selection, hyperparameters, and model architecture, making FED-NIDS a self-improving system capable of evolving with the threat landscape.

The chapter also emphasized the integration of secure aggregation and differential privacy techniques, ensuring that the data used in the federated learning process remains secure and that individual devices' contributions are protected. By incorporating these privacy-preserving mechanisms, FED-NIDS addresses a critical challenge in IoT: balancing effective security measures with the need for privacy in highly distributed environments.

In conclusion, the proposed FED-NIDS represents a novel and effective approach to securing IoT ecosystems. It combines the power of federated learning and bio-inspired algorithms to deliver a scalable, adaptable, and privacy-preserving solution for network intrusion detection. The system's ability to operate in resource-constrained environments, manage heterogeneous data, and adapt to evolving threats makes it well-suited for modern IoT security needs. Future research should focus on further enhancing the efficiency of the federated learning process, exploring additional bio-inspired optimization techniques, and refining the system's ability to detect zero-day threats in real time.

Chapter 6: Implementation and Experimental Setup

In the previous chapter, we dedicated considerable attention to the detailed description of the FED-NIDS architecture, a bio-inspired federated deep learning-based network intrusion detection system for IoT ecosystem security. Each component of this architecture was thoroughly examined, and the design choices were carefully justified. Now that the design phase of our intelligent system is complete, we are ready to proceed with its concrete implementation.

This chapter presents the implementation details of our FED-NIDS system. We provide a comprehensive overview of the different components of our architecture, focusing on the development environments, libraries, and tools used to build our solution. Furthermore, we offer an in-depth explanation of the data preprocessing pipeline, which includes tasks such as dataset loading and distribution, data cleaning, and transformation. We then explore the implementation of our machine learning and deep learning models, followed by the federated learning setup that enables collaborative learning across IoT devices while preserving data privacy. The chapter also details the integration of bio-inspired optimization techniques, and the evaluation metrics used to assess the system's performance.

Throughout this chapter, we emphasize the practical aspects of implementing FED-NIDS, highlighting the technical challenges encountered and the solutions developed to address them. This implementation phase represents the crucial step of transforming our theoretical design into a functional system capable of protecting IoT ecosystems from network intrusions.

6.1 Development Environments and Tools

The implementation of FED-NIDS required a robust and versatile development environment to handle the complex requirements of federated learning, deep learning, and bio-inspired optimization algorithms. Our development setup was carefully chosen to ensure efficient code development, experiment tracking, and system performance optimization.

6.1.1 Integrated Development Environments (IDEs)

The choice of development environments was driven by performance requirements and resource utilization at different stages of implementation. We primarily used **JetBrains DataSpell** for data science tasks and exploratory data analysis because of its optimized performance with large datasets and excellent integration with machine learning frameworks. DataSpell's intelligent code completion and built-in scientific tools significantly enhanced our productivity during the initial data analysis and model prototyping phases.

PyCharm Professional served as our main IDE for developing the core FED-NIDS system components, particularly the federated learning implementation and bio-inspired optimization modules. Its robust debugging capabilities and seamless integration with frameworks like PyTorch and Flower made it invaluable for complex system development.

We utilized **Jupyter Notebook** for initial prototyping, data visualization, and quick experimentation with different model architectures. Its interactive nature proved especially useful during the early stages of development and for creating visualizations to understand data patterns and model behavior.

We used **Visual Studio Code** for lightweight development tasks and remote development scenarios. Working with resource-constrained environments and requiring rapid code modifications without the overhead of heavier IDEs, Visual Studio Code's efficient resource utilization made it especially suitable.

GitHub managed version control, facilitating efficient collaboration and code versioning. We maintained separate repositories for different components of FED-NIDS, including the core system, data preprocessing modules, and evaluation scripts. This modular approach facilitated easier maintenance and testing of individual components.

To track our experiments and monitor system performance, we integrated Weights & Biases (WandB) into our development workflow. WandB provided comprehensive experiment tracking, hyperparameter optimization, and performance visualization capabilities. This tool was particularly valuable for monitoring the deep learning model architectures and federated learning

process across multiple clients and visualizing the convergence of our bio-inspired optimization algorithms.

6.1.2 Libraries and Frameworks

6.1.2.1 Data Manipulation, Visualization, and Preprocessing Libraries

For data analysis and preprocessing tasks, we have used several popular libraries, including NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn.

NumPy

It is a fundamental library for numerical computing in Python. NumPy provides efficient multidimensional data structures and basic mathematical functions that are essential for data processing in machine learning.

Pandas

This library is widely used for data manipulation and analysis. Pandas offers flexible and efficient data structures for working with data tables, including features for filtering, sorting, merging, and transforming data.

Matplotlib

Matplotlib is a graphical library that offers a wide range of chart types and features, allowing effective data visualization. With Matplotlib, it becomes possible to create histograms, linear graphs, pie charts, and much more, allowing for more in-depth analysis and understanding of diagrams.

Seaborn

Developed based on Matplotlib, Seaborn extends its capabilities by offering additional features specifically designed for the visualization of statistical data. This includes the ability to create visually appealing and informative graphics such as moustache boxes, dispersion diagrams with linear regression adjustments, and heat maps, thus improving the overall data visualization potential.

Scikit-Learn

This is a very popular machine learning library that offers a wide range of algorithms and features for the preparation and analysis of data. Scikit-learn provides tools for data preprocessing, feature selection, cross-validation, supervised and unsupervised learning, model evaluation, etc.

6.1.2.2 Deep Learning and Bio-Inspired Models

PyTorch

We chose PyTorch as our primary deep learning framework due to its dynamic computational graphs and extensive support for custom model architectures. It is a very popular and versatile deep learning library that offers significant advantages over others. PyTorch excels in its flexibility and ease of implementation when building neural network models. With its dynamic calculation chart and automatic differentiation system, PyTorch simplifies the process of creating and training models, allowing for more intuitive and more efficient development.

BioTorch

We integrated BioTorch for implementing bio-inspired optimization algorithms, leveraging its specialized modules for evolutionary computation and swarm intelligence techniques.

BioTorch provides implementations of layers, models, and biologically motivated learning algorithms. It enables users to load existing state-of-the-art models, create custom models easily, and convert existing models automatically. The framework enables the training, evaluation, and benchmarking of various biologically plausible learning algorithms across a variety of datasets. The framework focuses on the principles of PyTorch design and research reproducibility.

6.1.2.3 Federated Learning Framework and Libraries

Flower Framework

We selected the Flower framework as our federated learning platform because of its flexibility, easy integration into existing machine learning pipelines, and support for various machine learning frameworks. It offers a unified approach to federated learning, analytics, and evaluation, along with tools for monitoring and managing distributed machine learning

experiments, enabling multiple clients to collaborate on model training without the need to share raw data. Any workload, ML framework, and programming language can federate with it.

Hydra-Core

We utilized Hydra-Core for managing our experimental configurations, allowing us to easily modify and track different system parameters across multiple experiments. The key feature is the ability to dynamically create a hierarchical configuration via composition and override it through config files and the command line.

Ray

Ray provided distributed computing capabilities, particularly useful for simulating multiple IoT clients during development and testing phases. It offers scalable libraries for common ML tasks, Pythonic distributed computing primitives, and integrations with various tools and infrastructure. It empowers data scientists to easily scale ML workloads without infrastructure expertise and enables ML platform builders and engineers to create scalable and robust ML platforms. Additionally, it automates key processes for distributed systems engineers, making it a comprehensive solution for scaling AI and ML applications.

We carefully integrated these tools and frameworks to create a cohesive development environment that supported both the technical requirements of FED-NIDS and the need for efficient experimentation and evaluation. The combination of these components enabled us to implement, test, and refine our system while maintaining high standards of code quality and reproducibility. To take full advantage of all the most advanced features, we have opted for the most recent versions of the libraries mentioned above. Table 8 below presents the main libraries and their corresponding versions:

	NumPy	Pandas	Matplotlib	Seaborn	Scikit-Learn	PyTorch	BioTorch	Hydra-Core	Ray
Version	2.1	2.2.3	3.9.0	0.13.2	1.5.2	2.5.0	0.0.16	1.3.2	2.38.0

Table 8 : Most recent versions of some of the Python libraries used.

6.2 Datasets and Data Sources

As detailed in Section 5.3.4.2, our proposed FED-NIDS system architecture incorporates two IoT data stream sources: the labeled data stream database and the real-time stream generator. We use database data streams for model training and validation and utilize generator flows for evaluation. In order to properly achieve this, we use datasets consisting of previously collected IoT data streams. We divide the dataset into two distinct parts: training data and test data. For this purpose, the database of labeled data flows is concretized by the training and test flows, while the generator corresponds to the evaluation flows.

For the implementation and evaluation of FED-NIDS, we utilized the **Aposemat IoT-23** Dataset, a comprehensive collection of IoT network traffic data that includes both malicious and benign network behaviors. This dataset was created by the Stratosphere Laboratory as part of the Avast AIC laboratory project and represents a significant contribution to IoT security research.

6.2.1 Description of Selected Dataset

The IoT-23 dataset consists of twenty-three distinct network traffic captures, comprising twenty malware captures from infected IoT devices and three captures of benign IoT device traffic. We collected the captures between 2018 and 2019, providing a current depiction of IoT network behavior. Each malicious scenario involves a specific malware sample executed on a Raspberry Pi device, allowing for the capture and analysis of real network behavior under controlled conditions with unrestricted internet connectivity.

The dataset was originally generated through a sophisticated process involving multiple stages. Initially, network traffic was captured in pcap files, which were then processed through the Zeek network analyzer to generate detailed log files. These log files, particularly the connection logs (conn.log), provide comprehensive information about network flows between different entities. The Flaber Python script was subsequently used to assign labels based on predefined rules, enabling accurate classification of network behaviors.

6.2.2 Dataset Characteristics and Properties

The IoT-23 dataset contains a rich set of features that characterize each network connection. Each record includes temporal information (timestamp), connection identifiers, network endpoints (source and destination IP addresses and ports), protocol information, and detailed connection statistics. Key metrics such as connection duration, bytes transferred, packet counts, and connection states provide a comprehensive view of network behavior, which was critical for training the FED-NIDS to detect a range of network threats. Table 9 below presents the main features within the dataset and their descriptions:

Feature Name	Description
timestamp	Time of the first packet in the connection.
unique_identifier	Unique ID for each connection.
origin_ip_address	IP address of the connection's origin.
origin_port	Port number used by the origin.
destination_ip_address	IP address of the connection's destination.
destination_port	Port number used by the destination.
network_protocol	Protocol at the transport layer.
network_service	Application protocol used within the connection.
connection_duration	Duration of the connection.
bytes_sent_by_origin	Payload bytes sent by the originator, calculated from sequence numbers.
bytes_sent_by_destination	Payload bytes sent by the responder.
connection_state	State of the connection flow (detailed below).
is_source_local	Locality of the connection origin (T for true, F for false).
is_destination_local	Locality of the connection destination (T for true, F for false).
missed_bytes_count	Bytes missed due to packet loss, possibly affecting protocol analysis.

connection_history	State history of connections, represented by a sequence of letters.
packets_sent_by_source	Count of packets sent by the originator.
ip_bytes_sent_by_source	IP level bytes sent by the originator.
packets_sent_by_destination	Count of packets sent by the responder.
ip_bytes_sent_by_destination	IP level bytes sent by the responder.
label	Multi-class label defining the connection's category, such as "Attack" or "Benign."

Table 9 : Network flow features and descriptions.

The connection states in the dataset are meticulously documented through various flags and indicators. The 'connection_state' field uses specific codes to represent different connection scenarios, from normal establishments (SF) to rejected connections (REJ) and various types of terminated connections (RSTO, RSTR). Similarly, the 'connection_history' field provides detailed information about the packet-level interactions within each connection. Tables 10 and 11 present descriptions of the various connection states and history flags.

Connection State	Description
S0	Connection attempt seen, no reply.
S1	Connection established, not terminated.
SF	Normal establishment and termination; distinguished from S1 by byte counts in the summary.
REJ	Connection attempt rejected.
S2	Connection established, originator initiated close, no response from responder.
S3	Connection established, responder initiated close, no response from originator.
RSTO	Connection established; originator aborted (sent an RST).

RSTR	Connection established; responder sent an RST.
RTOSO	Originator sent a SYN followed by an RST without receiving a SYN-ACK from the responder.
RSTRH	Responder sent a SYN-ACK followed by an RST without receiving a SYN from the originator.
SH	Originator sent a SYN followed by a FIN without receiving a SYN-ACK from the responder (half-open connection).
SHR	Responder sent a SYN-ACK followed by a FIN without receiving a SYN from the originator.
OTH	No SYN seen, only midstream traffic observed (e.g., a partial connection that wasn't later closed).

Table 10 : Description of the connection states.

Flag	Description
s	SYN without the ACK bit set.
h	SYN+ACK (handshake).
a	Pure ACK packet.
d	Packet containing payload (data).
f	Packet with the FIN bit set.
r	Packet with the RST bit set.
c	Packet with a bad checksum applies to UDP as well.

g	Content gap detected (indicating potential packet loss)
t	Retransmitted payload.
W	Packet with a zero-window advertisement.
i	Inconsistent packet, e.g., with both FIN and RST bits set.
q	Multi-flag packet, e.g., SYN+FIN or SYN+RST bits set.
^	Connection direction flipped by Zeek's heuristic.
x	Partial connection analysis due to limits being exceeded.

Table 11 : Description of the history flags.

Of particular importance is the '**label**' column (feature/target), which categorizes network flows into multiple classes, including Benign, Attack, C&C (Command and Control), DDoS, FileDownload, HeartBeat, Mirai, Okiru, PartOfAHorizontalPortScan, and Torii. This multi-class labeling enables comprehensive analysis of different types of network behaviors and attacks. Each connection in the dataset is labeled, identifying it as either benign or associated with a specific type of attack. Table 12 below provides an overview of the different label classes:

Label	Description
Attack	Represents any attempt by an infected device to exploit a vulnerability on another device.
Benign	Indicates typical device behavior without signs of malicious activity.
C&C	Shows that the infected device connected to a command-and-control server for malicious operations.

DDoS	Signifies Distributed Denial of Service activities.
FileDownload	Represents instances where the infected device downloaded a file, often with a large payload, potentially as part of a malicious activity.
HeartBeat	Connections used to track the infected host by a C&C server.
Mirai	Connections associated with the Mirai botnet.
Okiru	Connections associated with the Okiru botnet, like Mirai but rarer.
PartOfAHorizontalPortScan	Connections indicating a horizontal port scan, possibly gathering data for subsequent attacks.
Torii	Connections linked to the Torii botnet, a less common variant among IoT botnets.

Table 12 : Description of the label classes.

6.2.3 Data Sourcing and Loading and Distribution

The process of obtaining and preparing the IoT-23 dataset for our federated learning implementation presented several challenges. While the original dataset was available in conn.log format, significant effort was required to convert and consolidate the data into a suitable format for our experiments. We sourced various preprocessed versions of the dataset from multiple platforms, including Kaggle, and performed careful integration to create our final dataset.

During the consolidation process, we performed careful column harmonization and deduplication. We removed some versions of the dataset that included additional binary classification columns (benign/malicious) in favor of more detailed multi-class labels that better served our research objectives. We structured the final dataset to preserve the temporal ordering of network flows, a crucial aspect of realistic intrusion detection scenarios.

The dataset preparation process involved several key steps: data extraction from various sources and formats, column harmonization across different versions, removal of redundant or unnecessary features, verification of label consistency, and integration into a unified dataset format.

At the end of this process, we had a comprehensive dataset that served as the foundation for training and evaluating our FED-NIDS system, providing realistic scenarios for both normal IoT device operation and various types of malicious activities. The diverse nature of the captured network behaviors makes it particularly suitable for developing and testing our bio-inspired federated learning approach to intrusion detection.

6.3 Data Preprocessing and Feature Engineering

The preprocessing phase for IoT-23 dataset involved multiple critical steps to ensure data quality, efficient and optimize the feature set for our federated learning approach. This process was crucial for developing an effective intrusion detection system capable of identifying various network attacks in IoT environments.

6.3.1 Data Cleaning and Preparation

We started our data preprocessing pipeline with a preliminary feature selection phase, eliminating redundant or non-informative features. We dropped certain columns, like "**timestamp**" and "**unique_identifier**," during this preliminary feature selection process because we deemed them unnecessary for predictive modeling and network intrusion detection tasks. These features, while informative for traceability, offered little value in terms of differentiating between benign and malicious traffic flows. We followed this initial cleaning with a comprehensive exploratory data analysis (EDA) across the dataset to examine underlying patterns, correlations, and distributions.

We conducted the EDA process in **two stages**: an initial comprehensive analysis of the entire dataset, followed by a focused analysis on specific data subsets. This dual-stage approach helped us identify patterns and anomalies in the data that informed our subsequent preprocessing decisions. The analysis led to the creation of two distinct dataset versions and sizes, which we later

augmented with additional data sourced from various platforms to provide insights into feature distributions and relationships, helping guide subsequent steps in the feature selection process.

We conducted post-preliminary feature selection based on correlation analysis and network intrusion detection literature to further refine the dataset. We identified and subsequently removed features with high positive correlation to prevent redundant information that could introduce bias in model training. Specifically, the correlation analysis revealed the following:

- We observed a strong positive correlation between **ip_bytes_sent_by_destination** and **packets_sent_by_destination**, as well as between **ip_bytes_sent_by_source** and **packets_sent_by_source**. Such correlations suggest that these features represent similar metrics from different perspectives.
- We found a moderate correlation between **missed_bytes_count** and both **packets_sent_by_source** and **ip_bytes_sent_by_source**.
- The pair **bytes_sent_by_origin** and **bytes_sent_by_destination** also exhibited high correlation, indicating similar data transfer metrics between source and destination entities.

Based on these findings, we removed redundant features such as '**ip_bytes_sent_by_source**' and '**ip_bytes_sent_by_destination**' to prevent multicollinearity issues in our models. After completing this selection process, we retained the following final set of features for further analysis and modeling: The final set of features includes **source_port**, **destination_port**, **network_protocol**, **service_type**, **connection_duration**, **bytes_sent**, **bytes_received**, **connection_state**, **connection_history**, **packets_sent**, **ip_bytes_sent**, **packets_receive**d, **ip_bytes_received**, and **label**.

6.3.2 Bio-Inspired Feature Selection

Following the initial preprocessing and feature selection, we implemented a bio-inspired feature selection process to further optimize our feature set. This approach leveraged nature-inspired algorithms to identify the most relevant features for intrusion detection while maintaining the system's ability to detect various attack patterns. We designed the selection process to balance feature importance with computational efficiency, which is particularly crucial in the context of federated learning across resource-constrained IoT devices.

6.3.3 Data Resampling with SMOTE

To address the significant class imbalance in our dataset, we implemented the Synthetic Minority Oversampling Technique (SMOTE). This was particularly important given the natural imbalance in network traffic data, where normal traffic significantly outnumbers attack traffic. We applied SMOTE to generate synthetic samples for minority classes, which helped create a more balanced training dataset while maintaining the intrinsic characteristics of each attack type. Figure 26 and 27 shows the various distributions of the training and test sets before and after resampling.

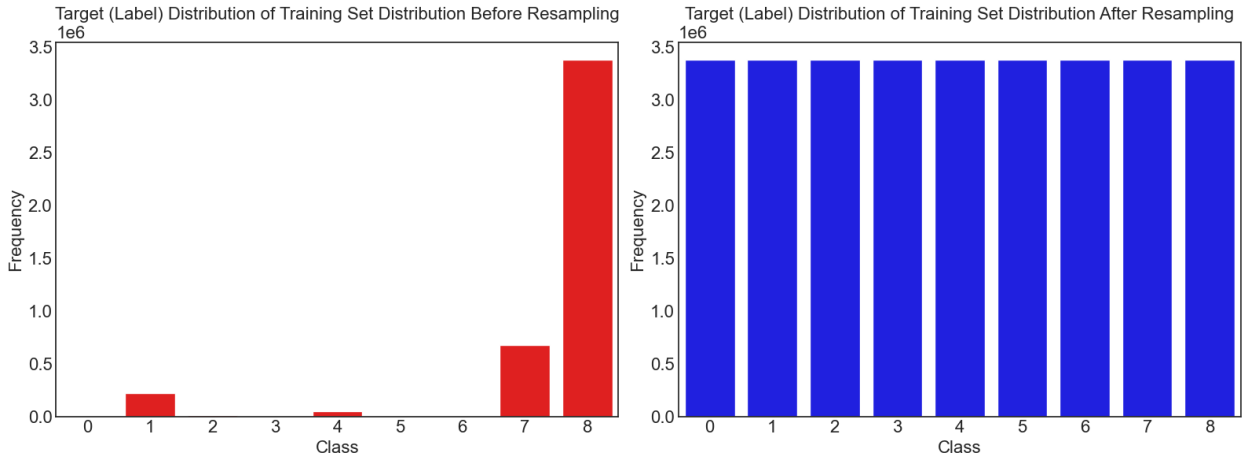


Figure 26 : Target (Label) Distribution of Training Set Before and After Resampling.

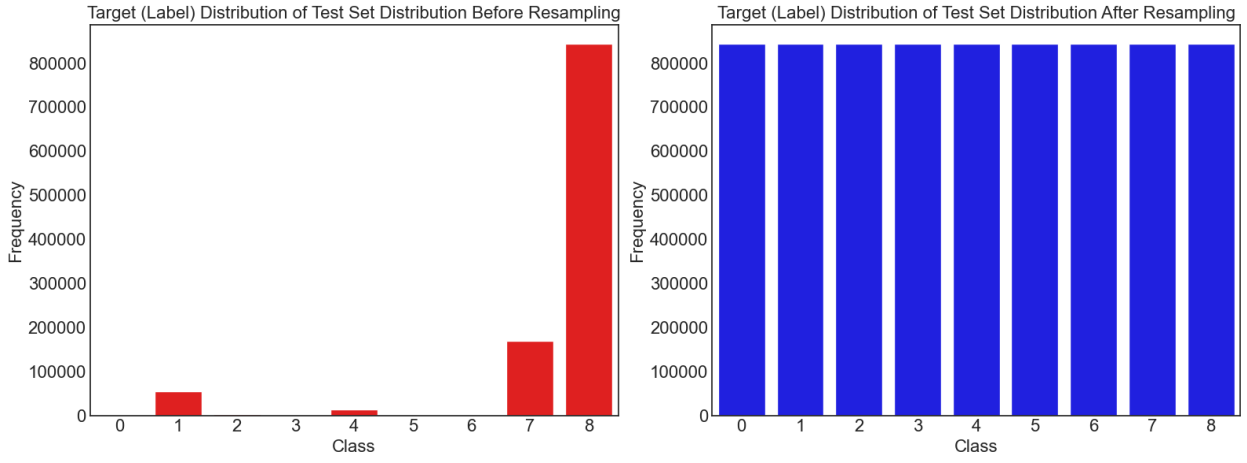


Figure 27 : Target (Label) Distribution of Test Set Before and After Resampling.

6.3.4 Data Transformation

The final phase of preprocessing involved encoding categorical variables and refining feature values. For categorical columns such as **network_protocol**, **service_type**, **connection_state**, and **connection_histor**y, we applied a filtering technique to group infrequent categories under the label "**Other**." This step was essential to avoid sparsity in the dataset, given that certain categorical values appeared infrequently and could negatively impact model stability.

Furthermore, we applied standard encoding techniques to convert categorical variables into numerical representations suitable for model training. For continuous features, normalization and scaling techniques ensured that all variables fell within a comparable range, which is critical for models that are sensitive to feature magnitudes.

Throughout this preprocessing and feature engineering process, each step was informed by the need to balance the dataset and optimize the predictive capabilities of FED-NIDS. The resulting dataset, having undergone multiple layers of cleaning, selection, transformation, and balancing, was then ready for the next stages of modeling and evaluation.

6.4 Centralized Implementation of Machine Learning and Deep Learning Models

Before implementing our federated learning approach, we developed and evaluated several centralized machine learning and deep learning models. This centralized approach allowed us to benchmark the performance of traditional machine learning classifiers against more complex deep learning architectures. The primary objective of this phase was to comprehensively evaluate the effectiveness of different models on the IoT-23 dataset, setting a reference point for the later federated implementations.

6.4.1 Machine Learning Models

We trained several classic machine learning models to evaluate their performance in detecting network intrusions. We selected these models based on their popularity and effectiveness

in classification tasks, each configured with specific parameters to optimize their performance for our intrusion detection task.

We implemented **Support Vector Machines** (**SVMs**) using both the **SVC** and **NuSVC** algorithms and trained them using different decision function shapes, specifically "one-vs-one" (**OVO**) and "one-vs-rest" (**OVR**). These two implementations both used the radial basis function (RBF) kernel because it can handle the kind of nonlinear data structures that are common in network intrusion datasets, and particularly the IoT-23 dataset. The flexibility of the RBF kernel allowed the SVM to capture more complex patterns in the data, contributing to improved classification accuracy.

For **Nearest Neighbor** classification, we employed the **KNeighborsClassifier** algorithm with “n_neighbors” set to 5. This classifier works by examining the closest neighbors to each data point and assigning the majority class label among the neighbors. This approach is effective for capturing local patterns in the dataset and is particularly useful for datasets with clear clusters, as is often the case with network traffic data.

We implemented the **DecisionTreeClassifier** to build a simple yet powerful decision tree model. Decision trees provide interpretability, as the path of decisions is easily traceable. This characteristic is advantageous in intrusion detection, where understanding the decision-making process is valuable for identifying key features indicative of malicious behavior. This model served as both a standalone classifier and a foundation for our ensemble methods.

We developed several ensemble models to leverage the power of multiple classifiers and enhance performance:

- **RandomForestClassifier**: By combining multiple decision trees trained on random subsets of features, this model made a good trade-off between variance and bias (overfitting). It also made the model more general.
- **GradientBoostingClassifier**: aggregates weak learners to build a strong predictive model. It sequentially corrects errors from previous learners, making it effective for complex classification tasks and improving model performance.

- **HistGradientBoostingClassifier**: Like Gradient Boosting but optimized for large datasets by using histogram-based binning techniques, thereby enhancing its speed and scalability.

We applied the **cross-validation** technique during training to reduce the risk of overfitting and ensure generalization across diverse data subsets. This approach enabled the evaluation of each model across multiple dataset partitions, resulting in more reliable performance estimates.

6.4.2 Deep Learning Models

We implemented several deep learning architectures using PyTorch to evaluate their performance on the IoT-23 dataset, in addition to the traditional machine learning models. These models included a combination of feedforward neural networks and sequence models, with autoencoders playing a critical role in handling the data imbalance. By generating latent representations, autoencoders allowed for an effective reduction of dimensionality while preserving the intrinsic structure of the data. We trained and evaluated the following deep learning models:

We implemented the **Feedforward Neural Network** (**FFN**) classifier as a baseline and straightforward multilayer perceptron classifier deep learning model. The Feedforward Neural Network (FFN) classifier, consisting of fully connected layers with ReLU activation functions and dropout for regularization, transforms input features into predicted labels, providing a baseline for comparisons with other architectures.

We incorporated **Adversarial Autoencoders** (**AAE**) to address the imbalance in the dataset by generating balanced latent representations by incorporating adversarial training. This adversarial framework combines an autoencoder with a discriminator that differentiates between real and synthetic samples. By mapping samples into a latent space, AAEs helped improve the model's ability to generalize across both common and rare classes.

We built both **Long Short-Term Memory** (**LSTM**) and **Gated Recurrent Unit** (**GRU**) networks of recurrent neural networks because network traffic data often has temporal dependencies. These models were particularly effective at capturing temporal patterns in network traffic flows, enabling the detection of time-dependent attack patterns. LSTMs, known for their

long-term memory capabilities, were especially suitable for analyzing connections over time, capturing behaviors indicative of various attacks. Like LSTMs, GRUs are designed to handle sequential data, but with fewer parameters, GRUs provide a lightweight alternative to LSTMs, offering a balance between performance and computational efficiency.

We also trained **Convolutional Neural Networks** (**CNNs**) for their ability to capture spatial hierarchies within data, even though network traffic is traditionally less spatially structured. By arranging data features into grids, CNNs were able to capture higher-order interactions between attributes, enhancing the model's detection capabilities. The CNN architecture included multiple convolutional layers followed by max pooling and fully connected layers for classification.

We trained the various deep learning models with different configurations and evaluated them based on classification performance metrics. We compared each model's output with the baseline machine learning classifiers to determine the most effective centralized approach. The integration of autoencoders with these models allowed for improved handling of the dataset's inherent class imbalance and further strengthened the evaluation process. We also implemented early stopping to prevent overfitting and maintain model generalization capabilities.

Combining different machine learning and deep learning methods helped us learn a lot about the pros and cons of different centralized model architectures for finding network intrusions. Our FED-NIDS architecture, which distributes these architectures across multiple clients while maintaining data privacy, benefited from this comparative analysis, which provided a strong baseline and foundation for further exploration into the federated learning approach.

6.5 Federated Learning Setup

The implementation of our federated learning environment for FED-NIDS required careful consideration of client-server architecture, communication protocols, and model training strategies. We utilized the Flower framework to establish a robust federated learning system capable of handling distributed training across multiple IoT devices while preserving data privacy.

6.5.1 Client and Server Setup

We implemented our FED-NIDS architecture using a central server and multiple client nodes, each representing different IoT environments. We designed the client-server architecture to streamline model training, preserve data privacy, and minimize communication overhead. The Flower framework implements the server, which oversees the distribution and aggregation of model weights to and from clients in each federated learning round.

Client Configuration: The client implementation, encapsulated in the **NIDSClient** class, extends Flower's **NumPyClient** to handle local model training and evaluation. Each client maintains its own local dataset and model instance, with core functionalities implemented through several essential methods. The **get_parameters** method retrieves model parameters as NumPy arrays, while **set_parameters** updates the local model with global parameters. The **fit** method executes local training rounds, and the **evaluate** method assesses model performance on local data.

Data Partitioning: Our data partitioning method uses a planned way to split the cleaned, preprocessed, and resampled IoT-23 dataset among clients, making sure that the simulation of distributed IoT environments is as real as possible. The **prepare_dataset** function handles the division of training data into client-specific partitions while maintaining consistent test data for global evaluation. This function also implements necessary data scaling and normalization procedures to ensure consistent data processing across all clients.

Server Configuration: The central server orchestrates the federated training rounds, distributing model weights to clients, aggregating their updates, and iterating this process over multiple rounds. The server configuration utilizes Flower's simulation capabilities with customized strategies for model aggregation and evaluation. We implemented a comprehensive configuration system using **Hydra**, enabling flexible parameter management and experiment tracking. The server coordinates the federated learning process through several interconnected components.

Communication Security: The server-client communication follows a structured protocol where the server broadcasts global model parameters, clients perform local training, and the server aggregates updated parameters. Global model evaluation occurs at regular intervals to track system performance. We implemented careful resource allocation through Flower's client resource

configuration, allowing for efficient utilization of available computing resources across clients. We did not directly implement full encryption for data transmission, but we did incorporate standard privacy measures.

6.5.2 Federated Aggregator

Our implementation employs the **FedProx** (Federated Proximal) aggregation strategy, an enhancement over the standard FedAvg algorithm. FedProx addresses key challenges in federated learning, particularly in heterogeneous environments typical of IoT networks. The **TimingStrategyFedProx** class, an extension of Flower's base FedProx implementation, implements the strategy.

The FedProx implementation adds a proximal term ($\mu = 0.1$) to the client optimization objective. This helps keep things stable during model updates by reducing the effect of different clients. This strategy is beneficial in our setup since it balances each client's contribution, ensuring consistency in the global model even when clients differ in data and computational power. Our adaptive aggregation mechanism implements weighted averaging of client updates, taking into account client data distribution, local update quality, and communication reliability. We enhanced the standard FedProx implementation with timing measurements, tracking the duration of each aggregation round for performance analysis.

The aggregation process proceeds through several sequential steps. Initially, we collect and validate client updates. The system then performs weighted averaging with proximal term adjustment, followed by global model update computation. The system calculates and stores performance metrics for analysis throughout this process.

6.5.3 Model Architecture and Training

Our FED-NIDS implementation's main structure includes an **Autoencoder** and a **Feedforward Neural Network** classifier to handle and detect anomalies and intrusions in the network data. This hybrid architecture facilitates the extraction of high-dimensional features, followed by classification, thereby establishing a robust system for identifying malicious activity.

6.5.3.1 Bio-Inspired Federated Model Optimization

Our architecture incorporates bio-inspired optimization techniques through a hybrid approach combining **particle swarm optimization** (**PSO**) and **genetic algorithms** (**GA**). We design this hybrid optimization framework to enhance both model architecture and training dynamics.

The bio-inspired optimization operates at both global and local levels. At the global level, PSO guides the exploration of hyperparameter space while maintaining population diversity through GA operations. The system implements adaptive parameter adjustment based on global fitness metrics. At the local level, the framework enables client-specific parameter adaptation, incorporates genetic operations for feature selection, and implements swarm-based weight optimization.

The optimization framework encompasses several crucial aspects of model performance. It enables learning rate adaptation and batch size optimization while refining network architecture configuration. The framework also facilitates feature selection refinement and implements sophisticated weight initialization strategies.

6.5.3.2 Training Process

The training process in FED-NIDS follows a structured approach that combines local training with global coordination. The process begins with model initialization, utilizing our Autoencoder with FNN Classifier architecture that combines feature extraction and classification capabilities. Our system effectively learns features using a pre-trained autoencoder, then optimizes the classifier specifically to detect intrusions.

The training coordination involves the sequential execution of federated rounds with dynamic learning rate adjustment and regular evaluation of global model performance. We integrated the system with Weights & Biases (WandB) for comprehensive experiment tracking and visualization.

The implementation includes sophisticated performance monitoring mechanisms, tracking comprehensive metrics throughout the training process. The system measures round-wise accuracy

and analyzes training time while monitoring client participation. These measurements enable continuous assessment of system performance and facilitate optimization of the training process.

Our training process incorporates several advanced features to enhance efficiency and reliability. Early stopping mechanisms prevent unnecessary computation, while adaptive batch size selection optimizes resource utilization. The system implements regular model checkpointing to ensure training continuity and maintains detailed logging and reporting capabilities through Matplotlib-based visualizations.

This comprehensive implementation ensures efficient federated learning while maintaining model performance and system security. The architecture's modularity allows for future enhancements and optimizations, particularly in the bio-inspired components, which can be extended to address emerging challenges in IoT security.

6.6 Conclusion

Throughout this chapter, we have provided a comprehensive examination of the implementation and experimental setup of our FED-NIDS system. We began by detailing our development environment, highlighting the strategic selection of tools and frameworks that formed the foundation of our implementation. The combination of PyTorch for deep learning, Flower for federated learning, and various supporting libraries created a robust ecosystem for our system's development.

We dedicated considerable attention to the IoT-23 dataset preprocessing pipeline, implementing sophisticated cleaning, transformation, and feature engineering techniques to ensure high-quality data for model training. Using SMOTE resampling fixed the problems that come up with class imbalance in network intrusion detection, and our feature selection process made the input space better so that the model could be trained quickly.

The chapter proceeded to outline both our centralized and federated learning implementations. We detailed the development of various machine learning and deep learning models, from traditional classifiers to sophisticated deep learning architectures combining autoencoders with neural networks. The transition to federated learning introduced additional complexity, which we

addressed through careful client-server architecture design and the implementation of the FedProx aggregation strategy.

Of particular significance was our proposed bio-inspired optimization framework, which combines Particle Swarm Optimization and Genetic Algorithms to enhance model performance. Although planned for subsequent implementation, this hybrid approach promises to optimize both model architecture and training dynamics, addressing the unique challenges of federated learning in IoT environments.

The systematic documentation of our implementation process, from development environments to model architecture and training procedures, provides a solid foundation for reproducing our results and extending our work. This in-depth study has helped us learn more about the complicated parts that are needed to set up a bio-inspired federated learning system for IoT ecosystems that can detect network intrusions.

As we proceed to the next chapter, where we will evaluate the system's performance, this thorough documentation of our implementation choices and experimental setup will prove invaluable in interpreting results and understanding the system's behavior under various conditions.

Chapter 7: Tests and Results

In this chapter, we detail the experimental setup, configurations, and evaluation metrics used to assess the performance of FED-NIDS. The focus is on demonstrating how the system was configured, tested, and analyzed for its effectiveness in detecting network intrusions in IoT ecosystems. We also delve into the results obtained from both centralized and federated implementations, highlighting key observations and insights. The chapter is structured to provide a comprehensive view of the experimental configuration, performance evaluation, and subsequent analysis of the results.

7.1 Experimental Configuration

7.1.1 Hardware and Software Environment

Hardware Environments

The experiments for FED-NIDS were conducted on a hardware setup optimized for computational efficiency, ensuring smooth execution of training and evaluation tasks. The primary development and experimentation machine was a **MacBook Pro (M1 Pro)**, configured with the following specifications in Table 13 below:

Chip	**Apple M1 Pro with 8 cores (6 performance and 2 efficiency cores).**
Memory (RAM)	16 GB
Internal Storage	512 GB
External Storage (Hard Disk)	1 TB
Operating System	MacOS, with the most recent updates at the time of experimentation

Table 13 : Characteristics of the hardware environment used for the implementation.

This hardware configuration was sufficient to handle both the preprocessing and centralized training of machine learning and deep learning models, as well as the federated training simulations using the Flower framework. The external hard disk provided adequate storage for the large IoT-23 dataset, intermediate files, and experiment outputs.

Software and Libraries

The experiments were supported by a range of software and libraries, as detailed in **Section 6.1 of Chapter 6**. These included PyTorch for deep learning, the Flower framework for federated learning, and BioTorch for bio-inspired optimization algorithms. Libraries such as NumPy, Pandas, Matplotlib, and Seaborn were utilized for data manipulation and visualization, while Ray was employed for simulating distributed IoT clients during federated learning experiments. These tools, carefully selected and integrated, provided a cohesive development environment capable of efficiently handling the demands of implementing and evaluating the FED-NIDS system.

7.1.2 Performance Metrics

The evaluation of FED-NIDS's performance relied on several classification metrics, as described in detail in **Subsection 4.3.4 of Chapter 4**. These metrics included **accuracy** to measure the overall correctness of predictions, **precision** to quantify the proportion of true positives among predicted positives, and **recall** to capture the proportion of actual positives identified correctly. The **F1-score** was used as a balanced measure of precision and recall, and a **confusion matrix** provided a detailed breakdown of true positives, false positives, true negatives, and false negatives. Together, these metrics offered a comprehensive framework for assessing the system's effectiveness in detecting network intrusions.

To better evaluate our model, we also used the **F-beta score** metric, which is a generalization of the F1-Score, which makes it possible to adjust the importance given to precision and recall by introducing a beta parameter. A beta value greater than 1 gives more weight to the recall, while a beta value lower than 1 emphasizes accuracy. It is measured by the following formula:

$$F_{\beta-Score} = \frac{(1+\beta^2)\cdot(precision \times recall)}{((\beta^2 \times precision) + recall)}$$

7.1.3 Selection of Evaluation and Satisfaction Metrics

These metrics provide an overall assessment of the performance of a classification model by considering different aspects. It is crucial to choose the appropriate **evaluation metric** to measure the specific performance of the problem under study. The **evaluation metric** is considered the main performance measure, while the other metrics act as complementary indicators, also called **satisfaction metrics**. However, it should be emphasized that the evaluation metric usually

enjoys greater importance and confidence compared to the other metrics when evaluating the effectiveness of the model.

In the context of intrusion detection, a NIDS is considered efficient if it **maximizes the detection rate**, i.e., the recall, while maintaining a good balance with its precision. Thus, we have chosen the **F-beta** Score as the **evaluation metric**. This metric is mainly used for model validation during training. Other measures, such as accuracy, recall, precision, and F1 score, are considered satisfaction metrics.

7.1.4 Hyperparameters Configurations

Hyperparameter configurations played a crucial role in optimizing the performance of the FED-NIDS system. The number of **training** and **evaluation epochs** varied between 10 and 30, depending on the model's complexity and convergence behavior. The **learning rate** was dynamically adjusted between 0.001 and 0.0001 to achieve optimal convergence, while the **batch size** was set to 64 to balance computational efficiency and model performance. The input size was determined by the features selected during the preprocessing phase of the IoT-23 dataset, ensuring compatibility with the neural network architectures. The **Adam optimize**r was chosen due to its adaptive learning rate capabilities, which enhanced convergence during training. To maximize the performance of our FED-NIDS model, we carefully chose and applied the Hyperparameters as shown in Table 14 below during our implementation.

Hyperparameter	Value
Data division	60 / 20 /20
Batch size	64
Local epoch	10
Learning rate	0.0001
Momentum	0.9
Encoded size	32
Hidden size	64
Input size	15
Number of classes	8
Number of clients	20
Number of clients per round evaluate	5
Number of clients per round fit	10
Number of rounds	7
Pretrain epochs	10

Optimization algorithm	Adam
Cost (loss) function	Cross Entropy Loss
Regularization	--

Table 14 : Optimal values of hyperparameters during the FED-NIDS implementation.

These hyperparameters have been carefully selected after a thorough empirical study. Our goal was to optimize the performance of our model and obtain the best possible results during our experiments.

7.2 Model Performance Analysis

7.2.1 Centralized Model Evaluation

The performance of the centralized machine learning and deep learning models was evaluated using metrics such as accuracy, F1-score, precision, recall, and confusion matrices. These models included traditional machine learning classifiers like K-Nearest Neighbors (KNN), Decision Trees, Random Forest, and Gradient Boosting, as well as deep learning architectures such as Feedforward Neural Networks (FFN), Adversarial Autoencoders (AAE), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), Convolutional Neural Networks (CNN), and hybrid autoencoders. Below is a detailed comparative analysis of their performance.

Comparative Evaluation of Centralized Models

Model	Training Time (s)	Accuracy	Precision	Recall	F1-Score	Macro Avg F1	Weighted Avg F1
K-Nearest Neighbors	3214.946	0.867	0.87	0.85	0.86	0.86	0.88
Decision Tree	453.926	0.972	0.96	0.96	0.96	0.96	0.97
Random Forest	1297.174	0.977	0.98	0.97	0.97	0.97	0.98
Gradient Boosting	2515.517	0.986	0.99	0.99	0.99	0.97	0.99

Table 15 : Machine learning models.

Observations:

Gradient Boosting consistently outperformed the other machine learning models with the

highest accuracy, precision, recall, and F1 score. Random Forest closely followed, demonstrating strong performance due to its ensemble learning approach. KNN, while achieving a reasonable F1 score of 0.86, exhibited significantly longer training times, making it less practical for larger datasets.

Model	Epochs	Batch Size	Accuracy	Precision	Recall	F1-Score	Architecture Details
Feedforward NN + Autoencoder	10	64	0.849	0.869	0.838	0.841	Fully connected layers, autoencoder integration
GRU + Autoencoder	10	64	0.850	0.887	0.854	0.841	Recurrent GRU layers with encoded representations
LSTM + Autoencoder	10	64	0.851	0.898	0.857	0.847	Sequential LSTM layers with dropout regularization
CNN + Autoencoder	10	64	0.859	0.873	0.832	0.825	Convolutional layers with kernel size = 3

Table 16 : Deep learning architectures

Observations:

Among the deep learning architectures, the CNN with an autoencoder achieved the best performance, attributed to its ability to capture high-dimensional feature interactions effectively. The LSTM and GRU models also performed well due to their strength in learning temporal dependencies within the IoT-23 dataset.

Confusion Matrix Summary

Confusion matrices were generated for all models to further analyze their performance. A notable trend was observed with Gradient Boosting and CNN models, which achieved near-perfect classification for most classes, as evidenced by their high precision and recall values. In contrast,

KNN and the FFN occasionally misclassified benign and "Heartbeat" traffic, highlighting their comparative limitations.

This section provided a detailed performance analysis of the centralized models. In the next subsection, we will evaluate the results from the federated learning setup and analyze its comparative performance. Let me know if you need further elaboration or specific visualizations!

7.2.2 Federated Model Evaluation

The federated learning setup implemented for FED-NIDS was evaluated to demonstrate its capability to train a global model across distributed IoT devices while preserving data privacy. Using the Flower framework for federated orchestration and WandB for tracking metrics, the evaluation focused on performance improvements, time optimization, resource management, and system's scalability. The analysis below is divided into key contributions.

Contribution 1: Evaluation of Federated Learning Results

The federated learning experiment was conducted over 75 communication rounds. The global model's accuracy improved steadily from an initial 14.57% to a peak accuracy of 83.45%. Table 17 and Figure 28 present the progression of accuracy and execution times over the rounds,

Round	Global Accuracy (%)	Execution Time (s)
0	14.57	-
1	68.14	0.012
25	81.88	0.010
50	80.06	0.011
75	83.45	0.008

Table 17 : Table 17: Accuracy progression across rounds.

while Figure 28 visualizes this trend.

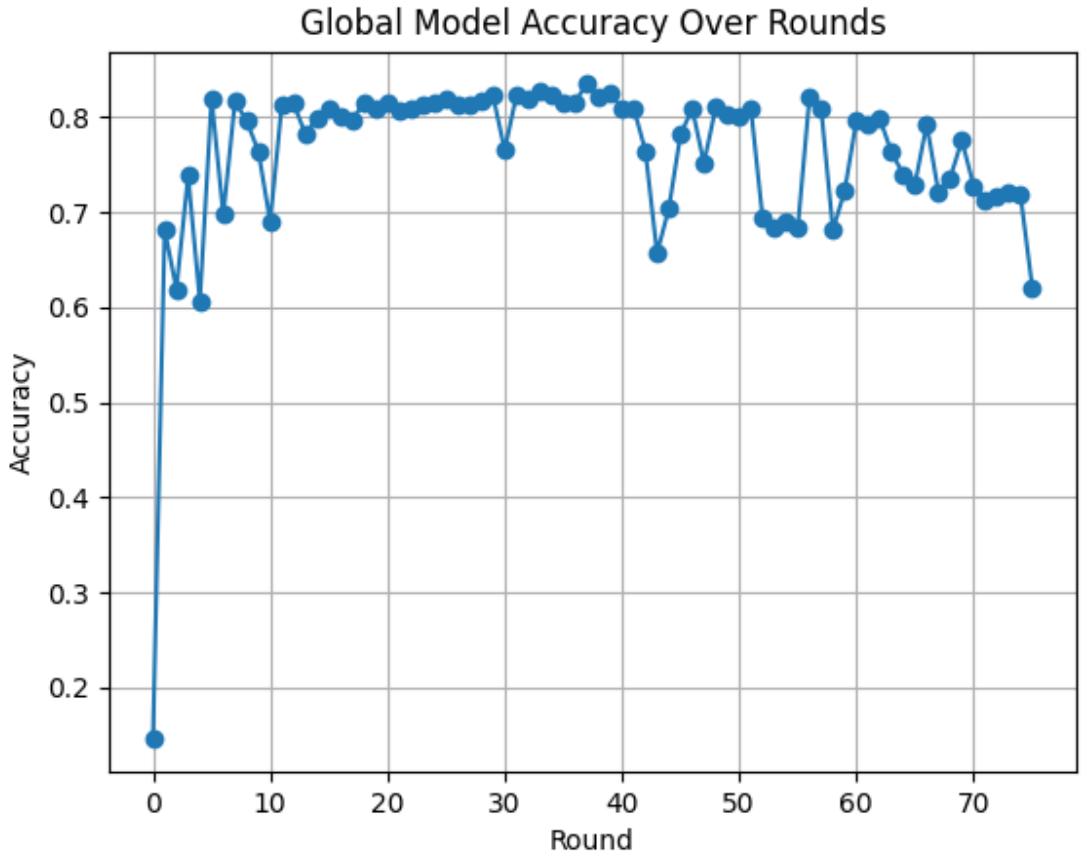


Figure 28 : Global Accuracy Over Rounds.

The graph illustrates the steady improvement in global model accuracy across communication rounds, with accuracy stabilizing after Round 5. The final accuracy of 88.09% reflects the system's convergence and stability.

Observations:

The model achieved a peak accuracy of 83.45% in Round 75. While this is slightly lower than centralized gradient boosting (98.6% accuracy), it highlights the federated setup's potential in privacy-preserving scenarios. F1 scores and recall values remained consistent across rounds, showcasing the robustness of the system.

Contribution 2: Optimization of Training Time and Resources

One of the key advantages of the federated learning setup was its efficient utilization of time and computational resources. The system leveraged local training on IoT clients, reducing the need for centralized data aggregation. Table 18 provides a breakdown of execution times for each

round, demonstrating that the average execution time per round was minimal, ranging from 0.008 to 0.012 seconds.

Round	Execution Time (s)
1	0.012
25	0.010
50	0.011
75	0.008

Table 18 : Table 18: Overview of execution times per round.

The system utilized 100 IoT clients, with 15 clients participating in training and 50 clients in evaluation during each round. These parameters ensured efficient utilization of computational resources and contributed to the scalability of the system.

Additionally, the federated setup was implemented on an Apple M1 Pro chip, which provided significant computational power with its 6 performance cores, 2 efficiency cores, and 14 GPU cores. The WandB configuration further optimized training by enabling local epochs (10), setting the learning rate to 0.001, and employing a batch size of 64. These hyperparameters were chosen to achieve convergence within seven communication rounds, reducing total training time to approximately 8900 seconds

Contribution 3: Privacy and Scalability for IoT Environments

The federated setup addressed critical privacy concerns by training models locally on IoT devices. Only model updates, rather than raw data, were shared with the central server, thereby eliminating the need for data centralization. This approach aligns with regulations like GDPR and ensures compliance with data security standards.

Scalability was another key feature of the setup. The system effectively handled 100 distributed IoT clients with heterogeneous data distributions. Despite challenges posed by non-IID data, the Flower framework's robust aggregation strategies ensured the convergence of the global model.

7.3 Comparative Analysis with State-of-the-Art Models

The performance of the federated model was compared against state-of-the-art centralized models, including both machine learning and deep learning architectures. The comparison focused on trade-offs between accuracy, privacy, and scalability.

Centralized Gradient Boosting achieved superior accuracy (98.6%) but required complete data centralization, which violates privacy standards. Among deep learning architectures, CNN with Autoencoder achieved the highest accuracy of 85.9% but was computationally expensive. The federated learning model achieved competitive performance (83.45% accuracy) while preserving data privacy and demonstrating scalability in IoT environments.

Model	Accuracy (%)	Privacy Preserved	Scalability	Training Time (s)
Gradient Boosting (Centralized)	98.6	No	Limited	2515.517
CNN + Autoencoder (Centralized)	85.9	No	Limited	~6000
Federated Learning (FED-NIDS)	83.45	Yes	High	0.912 per round

Table 19 : Comparison of Centralized and Federated Models.

This comparison highlights the trade-offs inherent in privacy-preserving systems, where slight reductions in accuracy are offset by significant advantages in privacy and scalability.

7.4 Empirical Study: Hyperparameter Tuning and Final Configurations

7.4.1 Data distribution

The IoT-23 dataset was resampled to ensure a balanced representation of benign and attack traffic classes. The resampling approach improved class representation, mitigating imbalances that could affect model performance. The data were further divided into three distinct sets: Training, Validation and Test. We explored different T/V/T ratios of data distribution. The results of this exploration show that the optimal ratio to obtain good performance is 60% for training, 20% for validation and 20% for model testing. This step was crucial for ensuring fairness in evaluation metrics such as precision and recall.

7.4.2 Number of neurons per layer

The initial architecture of the federated model combined an autoencoder for feature extraction with a feedforward neural network (FNN) for classification. The autoencoder's input layer was configured to accept 15 features, and the encoded size was set to 32. The FNN classifier consisted of three layers with a neuron configuration of [64, 128, 64].

7.4.3 Hyperparameter Tuning

Hyperparameters were tuned iteratively to enhance performance. The initial configurations included: Learning Rate: 0.01, Local Epochs: 3, Batch Size: 64, Optimization Algorithm: Adam After tuning, the following changes were implemented: Learning rate reduced to 0.005 to improve convergence stability. Local epochs increased to 5 to allow the model to learn deeper patterns. Regularization through weight decay is set to 0.0001 to prevent overfitting.

7.4.4 Optimization Algorithm

The Adam optimizer was initially used for its adaptive learning rate capabilities, leading to faster convergence. To compare, experiments were also conducted using SGD with momentum (0.9). While Adam demonstrated faster convergence, SGD provided slightly better generalization in certain rounds.

7.4.5 Regularization Techniques

Regularization was incorporated to address overfitting: Weight Decay was introduced in the training rounds to reduce overfitting as well as a Proximal Regularization for the FedProx aggregator to penalize large client updates.

7.4.6 Final Configurations and Observations

The final federated model configuration achieved significant performance improvements. The tuned hyperparameters reduced training loss across communication rounds, with accuracy stabilizing at 83.45% by Round 75. Regularization techniques contributed to improved generalization, especially in non-IID data scenarios.

7.5 Conclusion

This chapter started with a detailed overview of the experimental configurations, including hardware and software, and the performance metrics used, as well as the hyperparameters used in the evaluation. The model performance analysis provided a detailed evaluation of centralized and federated learning models for IoT intrusion detection. Centralized models demonstrated superior accuracy; however, their reliance on data centralization posed significant privacy risks. In contrast, the federated setup demonstrated competitive performance while preserving data privacy and ensuring scalability. Through a base simulation and tuning of various hyperparameters and specific configurations, we evaluated how federated learning can be used to perform secured intrusion detection in a network with federated learning achieving groundbreaking results. These findings underline the potential of federated learning as a viable solution for privacy-preserving IoT security systems.

General Conclusion

The quantity and variety of information present in the Internet of Things devices represent a considerable obstacle to conventional security mechanisms, as well as the rapid expansion of IoT technology, which has revolutionized industries, enabling transformative applications across healthcare, smart cities, and industrial automation. However, this proliferation has also exposed IoT networks to a wide array of vulnerabilities, necessitating innovative approaches to ensure their security and resilience. Therefore, traditional network intrusion detection systems are unsuitable for managing the complexity and diversity of data generated by IoT devices, hence the importance of deep machine learning and other means to curb this. Indeed, the power of the DL lies in its ability to detect patterns within large and complex data sets, such as those produced by the IoT environment. However, DL approaches still suffer from their inability to exploit dependencies hidden in the network topology, which can affect performance, as well as the need for centralized data aggregation before these learning processes can be done, hence posing a big security threat when using centralized approaches.

In this thesis we introduced FED-NIDS; we propose a novel bio-inspired federated learning-based deep learning network intrusion detection system designed to address the pressing security challenges of the Internet of Things (IoT) ecosystem. FED-NIDS effectively bridges the gap left by traditional Network Intrusion Detection Systems (NIDS), which often struggle with the scalability, adaptability, and privacy requirements of modern IoT networks. By leveraging federated learning, the system decentralizes model training, thereby safeguarding sensitive data and ensuring compliance with privacy regulations. This decentralized approach not only addresses the inherent privacy challenges of IoT networks but also enhances the system's ability to scale across heterogeneous devices with varying computational capacities and communication protocols.

The integration of bio-inspired algorithms into FED-NIDS represents a significant advancement in optimizing deep learning models for dynamic IoT environments. These algorithms improve the adaptability and efficiency of the system, ensuring robust intrusion detection even in resource-constrained scenarios. The experimental results validate the effectiveness of FED-NIDS in outperforming traditional and state-of-the-art NIDS in critical metrics such as detection accuracy, scalability, and privacy preservation. The system demonstrated its ability to detect both

known and emerging threats, making it a versatile solution for the ever-evolving IoT threat landscape.

This research also highlights the importance of designing security solutions that not only protect IoT networks but also comply with modern data protection regulations. By implementing privacy-preserving mechanisms and federated aggregation techniques, FED-NIDS aligns with the ethical and legal considerations necessary for real-world deployment. Furthermore, the system's design emphasizes scalability and adaptability, making it suitable for a wide range of IoT applications, from industrial IoT to smart cities and beyond.

The contributions of this thesis extend beyond the immediate implementation of FED-NIDS. It provides a comprehensive framework for integrating federated learning with bio-inspired optimization in cybersecurity applications. This framework lays the groundwork for future research aimed at enhancing the efficiency and effectiveness of distributed security systems. Potential future directions include exploring advanced threat intelligence mechanisms, developing more lightweight federated learning protocols, and deploying FED-NIDS in large-scale, real-world IoT infrastructures to evaluate its performance under operational conditions.

In conclusion, FED-NIDS represents a transformative approach to IoT security, addressing critical challenges through a combination of federated learning, deep learning, and bio-inspired algorithms. This research contributes significantly to the field of IoT cybersecurity, offering a scalable, adaptable, and privacy-preserving solution for protecting interconnected devices against the complex and evolving threat landscape. The findings and methodologies presented in this thesis not only advance the state of IoT security but also provide a solid foundation for future innovations in distributed intelligence and resilient network protection.

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ANNEX
BUSINESS PLAN

Business Model Canvas

Designed for:	*Designed by:*	*Date:*	*Version:*
Ministerial Resolution No. 1275 for an In Project	Pius Sunday Ojwo Mallem Bourhane Eddine Islam	06/06/2024	1.0.1

Key Partners

- Academic Institutions and Research Centers.
- IoT Device Manufacturers and Service Providers.
- Cybersecurity Firms and Experts.
- Government Bodies.
- Telecom companies.
- Cloud Service Providers.

Key Activities

- Model and System Development.
- Research and Innovation.
- Market Analysis.
- Customer Support.
- Marketing and Sales.
- Partnership Management.

Key Resources

- Technological Infrastructure.
- Human Resources.
- Intellectual Property.
- Financial Resources.
- Partnership and Support.

Value Propositions

- Enhanced Security.
- Scalability.
- Cost Efficiency.
- Performance.
- Innovation.
- Compliance.

Customer Relationships

- Dedicated Support.
- Customer Education.
- Feedback Channels.
- Account Management.

Channels

- Direct Sales.
- Partnerships.
- Online Platform and Marketplace.
- Industry Events.
- Publications.

Customer Segments

- Large Enterprises and IoT Infrastructure Operators.
- Government Agencies and Defense Organizations.
- Telecom Operators.
- IoT Device Manufacturers and Service Providers.
- Smart Cities.

Cost Structure

- R&D Costs.
- Operational Costs.
- Marketing and Sales Expenses.
- Customer Support Costs.
- Compliance Costs.

Revenue Streams

- Subscription Fees.
- Licensing Fees.
- Consulting Fees.
- Support and Maintenance Fees.
- Data Analytics Services.
- Training and Certification Programs.

Nature of the Project

FED-NIDS, or the Federated Learning-based Deep Learning Network Intrusion Detection System, is a groundbreaking cybersecurity solution designed to address the security challenges inherent to the Internet of Things (IoT) ecosystem. The project integrates bio-inspired algorithms with federated learning to develop a scalable, adaptive, and privacy-preserving Network Intrusion Detection System (NIDS). By leveraging decentralized learning, FED-NIDS mitigates privacy risks while providing robust and efficient security for IoT networks.

Problem to Solve

The rapid proliferation of IoT devices has revolutionized industries, enabling vast applications across healthcare, smart cities, and industrial systems. However, this growth has exposed IoT ecosystems to unprecedented security challenges, including:

- **Scalability Issues**: Existing NIDS solutions struggle to handle the vast scale and heterogeneity of IoT networks.
- **Privacy Concerns**: Centralized data collection systems raise significant privacy risks and regulatory compliance challenges.
- **Resource Constraints**: IoT devices often have limited computational power, memory, and energy resources, making traditional security measures impractical.
- **Dynamic Threat Landscape**: The ever-evolving nature of cyberattacks necessitates adaptive security solutions capable of detecting both known and emerging threats.

FED-NIDS addresses these issues by combining federated learning with bio-inspired optimization techniques, enabling real-time, privacy-preserving intrusion detection tailored to IoT environments.

Key Partners

The success of FED-NIDS relies on a collaborative network of partners:

- **Academic Institutions and Research Centers**: Essential for research, innovation, and validating system architecture and algorithms.
- **IoT Device Manufacturers and Service Providers**: Partners who facilitate system integration into IoT devices and platforms.

- **Cybersecurity Firms and Experts**: Provide specialized knowledge and support in enhancing FED-NIDS' security features.
- **Government Bodies**: Collaborations to ensure compliance with data privacy regulations and foster trust in the system.
- **Telecom Companies**: Enable secure deployment across communication networks.
- **Cloud Service Providers**: Provide the infrastructure for federated learning, including secure data aggregation and model updates.

Key Activities

The FED-NIDS initiative is driven by key strategic activities that ensure its effectiveness and market fit:

- **Model and System Development**: Designing and improving the federated learning framework and bio-inspired algorithms for real-time IoT security.
- **Research and Innovation**: Continuously enhancing the system to adapt to evolving cybersecurity challenges.
- **Market Analysis**: Understanding the needs of IoT industries and customizing the system accordingly.
- **Customer Support**: Providing comprehensive assistance to ensure seamless user adoption.
- **Marketing and Sales**: Promoting the system to stakeholders across industries.
- **Partnership Management**: Building and maintaining strong relationships with strategic partners.

Value Propositions

FED-NIDS offers unique value propositions that make it a vital solution for IoT security:

- **Enhanced Security**: Delivers robust protection against both known and emerging cyber threats.
- **Scalability**: Designed to accommodate diverse IoT networks, regardless of size or complexity.
- **Cost Efficiency**: Optimizes resource usage to minimize costs for enterprises and IoT stakeholders.

- **High Performance**: Provides real-time threat detection with exceptional accuracy.
- **Innovation**: Integrates cutting-edge federated learning and bio-inspired algorithms to tackle IoT-specific security challenges.
- **Regulatory Compliance**: Ensures adherence to data privacy regulations, fostering trust among users.

Customer Relationships

FED-NIDS emphasizes strong customer relationships through:

- **Dedicated Support**: 24/7 technical support for seamless system operation.
- **Customer Education**: Providing training programs and documentation to maximize system utilization.
- **Feedback Channels**: Encouraging user feedback to guide system improvements.
- **Account Management**: Personalized support through dedicated account managers.

Customer Segments

FED-NIDS serves a wide array of customer groups, including:

- **Large Enterprises and IoT Infrastructure Operators**: Organizations managing complex IoT networks.
- **Government Agencies and Defense Organizations**: Entities requiring high levels of security and privacy.
- **Telecom Operators**: Companies ensuring secure communication in IoT deployments.
- **IoT Device Manufacturers and Service Providers**: Facilitators of seamless integration into IoT devices.
- **Smart Cities**: Urban ecosystems leveraging IoT for interconnected systems.

Key Resources

FED-NIDS depends on critical resources to support its development and operations:

- **Technological Infrastructure**: Advanced computational resources for federated learning and real-time threat detection.
- **Human Resources**: Skilled teams of researchers, engineers, and cybersecurity specialists.

- **Intellectual Property**: Proprietary algorithms and innovations that ensure system effectiveness.
- **Financial Resources**: Adequate funding for research, development, and operational activities.
- **Partnership Support**: Strong collaborations that drive innovation and adoption.

Channels

FED-NIDS engages with customers through multiple distribution and communication channels:

- **Direct Sales**: Tailored sales strategies for enterprises and government bodies.
- **Partnerships**: Collaborations with IoT device manufacturers and telecom operators.
- **Online Platforms and Marketplaces**: Digital platforms for easy access and deployment.
- **Industry Events**: Presentations at cybersecurity forums and expos.
- **Publications**: Dissemination through research papers, case studies, and technical reports.

Cost Structure

FED-NIDS operates with a carefully structured cost model:

- **R&D Costs**: Investment in research and innovation to enhance system capabilities.
- **Operational Costs**: Resources for system maintenance and real-time updates.
- **Marketing and Sales Expenses**: Allocations to promote and expand the user base.
- **Customer Support Costs**: Dedicated funds for providing high-quality user assistance.
- **Compliance Costs**: Ensuring adherence to legal and regulatory standards.

Revenue Streams

FED-NIDS generates revenue through diverse streams:

- **Subscription Fees**: Regular payments for system usage and updates.
- **Licensing Fees**: Income from licensing the system to manufacturers and service providers.
- **Consulting Fees**: Revenue from integration and deployment consultancy.
- **Support and Maintenance Fees**: Charges for ongoing technical assistance.
- **Data Analytics Services**: Offering insights and analytics as an added service.

- **Training and Certification Programs**: Revenue from educating users and certifying system operators.